A 10-Bit 500-MS/s Pipelined SAR ADC With Feedback Factor Compensation in 6-nm FinFET

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***Abstract*—This brief introduces an energy-efficient 10-bit 500-MS/s pipelined SAR ADC that uses feedback factor compensation in 6-nm FinFET technology. The design challenges of residue amplifier in FinFET technology are adequately addressed by incorporating feedback factor compensation. This includes a dynamic negative capacitance circuit at the virtual ground that compensates for the feedback factor and relaxes the requirements of the residue amplifier. This enables the use of an inverter-based residue amplifier that can achieve a high-speed operation of 500-MS/s at a low supply voltage of 0.9 V. The prototype ADC is fabricated in a 6-nm FinFET and occupies 0.014 mm^2. With a Nyquist input signal, it achieves an SNR of 54.2 dB and an SNDR of 53.6 dB, while consuming 2.7 mW from a 0.9 V supply voltage. This brief achieves a competitive Walden figure of merit (FoM) of 13.8 fJ/conv.-step.**

***Index Terms*—Analog-to-digital converter (ADC), 6-nm FinFET process, pipelined SAR ADC, feedback factor compensation, negative capacitance (NC), NC-assisted residue amplifier.**

I. Introduction

IN ADVANCED process nodes, e.g., sub-10 nm process, the level of integration can be significantly increased, but this introduces several challenges for ADC design. For instance, it becomes increasingly difficult to achieve high open-loop gain (A_{OL}) due to the reduction in supply voltage. In addition, design constraints arise from the significant increase in parasitic components. As shown in Table I, the advanced nodes of FinFET technology exhibit a significant increase in parasitic resistance R_P due to additional layers in the fabrication process [1]. An alternative is to utilize wider metal interconnects or additional fins; however, this comes at the cost of increased parasitic capacitance C_P. Consequently, designing using sub-10 nm process poses significant challenges.

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TABLE I
INTERCONNECT PARASITICS WITH TECHNOLOGY SCALING

	40nm	28nm	16nm	7nm	5nm
C_P	1	0.95	0.83	1.05	1.07
R_P	1	1.5	6	20	63

Due to the limited voltage headroom, a cascade approach has become inevitable. In response, a multi-stage amplifier, such as a ring amplifier, has been developed that achieves high energy efficiency by exploiting slew-based charging [2], [3], [4], [5], [6]. Nevertheless, the multi-stage amplifier requires precise stabilization techniques to ensure stability within the closed-loop configuration. An alternative approach is to use open-loop dynamic or G_m-based amplifiers for the residue gain. These have been shown to achieve high energy efficiency by circumventing the stringent requirements on the closed-loop residue amplifiers [7], [8], [9], [10], [11]. However, the open-loop configuration is susceptible to variations in process, voltage, and temperature (PVT), requiring additional calibration that increases the design complexity.

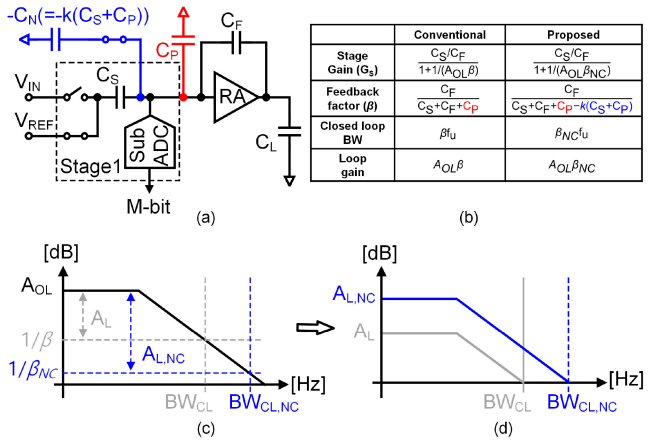
On the other hand, a feedback factor (β) compensation technique using a reference buffer or negative capacitor (NC) at the virtual ground of the residue amplifier has been proposed [12], [13]. This offers a clear advantage in terms of reducing the requirements on the main amplifier, resulting in an improved energy efficiency. However, the main amplifier of these ADCs still uses a high-performance amplifier with a telescopic OTA [13] or two-stage OTA [12]. In this brief, by using the dynamic NC for β compensation, a design optimization of the OTA is described to achieve an improved energy-efficiency. Implemented in a 6-nm FinFET process, a 10-bit 500-MS/s pipelined SAR ADC using a single-stage inverter-based OTA achieves 53.6 dB SNDR and 66.9 dB SFDR for Nyquist input, respectively, while consuming only 2.7 mW, resulting in a Walden figure-of-merit (FoM$_W$) of 13.8 fJ/conv.-step. The remaining of this brief is organized as follows. Section II describes the proposed pipelined SAR ADC, while Section III summarizes the measurement results. Finally, this brief ends with conclusions in Section IV.

II. Proposed Pipelined SAR ADC

A. β Compensation Technique

Fig. 1(a) presents the closed-loop residue amplification incorporating the β compensation technique [12]. During the amplification phase, the NC ($-C_N$) is connected to the virtual

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	Conventional	Proposed
Stage Gain (G_s)	$\frac{C_S/C_F}{1+1/(A_{OL}\beta)}$	$\frac{C_S/C_F}{1+1/(A_{OL}\beta_{NC})}$
Feedback factor (β)	$\frac{C_F}{C_S+C_F+C_P}$	$\frac{C_F}{C_S+C_F+C_P-k(C_S+C_P)}$
Closed loop BW	βf_u	$\beta_{NC} f_u$
Loop gain	$A_{OL}\beta$	$A_{OL}\beta_{NC}$

Fig. 1. (a) The closed-loop residue amplification with the proposed β compensation, (b) comparison of conventional and proposed scheme, Bode plot for (c) A_{OL} and (d) A_L.

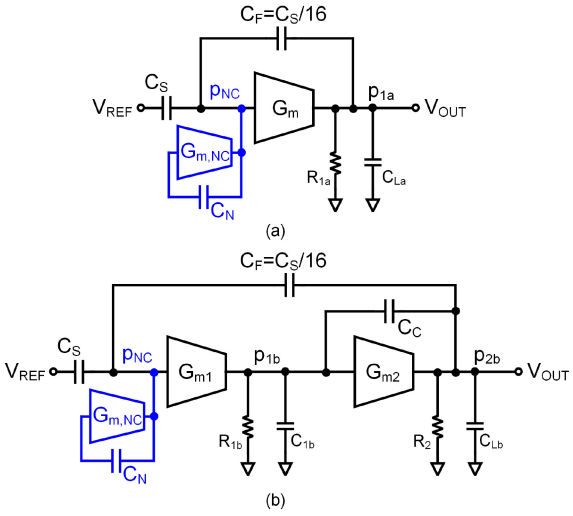


Fig. 2. Closed-loop NC-assisted residue amplifier ac model (a) 1-stage OTA and (b) 2-stage OTA.

ground to provide β compensation. This serves to mitigate the impact of C_P and C_S. As the compensation coefficient k approaches 1, ideal compensation is achieved ($\beta_{NC} = 1$). Fig. 1(b) illustrates that the β compensation enhances the loop gain (A_L) and the closed-loop bandwidth (BW_{CL}) residue amplifier by an x-factor of β_{NC}/β. Fig. 1(c) and (d) visualize the summary of (b) in the form of Bode plots for the A_{OL} and the adjusted A_L ($= A_{OL}\beta$).

B. Comparison of the NC-Assisted Residue Amplifiers

In prior work using a 28-nm CMOS process, the A_{OL} requirement could be reduced to 50 dB by applying β compensation technique and an inverter-based 2-stage OTA is used as a residue amplifier [12]. However, the use of 2-stage OTA requires additional compensation to ensure stability, which adds complexity to the design and limits its bandwidth, also resulting in higher power consumption. Fig. 2 depicts both the 1-stage and 2-stage OTAs with NC. The load capacitance (C_{La} and C_{Lb}) includes the parasitic capacitance resulting from metal routing associated with the residual amplifier,

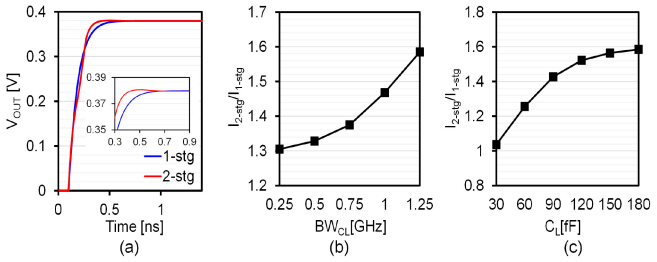


Fig. 3. (a) Step response of two amplifiers, simulated current consumption ratio of two amplifiers vs (b) BW_{CL} and (c) C_L.

	A_L	G_m @ 1-stg OTA (=G_m)	G_m @ 2-stg OTA (=G_{m1}+G_{m2})
w/o NC	25.4dB	26.2mS	34.1mS
w/ NC	37.5dB	8.0mS	15.6mS
X-factor		**3.3X**	**2.2X**

(a)

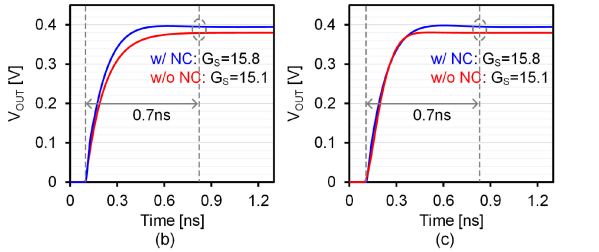


Fig. 4. Comparison of two schemes with NC (a) Summary of the simulated results, step response of (b) 1-stage OTA and (c) 2-stage OTA.

assuming C_{1b} to be one-third of C_{Lb}. Fig. 3(a) illustrates the step response of the two amplifiers, each with a similar A_{OL} (= 50 dB) and BW_{CL} (= 1.25 GHz). The absence of negative capacitance (NC) has resulted in an A_L of only 25 dB due to β, with the stage gain (G_S) failing to reach the desired 16×, thus resulting in a gain error.

A detailed examination of the zoomed-in waveform reveals that the 1-stage OTA exhibits a relatively improved stability. Fig. 3(b) shows that the current consumption of the 2-stage OTA increases compared to the 1-stage OTA as the required BW_{CL} rises. Fig. 3(c) indicates that the energy efficiency of the 2-stage OTA decreases with increasing C_L. Since the C_P in the FinFET process significantly enlarges the C_L, minimizing the number of amplifier stages is beneficial. Therefore, employing the 1-stage OTA as the main amplifier with NC to compensate for the deficient A_{OL} and BW_{CL} is a promising strategy to improve energy efficiency.

Fig. 4(a) depicts the A_L and required transconductance (G_m) for the combination of the NC and the two different OTAs. Although the two OTAs depicted in Fig. 3 necessitate a G_m of 26.2 mS and 34.1 mS, respectively, the A_L without the NC is limited to a 25.4 dB, and the G_S turns out to be insufficient (∼15.1). The NC-assisted OTA boosts the A_L to 37.5 dB, thereby achieving the resolution requirements for the subsequent stage (6 bits). The NC-assisted OTA achieves the required G_S (=15.8), while reducing the G_m of the main OTA to 8 and 15.6 mS, respectively. The requirements of the 1-stage and 2-stage OTAs can be reduced by 3.3× and 2.2×, respectively. As can be seen, the 1-stage OTA is 1.5 times more efficient than the 2-stage OTA. In the case of the 2-stage

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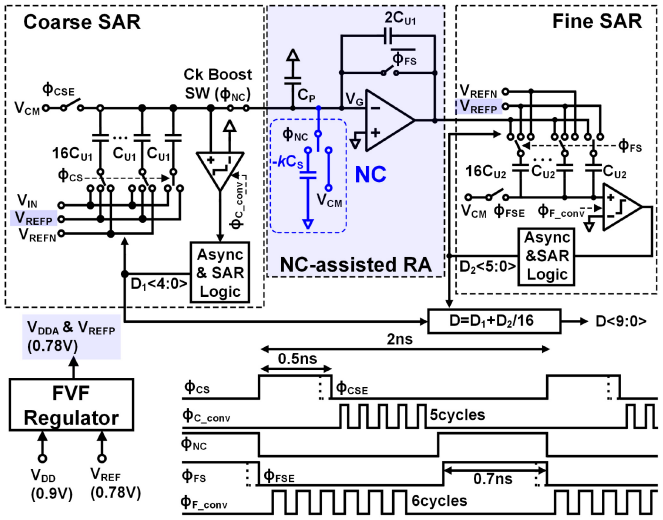


Fig. 5. Schematic and timing of proposed pipelined SAR ADC.

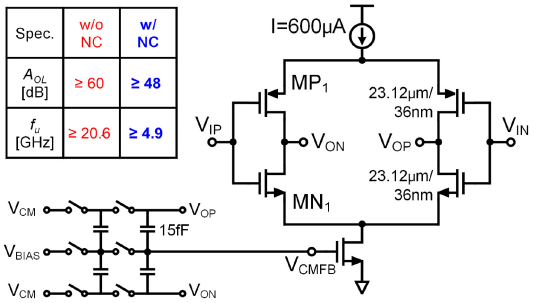


Fig. 6. Schematic of the main amplifier and its specification.

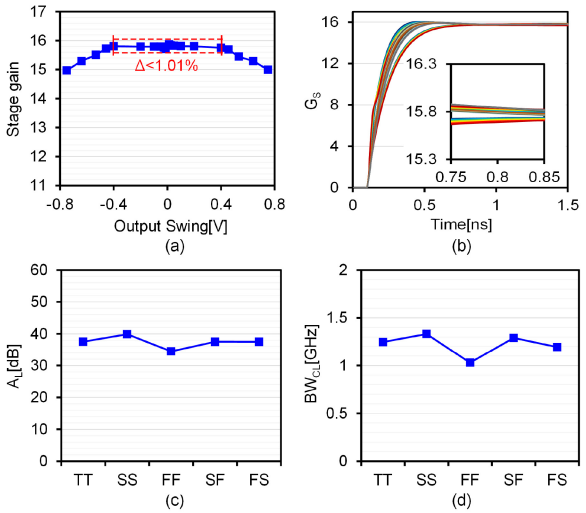


Fig. 7. Proposed dynamic NC assisted residue amplifier's simulation results, (a) Stage gain vs Output swing, (b) Step-response vs corner (TT, SS, FF, FS, SF) and temperature (0°C - 80°C), (c) A_L, and (d) BW_{CL}.

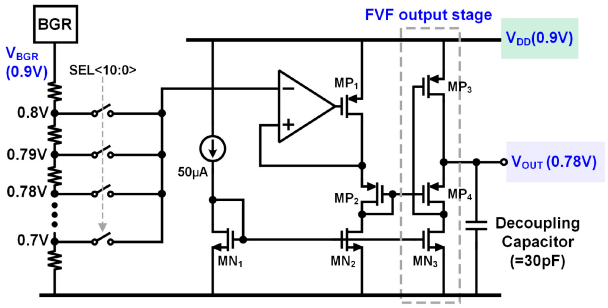


Fig. 8. Schematic of the on-chip regulator.

OTA, the pole introduced by NC (p_{NC}) is positioned between nodes p_{1b} and p_{2b}, additional current is required to drive p_{2b} at a high frequency to ensure stability.

C. ADC Implementation

Fig. 5 shows the schematic and timing diagram of the proposed pipelined SAR ADC. The C_{U1} of the 5-bit coarse SAR is 8.375 fF, and with a total sampling capacitance of 268 fF and a C_F of 16.75 fF, the G_S is 16×. The NC is of the dynamic type, as implemented in prior work [12]. The use of the 6-nm process introduces a significantly higher C_P at the virtual ground node compared to previous technology nodes, raising concerns about C_P variation and stability during β compensation. Consequently, the NC's compensation coefficient k is set to 0.8, with the NC designed to be 295 fF to compensate for 368 fF, which includes the C_P (≈100 fF) of the virtual ground node. This adjustment increases β from 0.059 to 0.24, relaxing the requirements for A_{OL} and f_u by a factor of 4.07. The inverter-based residue amplifier, shown in Fig. 6, must achieve the G_S of 16× within 700 ps, which requires the A_L of 36 dB and the BW_{CL} exceeding 1.2 GHz. The use of NC allows the required A_{OL} and f_u to be relaxed to 48 dB and 4.9 GHz, respectively. Fig. 7 presents the simulation results of the proposed dynamic NC-assisted residue amplifier. As shown in Fig. 7(a), it maintains approximately 16× gain with a variation of 1.01% within ±0.4 V differential output swing range. Fig. 7(b) illustrates the simulation results under temperature variations from 0 to 80°C across three process corners, which remains within a specified timing range within a maximum error of 0.5 LSB. The NC value shows minimal variation across the corners, with the highest value occurring at the SS corner and the lowest at the FF corner. Fig. 7(c) and 7(d) demonstrate that both the A_L and BW_{CL} are smallest at the FF corner due to the reduced NC value. Nevertheless, this reduction is not significant enough to pose any operational issues.

The 6-nm process exhibits significant C_P and R_P inherent to the implementation. In particular, the R_P in the lower metal layer is large enough to affect the performance of digital circuits composed of basic logic gates. To address this, the power supply is split, delivering 0.78 V to the analog section and 0.9 V to the digital section. Since the analog portion powers the residue amplifier and the V_{REFP}, a clean and stable voltage is required. To accomplish this, an on-chip flipped voltage follower (FVF) regulator is integrated. The 0.9 V supply powers both the FVF regulator and the digital components of the ADC, while the stabilized 0.78 V output is provided to the analog section. Fig. 8 shows the schematic of the on-chip regulator, which comprises a bias circuit, an error amplifier, and an FVF output stage [15]. The FVF output stage enables fast transient performance by leveraging a local feedback loop to reduce output impedance and enhance current

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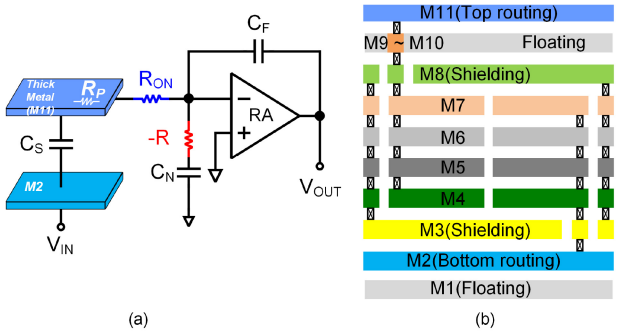


Fig. 9. (a) Routing of the C-DAC and Residue amplifier at virtual ground node and (b) encapsulated MOM capacitor layout and routing.

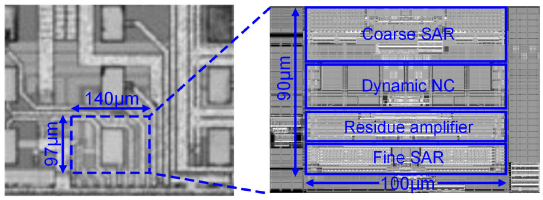


Fig. 10. Chip micrograph.

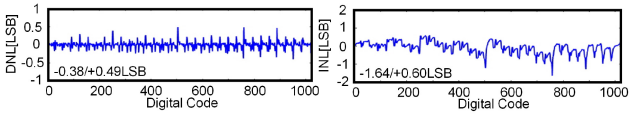


Fig. 11. Measured DNL/INL.

sourcing capability. A 30 pF decoupling capacitor is placed in the available idle space to ensure stable operation.

As shown in Fig. 9(a), the R_P of the virtual ground node must be minimized, as it can induce fluctuations in the R_{ON} used to compensate for the negative resistance (-R) of the NC. Fig. 9(b) illustrates the encapsulated MOM capacitor and its routing for the unit capacitor, which simplifies the application of the concept in Fig. 9(a). The unit capacitance is formed using metal layers M4 through M7, with bottom routing to M2 and top routing to M11. To reduce C_P between the bottom plate (M4) and routing (M2), M3 is used as a shield, while the lower metal layers M1 and M0 are left unused and floating, as lower layers typically exhibit higher R_P. To minimize the R_P of the top plate, thick metal (M11) is employed, effectively reducing R_P but at the cost of increased C_P. To address this, shielding is provided using M8, and layers M9 through M10 are left floating. Additionally, the capacitors are encapsulated by M4 through M7 to reduce capacitive coupling between adjacent capacitors. All shielding layers are grounded to maintain a stable potential.

III. Measurement Results

The proposed 10-bit 500-MS/s ADC prototype is fabricated in a 6-nm FinFET process. It occupies an active area of 0.014 mm^2, as shown in Fig. 10, where an active area excluding the integrated regulator and decoupling capacitor is only 0.009 mm^2. As shown in Fig. 11, the measured differential nonlinearity (DNL) and integral nonlinearity (INL) are −0.38/+0.49 and −1.64/+0.60, respectively. Fig. 12 shows

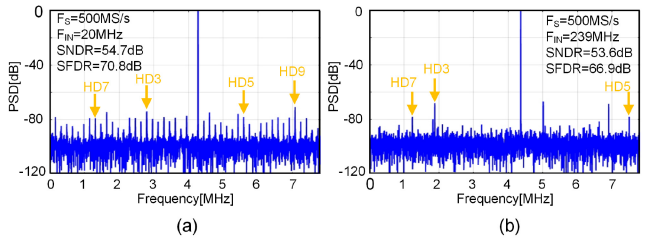


Fig. 12. Measured output spectrum with (a) 20-MHz, (b) 239-MHz.

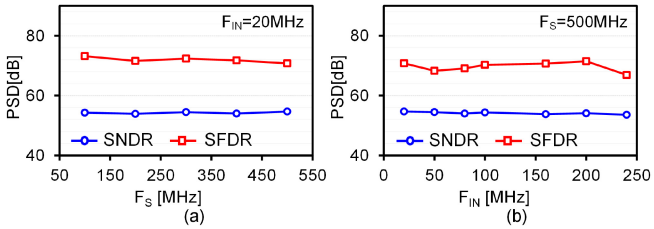


Fig. 13. Measured SNDR/SFDR vs (a) F_S, (b) F_{IN}.

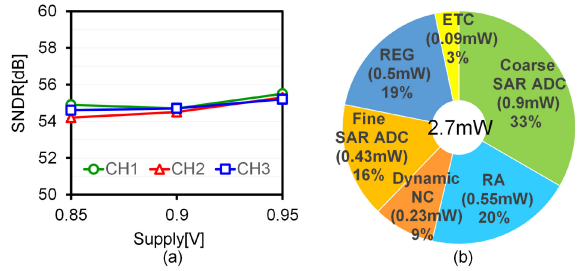


Fig. 14. (a) Measured SNDR with supply and chip variations and (b) power breakdown.

the measured power spectral densities (PSDs) decimated by a factor of 32. For the 20-MHz input, the SNDR and SFDR are 54.7 dB and 70.8 dB, respectively. For the near Nyquist input of 239-MHz, the measured SNDR and SFDR are 53.6 dB and 66.9 dB, respectively. Fig.13(a) shows the measured performance for different sampling frequencies at the 20-MHz input, where the variations of SNDR and SFDR are in the range of 0.8 dB and 2.4 dB, respectively. Fig. 13(b) demonstrates the measured performance for different F_{IN} at a sampling rate of 500-MS/s, which confirms that the performance remains stable up to the Nyquist frequency. By using the integrated regulator, the measured results for the supply variations of the three chips remain stable, as shown in Fig. 14(a). Fig. 14(b) shows the power breakdown of the ADC. The overall power consumption is only 2.7 mW. By using only 9% of the total power consumption with the dynamic NC, the power consumption of the residue amplifier can be reduced to 0.55 mW (∼20%).

Table II provides a performance summary and comparison with prior state-of-the-art works fabricated in a 28-nm process [12], or below [3], [5], [6], [14]. The use of an external reference regulator further enhances the FoM_W (= 9.6 fJ/conv.-step), as eliminating the power consumption of the integrated regulator and the use of a low-noise external regulator (LT3045) allow an improvement in SNDR. Compared to [12], this brief achieves both a better FoM_W

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TABLE II
PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART WORKS

	This work		A-SSCC' 23 Kwon [12]	ISSCC' 21 Baek [14]	ISSCC' 24 Lim [6]	ISSCC' 19 Hershberg [3]	JSSC' 22 Lagos [5]	
Process[nm]	**6**		28	7	8	16	16	
Architecture	**Pipelined SAR**		Pipelined SAR	SAR + TI-DSM	Pipelined SAR	Pipeline	Pipelined SAR	
Residue Amplifier	**Inverter-based 1-stage OTA**		Inverter-based 2-stage OTA	No	Ring Amplifier	Ring Amplifier	Ring Amplifier	
Amplifier Auxiliary technique	**NC-assisted β Compensation**		NC-assisted β Compensation	Residue path Replica	Dynamic bias	Gain calibration	Dither-based Gain calibration	
Resolution[bits]	**10**		11	12	12	11	12	
F_S[MS/s]	**500**		360	600	400	600	500	
Regulator	**Yes**	**No**	No	No	No	No	Yes	No
Area[mm^2]	**0.014**	**0.009**	0.016	0.037	0.017	0.037	0.047	0.015
Supply[V]	**0.9**	**0.78(A)/0.9(D)**	0.95	0.8	0.9	0.85	0.9	
Power[mW]	**2.7**	**2.1**	3.9	13	2.1	6	3.3	2.8
SNDR[dB]	**53.6**	**54.5**	58	55.3	62.8	60.2	62.3	62.9
SFDR[dB]	**66.9**	**68.3**	77.9	67.8	68.6	78.3	75.5	75.5
*FoM_W[fJ/c.-s]	**13.8**	**9.6**	16.7	45.6	4.6	12.0	6.2	4.9

*FoM_W = Power/($2^{ENOB}\times 2\times$BW)

with a higher operation speed (1.38×). Compared to an ADC that does not use a residue amplifier in a 7-nm FinFET [14], this brief achieves significantly improved energy efficiency at similar resolution and speed. Compared to ADCs with FinFET process that use a ring amplifier [3], [5], [6], this brief achieves competitive energy efficiency and operation speed.

IV. CONCLUSION

This brief presents a pipelined SAR ADC featuring a dynamic NC-assisted residue amplifier fabricated in a 6-nm FinFET process. The dynamic NC enables high energy efficiency by exploiting a single-stage inverter-based OTA, required for low supply voltage operation. The ADC prototype achieves a competitive FoM_W of 13.8 fJ/conv.-step when operating at 500-MS/s, and is robust against PVT variations due to its closed-loop configuration. The combination of a single-stage OTA with a dynamic NC represents a promising solution for energy-efficient pipelined SAR ADCs in advanced process nodes.

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A 112-Gb/s PAM-4 Long-Reach Wireline Transceiver Using a 36-Way Time-Interleaved SAR ADC and Inverter-Based RX Analog Front-End in 7-nm FinFET

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***Abstract*— A 36-way time-interleaved 56-GS/s 7-bit ADC is designed to realize 112-Gb/s pulse-amplitude modulation (PAM-4) transceiver in a 7-nm FinFET CMOS. The receiver analog front-end stages and the ADC track-and-hold (T/H) buffers are implemented using inverter-based Gm/inverse-Gm-load cells. A distributed inductor peaking network and multi-phase clock calibration is implemented in the quarter-rate transmitter. The transceiver achieves <1E-8 pseudorandom binary sequence (PRBS)-31 PAM-4 bit error rate (BER) over a channel with 37.5-dB loss at 28 GHz while dissipating 602 mW per channel, excluding DSP.**

***Index Terms*— 112 Gb/s, 7-nm FinFET, CMOS, DSP, inverter-based amplifier, pulse-amplitude modulation (PAM-4), time-interleaved SAR ADC, wireline transceiver.**

I. Introduction

INCREASING demand for more bandwidth from data center and communication applications has been accelerating the development of 112-Gb/s transceivers in recent years [1]–[5]. Pulse-amplitude modulation (PAM-4) signaling remains the dominant choice of modulation scheme due to its superior spectral efficiency when compared with nonreturn-to-zero (NRZ) modulation and its simpler implementation when compared with more sophisticated schemes. In many applications, a multitude of these transceivers are monolithically integrated in ASICs and FPGAs to realize high aggregate throughput per chip, therefore mandating the power and area efficiency to be the primary design optimization goal. In addition, these high-throughput VLSIs tend to adopt more advanced process design kits (PDKs) to benefit from the technology scaling that enables compact and energy-efficient implementations of their processing logic cores. While the logic cores benefit from advanced process nodes optimized for digital designs, high-speed transceiver design flows need to commence with a consideration of the constraints and capabilities of a given PDK to result in an efficient solution.

This article reports a long-reach capable 112-Gb/s PAM-4 transceiver designed in a 7-nm FinFET process, extensively using CMOS-style circuit implementations of high bandwidth linear and nonlinear functions needed for data paths and clock paths. Specifically, the entire receiver analog front-end (AFE) and ADC sampling front-end (SFE) are realized by inverter-based gain stages and complementary CMOS pass-gates, obviating the need for current sources and boot-strapped switches. High-speed clock distribution, multi-phase clock generation, duty cycle distortion (DCD)/IQ correction, and phase rotation circuits are also implemented exclusively in the CMOS style. The sub-ADC is implemented in the digital-process-friendly SAR ADC topology. The decision of sub-ADC circuit specification and time-interleaving architecture of the ADC SFE is optimized to fully use the device f_T of the 7-nm FinFET process, to achieve an overall area and power efficiency customized for ultra-high-speed wireline receiver applications.

II. Transceiver Architecture

Fig. 1 illustrates the top-level block diagram of the transceiver, consisted of a shared LC-phase-locked loop (PLL), a transceiver channel, and a control block. Each channel includes a TX, an RX, and a shared channel clock generation block. A shared bias generation and a resistor calibration block are also included. The control block generates a pseudorandom binary sequence (PRBS) pattern, sends 875-MHz 128-b-wide TX data, receives 778-MHz 144-b-wide RX data to and from transceiver channels, and calculates the bit error rate (BER). It includes a raw ADC data capture buffer and interfaces with an off-chip FPGA controller running a background RX adaptation and various calibration loops.

The LC-PLL synthesizes a 14-GHz differential clock from a 583.33-MHz external reference clock. The PLL output clock is

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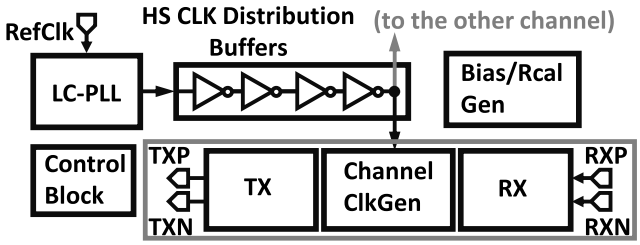


Fig. 1. Transceiver top-level block diagram. (One transceiver lane is shown).

then delivered to each of the two channels via supply-regulated CMOS high-speed clock distribution buffers. The third-order analog LC-PLL uses a compact wide-range transformer-based LC-voltage-controlled oscillator (VCO) [9], optimized for a lower jitter needed for 112-Gb/s operation.

The transceiver design is completed with production specification verified and signed off, including process, voltage, and temperature (PVT) tolerance, timing margin, and reliability checks (electro-migration, device self-heating, bias temperature instability (BTI), HCI, time-dependent dielectric breakdown (TDDB), and electrostatic discharge (ESD), latch-up). Notably, high-bandwidth RX AFE and ADC SFE circuits use thin-oxide MOSFETs for best gain-bandwidth product and f_T available from the 7-nm FinFET PDK. Their device reliability is guaranteed by carefully designing dc bias points both in the normal operation mode and in power down mode such that device terminal voltage differences do not exceed the allowed maximum rating for thin-oxide devices.

III. Transmitter

The TX architecture shown in Fig. 2 adopts a four-tap feed-forward equalizer (FFE) (for two pre-cursor and one post-cursor taps) and a quarter-rate clocking scheme with duty cycle and I/Q mismatch calibration to enable the use of a power-efficient CMOS 4:1 MUX [4]. A current-mode logic (CML) pad driver topology [4], [5] is chosen to achieve 3% finite-impulse response (FIR) resolution without incurring an excessive power penalty from the large number of unit cells containing high-bandwidth node data switches as would be required in a voltage-mode (VM) driver. An array of tail-less CML driver cells with source node switching topology allows the use of more power-efficient CMOS pre-driver and 4:1 MUX. The data path comprises a 128 to 32 serializer, a four-tap FFE data generation, and a 32 to 8 serializer, followed by the final 4:1 MUX. A retimer array is needed to close the timing between the 32:8 serializer and the 4:1 MUX array due to the relatively long routings between the two blocks.

A. Final 4-to-1 MUX

The final 4:1 MUX stage is a performance-critical block in quarter-rate TX designs as it has the tightest timing paths, hence it largely determines the transmitter output jitter performance. Consequently, the block contributes a significant amount to the total TX power consumption. We implemented the 4:1 MUX using a simple CMOS pass-gate-based two-stage MUX clocked by four-phase 14-GHz clocks as shown in Fig. 3. The use of CMOS MUX is motivated by its compact layout and superior energy efficiency.

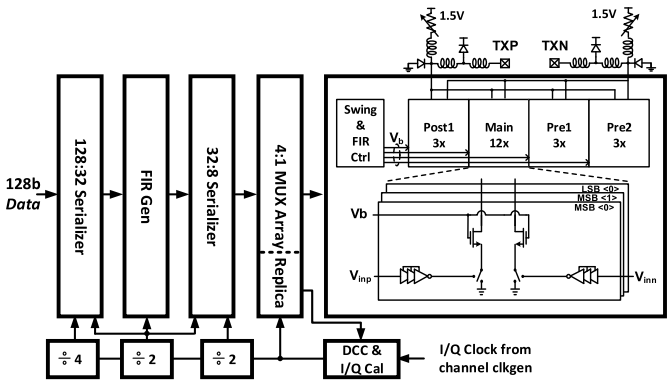


Fig. 2. TX sub-system.

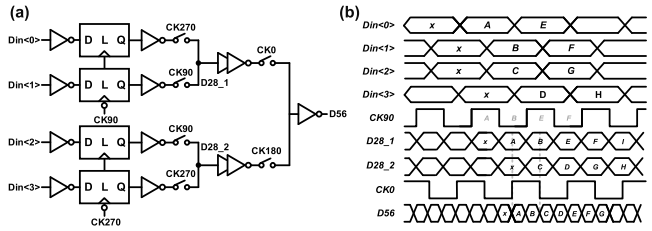


Fig. 3. (a) TX final 4:1 MUX. (b) Timing diagram.

The 14-GHz I/Q clock is produced by the channel clock generation block shared by TX and RX and distributed to TX in full CMOS swing via a regulated CMOS buffer chain. A TX clock calibration block for duty cycle correction (DCC) and I/Q error compensation is needed to correct the duty cycle distortion and I/Q phase errors caused by device mismatches from the MUX circuit so that a low deterministic jitter can be achieved at 112 Gb/s. The phase errors are sensed at the output of the replica MUXs and are background calibrated with smaller than 100-fs resolutions.

B. Pad Driver

A tail-less CML driver [5] is chosen in this design to achieve four-tap FIR (two pre-cursors and one post-cursor) with less than 3% step size. The CML driver is known to consume 4 times more current than the VM driver for the same output swing. However, a CML driver can attain a fine FIR resolution more easily by controlling its DC biasing without affecting high-speed signal paths. In contrast, a slice-based VM driver [4] implements the FIR by configuring the input data of each unit slice, which limits the minimum number of required 4:1 MUXes. Consequently, the amount of power saved in the VM pad driver is not compensated enough for the power overhead of the 4 to 1 MUX and clocking circuits in a quarter-rate architecture with a fine FIR resolution.

In this work, a distributed peaking network is designed to compensate for over 200-fF device and parasitic capacitance at the driver current summing node. A T-coil with an additional series inductor effectively separates the N- and P-type ESD diodes, C4 bump pad, and on-die termination (ODT) resistor array, implementing a lumped transmission line for bandwidth extension and improved return loss. With the distributed peaking network, a 46-GHz bandwidth is achieved at the TX output.

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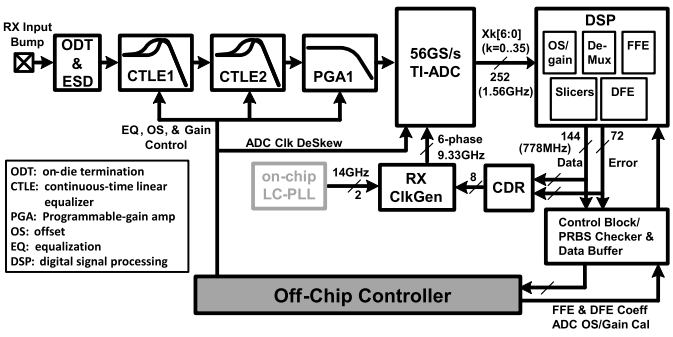


Fig. 4. RX sub-system.

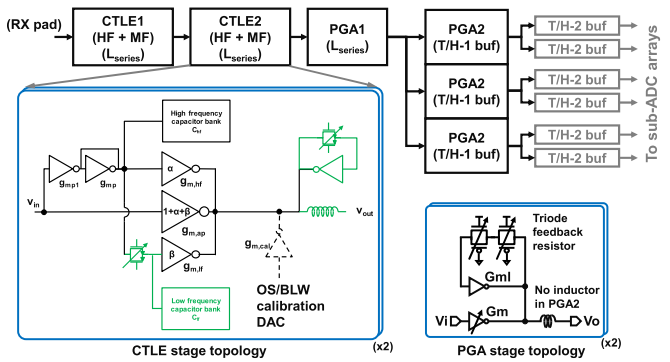


Fig. 5. RX analog front-end implemented using inverter-based Gm/Gml stages.

IV. Receiver

Fig. 4 depicts the RX top-level functional diagram. The PAM-4 signal received at the RXP/RXN input pads is terminated by calibrated ODT resistors. T-coil inductors are used for impedance matching to improve return loss and bandwidth by neutralizing parasitic capacitances caused by input pads, ESD diodes, ODT, and continuous-time linear equalizer (CTLE) input capacitance. The all CMOS RX AFE, consisting of two stages of CTLE, equalizes the input signals by boosting the Nyquist frequency (HF) and mid-range frequency (MF) bands separately. The following two stages of the programmable-gain amplifier (PGA) adjust aggregated broadband gain for automatic gain control adaptation that controls the output signal range for the subsequent ADC to digitize with fully used full scale range (FSR).

The ADC SFE realizes analog signal de-muxing by two track-and-hold (T/H) stages. The second PGA stage acts as the first-rank T/H buffer. A two-rank 6-by-6 time interleaving serially breaks down the ADC input signal into 36 sub-ADCs, which subsequently digitize the signal into 7-b symbols, xk[6:0] ($k = 0..35$). The 6×6 interleaver architecture is chosen with considerations of tradeoffs among the SFE, clocking circuitry, and sub-ADCs' sampling speeds, which will be discussed in further detail.

The DSP retimes ADC output symbols, xk[6:0] ($k = 0..35$), and performs offset and gain correction for the 36 individual sub-ADCs. The 72 deserialized xk are fed into subsequent 31-tap FFE and one-tap speculative decision-feedback equalizer (DFE) stages for ISI cancellation before being converted to produce data and error bits.

An on-chip baud-rate clock and data recovery (CDR) logic detects the optimal ADC sampling aperture phase (T/H-1 clock falling edge) by accumulating the Mueller–Muller [8] phase detector output to update an 8-b phase interpolator (PI) code.

A. Data Path

The RX AFE in Fig. 5 consists of a two-stage CTLE followed by two PGA stages. The CTLEs implement high-frequency peaking with a feedforward pole-subtractive topology, and hybrid peaking (i.e., both HF and MF) is realized in both the stages [8]. At the max peaking setting, the CTLE provides more than 16 dB of peaking at the Nyquist frequency. MF tuning provides additional long-tail ISI cancellation, which helps reduce the post-cursor FFE coefficient ranges. To further reduce the area of the device capacitors for MF path, a series triode device is inserted to allow resistance tuning for the zero frequency. A combination of active-inductor Gm load (Gml) and passive inductor effectively boosts the bandwidth of each stage of CTLE and PGA1. Compact inverter-based amplifier stages consisted of a Gm cell and inverse-Gml, with inherently good linearity due to ratio-metric design [8], are used to implement CTLE, PGA, and ADC T/H buffers. The active inductor is realized by complementary pass-gates in the triode mode, which are used as feedback resistors instead of the conventional TiN resistors, dramatically reducing the cell size and parasitic wire loading. All AFE circuit elements (Gm, Gml, triode-mode feedback resistors, zero resistors, and zero capacitors) are laid out using the same minimum design rule poly-to-poly pitch to achieve high area efficiency and more process uniformity. An overall 70% area reduction (passive inductor area not included) is achieved compared with [8] for each CTLE stage.

The time-interleaved ADC architecture is carefully chosen to optimize the overall power and area efficiency under various constraints of the design. Due to the CMOS nature of the AFE and ADC, the fan-out of each can be decided in a "digital" fashion. Once the design of the sub-ADC is optimized, its input capacitance propagates eventually back to the RX input, determining along the way the drive strengths of the preceding stage given their respective bandwidth requirements. Therefore, we first evaluate the number of required sub-ADCs to minimize fan-out and reduce the overhead incurred by layout complexity and parasitics. To fully take advantage of technology scaling provided by the 7-nm FinFET, the sub-ADC is optimized for sampling speed and area.

The metastability impact on the sub-ADC performance could become the gating factor of the overall system performance. Therefore, an in-depth study was done to understand the effect of metastability-induced errors in asynchronous SAR ADCs [10]. While pushing the sub-ADCs to their speed limit can significantly increase the probabilities of metastability events, the generated errors still could be tolerable in SerDes applications. An intuitive understanding of the detailed analysis in [10] could be established with the following metastable conditions. In a high-speed asynchronous SAR ADC, it is more probable for the LSB bit position to run out of time for evaluation creating an LSB metastable event, but the error created is small (i.e., 0 or 1 LSB code error) and

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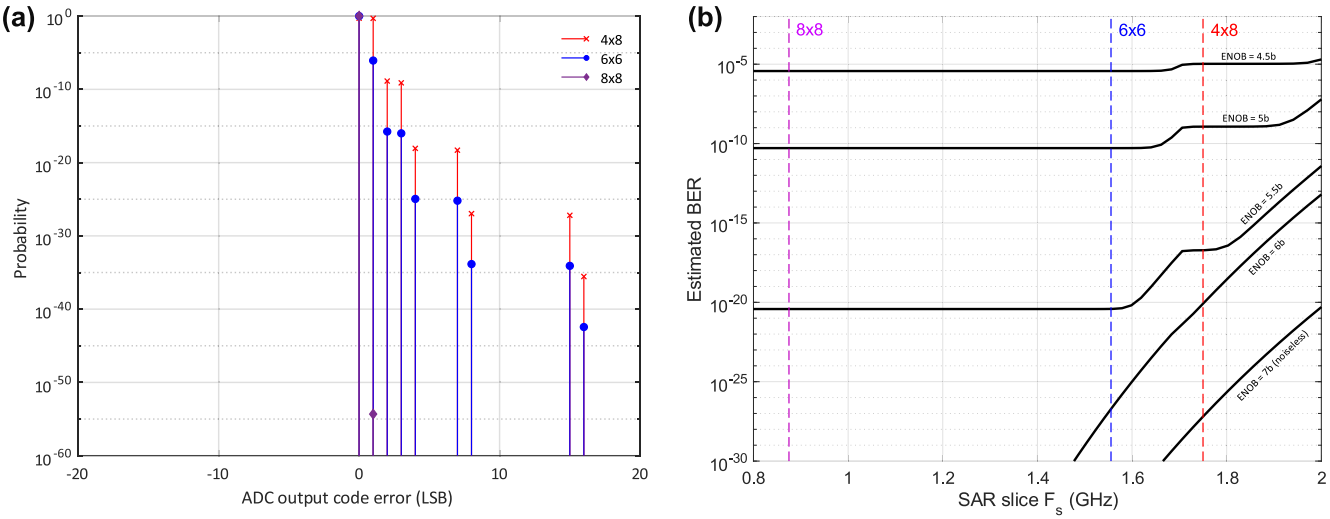


Fig. 6. (a) Simulated 7-b SAR ADC metastability error probability versus error magnitude for 4 × 8, 6 × 6, and 8 × 8 time-interleaving. (b) Simulated system BER versus sampling speed for five cases of thermal noise level.

typically below the thermal noise floor, thus having minimal impact on the BER of the overall system. On the other hand, an MSB bit position metastability event could yield much larger code errors, which almost certainly causes bit/symbol errors. However, these are exponentially unlikely events by construction of asynchronous SAR ADCs. Taking the 7-nm process capabilities and metastability impact on the system BER into account, we concluded that the sub-ADC design in this work can operate at around 1.6 GS/s across different PVT corners with a sufficient margin.

As an illustration, three possible time-interleaving architectures for 56 GS/s are considered for BER impact simulation: 4 × 8, 6 × 6, and 8 × 8, corresponding to sub-ADC sampling speed (F_S) of 1.75 GS/s, 1.56 GS/s, and F_S = 875 MS/s, respectively. Fig. 6(a) shows the simulated metastability error probability versus error magnitude in the unit of LSB for the three cases. The 4 × 8 case shows 1 LSB magnitude error probability almost at the same level as zero error, implying that the LSB bit resolution during the SAR loop almost always runs out of time. Also, 2 to 4 LSB error probabilities are high enough (∼1E-8) to impact on the system BER. On the other hand, the 8 × 8 case shows almost no metastability code errors with less than 1E-50 chance of failing with one LSB error. The 6 × 6 architecture achieves the balance between metastability error probabilities and pushing sub-ADC speed. The one LSB error probability is much smaller compared with that caused by the thermal noise, while higher impact 2 to 4 LSB errors are sufficiently rare (less than 1E-15 chance). Fig. 6(b) shows the simulated system BER versus F_S for five cases of system noise level as modeled by equivalent ENOB ranging from 4.5 to 7 b (noise-less). Three vertical dashed lines mark F_S corresponding to 8 × 8, 6 × 6, and 4 × 8 interleaving architectures, respectively. For 8 × 8, the system BER is always thermal noise limited (i.e., independent from F_S variation) and each SAR ADC slice is far away from their speed capabilities, under-utilizing 7-nm device speed limit. For the other extreme of 4 × 8 architecture, the metastability errors cause significant degradations in system performance, especially at lower baseline BER. For 6 × 6, we are operating closer to the metastability BER corners while the system performance is still dominated by thermal noise.

Given the suitable sampling speed of sub-ADC, various interleave architecture candidates are compared in Fig. 7. The 4 × 8 architecture stresses the sampling bandwidth of the front-end, clocking, and sub-ADC sampling speed despite its area advantage. An 8 × 8 architecture [2] significantly relaxes the speed requirement of each ADC component, but in turn wastes the extra speed offered by the process technology (e.g., concluded 1.6 GS/s capability of the sub-ADC with respect to the required 875-MS/s sampling speed). The area penalty and the associated floor-planning challenge of the 8 × 8 interleaving are also non-negligible due to a larger number of sub-ADC and clock generation. Therefore, the 6 × 6 architecture serves as a good trade-off between effective utilization of each key block design without over-stressing design margins and layout complexity. The performance-critical rank 1 T/H will sample at 9.333 GHz, which is a reasonable frequency for both the analog amplifiers and clock generation and distribution.

Fig. 8(b) illustrates detailed schematics of the 6 × 6 SFE. The SFE requires 3× rank 1 T/H buffers, each sampling with ∼50% non-overlapping clocks. The T/H-1 buffers also act as second-stage PGAs, which are a scaled down version of the inductor-less PGA1. Each T/H-1 buffer drives 2× rank 2 T/H unity gain buffers. All SFE stages use the same inverter-based circuits (Gm and active inductor triodes) as the CTLEs. The output common-mode of the Gm/Gml cells are compatible with the complementary CMOS pass-gates used for all T/H switches in the ADC, obviating the need for bootstrapping circuits and significantly reducing area and clock power consumption. A correct bias voltage is needed at the interface of rank 1 and rank 2 T/H buffers due to the nature of inverter-based T/H-2 Gm cells, and thus clock/charge injection switches are added for the T/H-1 switches. Since the StrongArm sense-amp comparator in

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	4x8	8x8	6x6 [this work]
Total # of sub-ADCs	32	64	36
Sub-ADC sampling speed	1.75GS/s	875MS/s	1.56GS/s
Rank1 sampler frequency	14GHz	7GHz	9.33GHz
Rank1 settling BW (7b)	~28GHz	~13GHz	~18GHz
# of T/H amplifiers	2 + 4	4 + 8	3 + 6
Area penalty	Small	Large	Medium
Clock distribution	Easy	Difficult	Moderate
Challenges	❑ Sub-ADC & rank1 speed ❑ Rank1 settling BW requirement	❑ Area & power penalty ❑ Clock distribution	❑ Manageable area & power penalty (compared to 4x8)
Advantages	❑ Compact layout ❑ Easy clocking	❑ Relaxed BW & timing requirement	❑ Good trade-off between process f_T & design margin

Fig. 7. Design tradeoffs for 4 × 8, 8 × 8, and 6 × 6 time-interleaving architectures for ADC SFE.

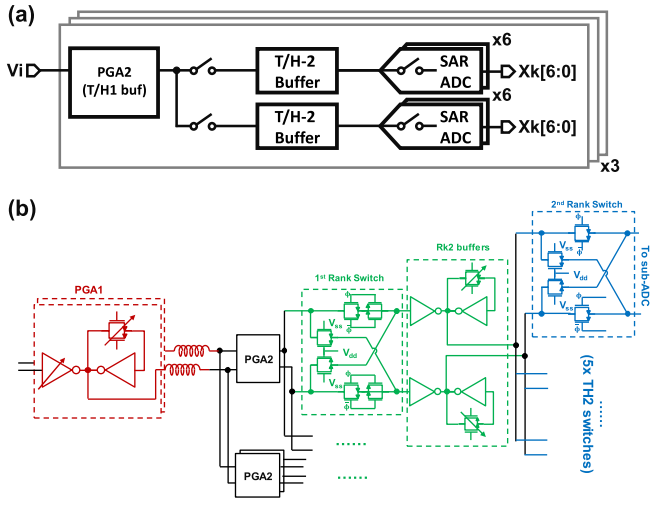


Fig. 8. (a) ADC sub-system and (b) circuit implementation of its sampling front-end.

sub-ADC can operate with a slightly wider common-mode range, the charge injection cancellation devices of T/H-2 switches are removed to reduce the parasitic capacitance at the capacitive digital-to-analog converter (CDAC) node. Both the stages have dummy cross-coupling devices for >50 dB hold mode feedthrough cancellation. The sub-ADC adopted a speed-optimized asynchronous SAR ADC architecture using a binary CDAC with an estimated unit capacitance of 0.5 fF.

B. Clock Generation

The RX clock generation architecture is driven from the top down by some key system constraints and optimizations. The use of a quarter rate architecture in the transmitter gives rise to the need for non-rotating 14-GHz quadrature input clocks, along with an LC PLL running at this frequency. The tradeoffs outlined regarding the speed of each sub-ADC and the number of sub-ADCs to optimize for area, power, and bandwidth result in the RX requirement for six equally spaced phases at 9.333 GHz, with optional phase rotation to track frequency differences between an external TX and the on-chip PLL. Generating the RX clocks from the same clock source as the TX obviates a need for a second clock source. The time-stamp information for a 9.333-GHz clock is already present in a 14-GHz clock, provided the duty cycle is accurate.

The architecture is given in Fig. 9. The 14-GHz clocks from the PLL are used as the injection for a four-stage CMOS 1:1 injection-locked oscillator (ILO). The four-stage ILO uses XOR-based phase sensing with supply regulation [7]. The outputs of this ILO drive both the TX and RX. At the RX, the eight equally spaced output phases at 14 GHz are used to drive an all CMOS 8-bit PI producing a rotating 14-GHz output. This differential clock has variable duty cycle control. The DCD-corrected output provides the source for an IQ-divide-by-6 circuit. The division by 6 is a cascade of a divide-by-3 and a divide-by-2, with one each in parallel for the I- and Q-paths. The quadrature is achieved by releasing the I-path first, followed by the Q-path 1.5 cycles later. The four-phase output at 2.333 GHz contains fewer transitions than a differential 14-GHz clock, thus being advantageous from a distribution power perspective. It is important to note that any DCD in the output of the PI will translate into the IQ error at the output of the divide-by-6 block. As such, IQ correction of the 2.333-GHz clocks is carried out by adjusting the duty cycle after the PI. An analog-style XOR detection is used to generate an error voltage which after processing with a dynamic comparator is used as an input to a digital control algorithm as illustrated in Fig. 9(c).

As illustrated in Fig. 10, the 2.333-GHz IQ clocks are now progressively processed to realize a multi-phase 9.333-GHz clock. The first step is a 4-to-1 self-timed multiplexer used to cyclically select the four 2.333-GHz clocks. This creates a 9.333-GHz pulse train with well-defined low jitter positive edges, but poorly controlled duty cycle. Passing this signal through a pulse generation circuit with adjustable pulse width, as shown in Fig. 10(a), generates a signal suitable for injecting into a ring oscillator pre-calibrated to near

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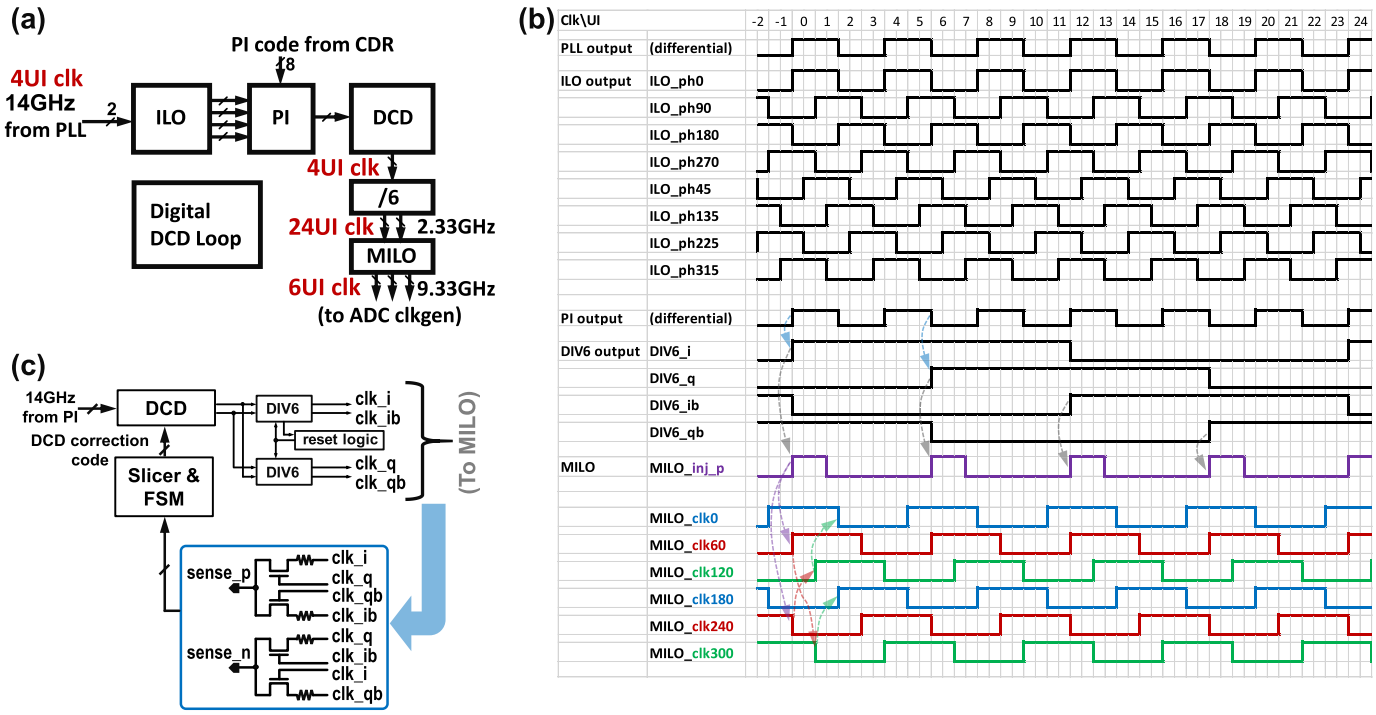


Fig. 9. (a) RX clock generation sub-system. (b) Timing diagram (only cycle-wise accurate). (c) DIV6 and DCD correction loop.

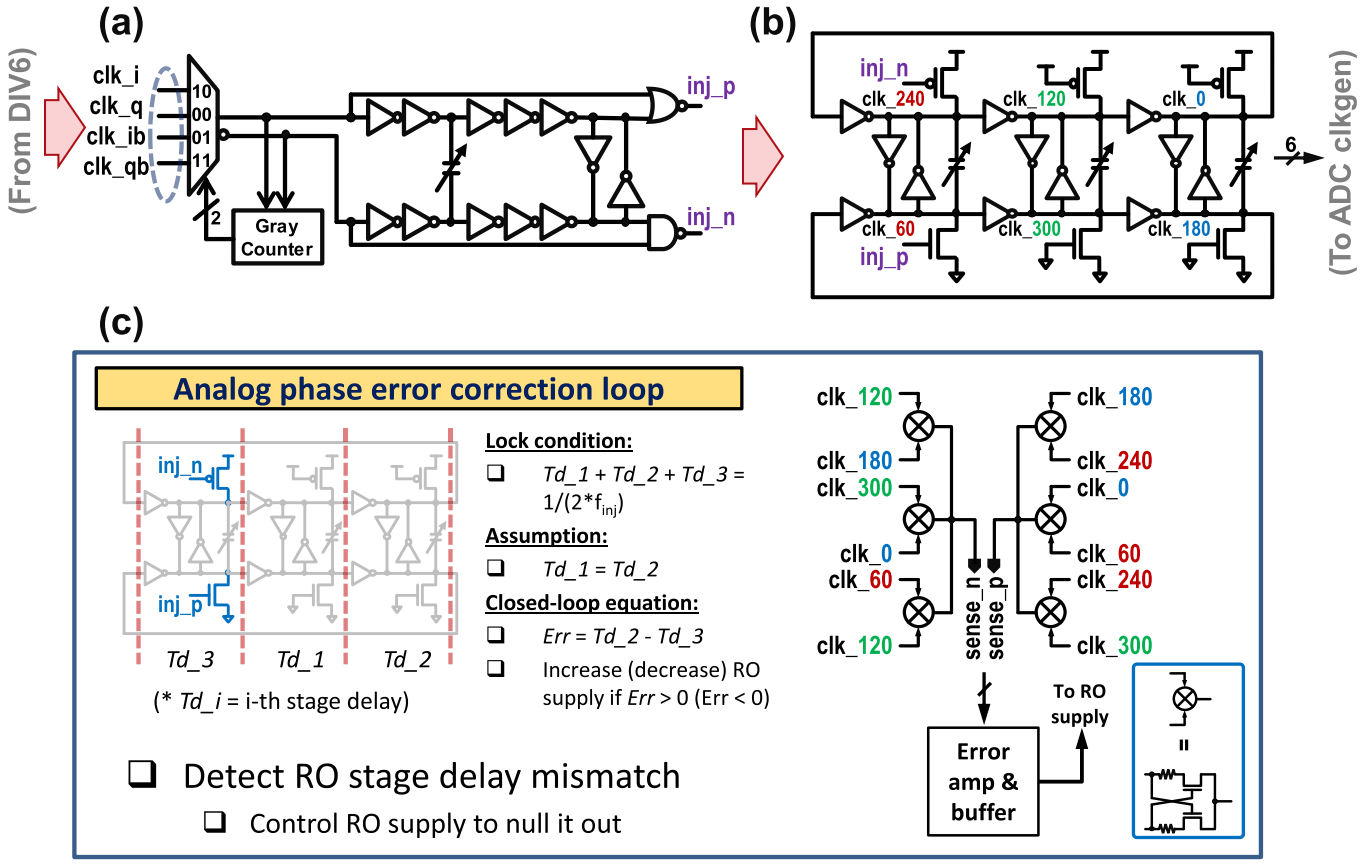


Fig. 10. Multiplying ILO circuits. (a) Pulse generator. (b) Injection-locked ring oscillator. (c) Phase error detector and correction loop.

9.333 GHz [Fig. 10(b)]. The combination of the multiplexer, pulse generator, and tuned ring oscillator together forms a multiplying-ILO (MILO) [11].

Fine control of the natural oscillation frequency of the MILO is required to achieve low phase spacing error at the six-phase output. The supply to the MILO is regulated by

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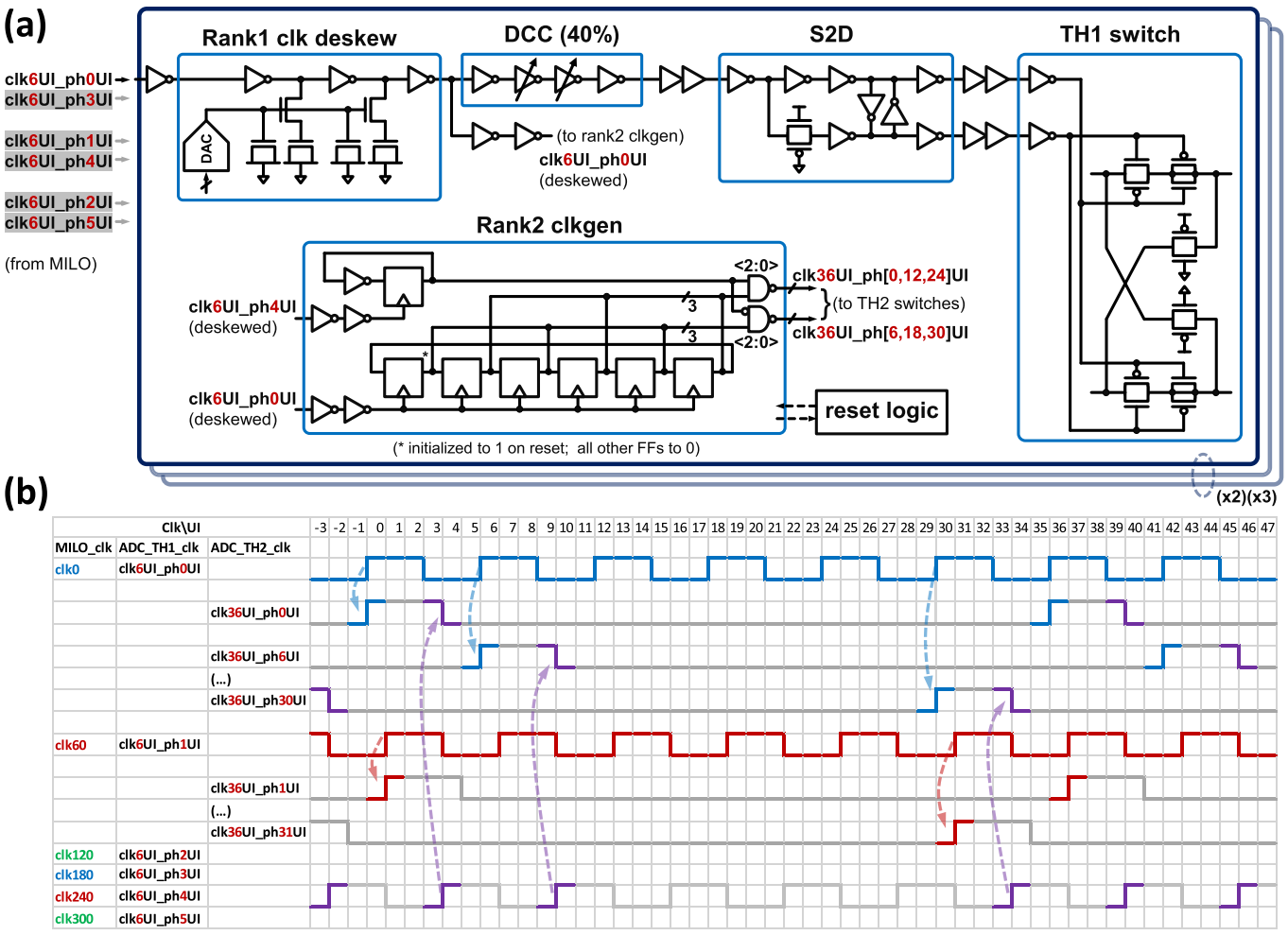


Fig. 11. ADC clock generation for rank 1 and rank 2 clocks. (a) Circuit implementation. (b) Timing diagram.

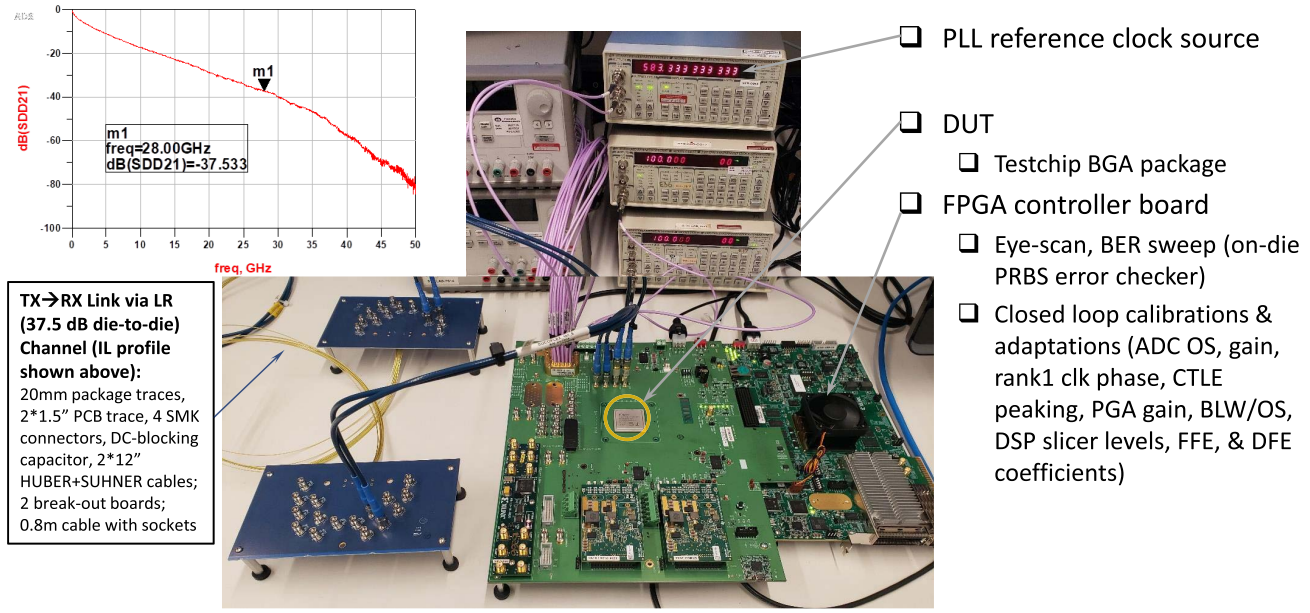


Fig. 12. Bench setup shown with TX to RX long-reach link performance test.

combining the six output phases to produce an error signal which is then integrated and filtered as illustrated in Fig. 10(c). The regulation has the effect of adjusting the natural oscillation frequency of the ILO such that its output period matches the input period. It is important to minimize the offset errors at the phase combination and error amplifier blocks as not all errors

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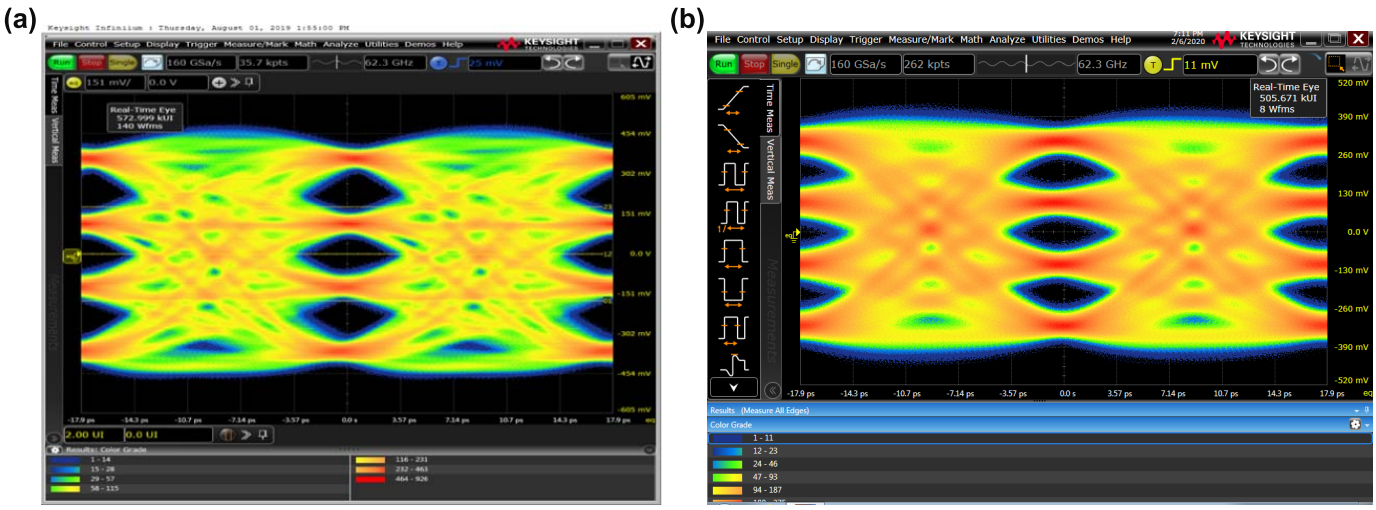


Fig. 13. 112-Gb/s PAM-4 TX output eye diagrams of (a) PRBS-7 and (b) PRBS-31 patterns.

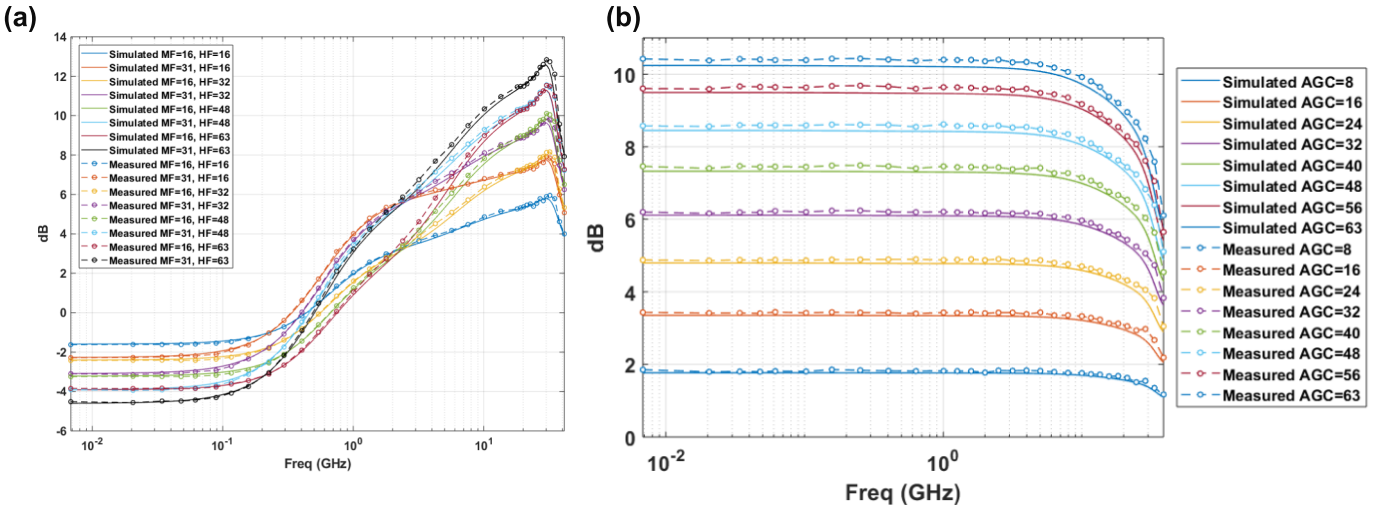


Fig. 14. RX AFE transfer function measurement versus simulation for (a) CTLE and (b) PGA.

accumulated are static or correctable. With a single control point, the average phase error at the MILO is minimized, but care is taken with the physical design to equalize the loading on the ring oscillator nodes to reduce the residual mismatch-induced errors.

The MILO method has the advantage that multi-phase clocks can be generated close to the locality of final use, lowering the distribution power. It should be noted for jitter purposes, however, that while the injected phases can have low-frequency jitter approaching that of the reference clock, high-frequency jitter (beyond the ILO bandwidth) will tend toward that of the intrinsic ring oscillator, and the non-injected phases will have further added jitter. Additional power-supply-induced jitter also requires careful budgeting.

It can be noted that instances of the pulse generation circuit being driven directly by the 2.333-GHz clocks would also produce a series of useful pulses, but that in such a case, multiple injection devices would be required. Instead, the use of fewer but higher rate injection devices lowers the loading on the ring oscillator and reduces multi-path mismatch.

While the MILO does have inherent suppression of timing errors at the injection point, residual offsets and random variations in both the MILO itself and downstream processing require further calibration. As such, the clocks are corrected for both skew and duty cycle, prior to application to the T/H switches in rank 1 or rank 2 ADC time interleavers.

The ADC clock generation circuit in Fig. 11(a) receives 6UI period six phase clocks from the MILO and processes them to produce T/H-1 and T/H-2 switching clocks. The skew correction is carried out using a DAC to adjust an MOS-based load on the clock paths. This affects both the rising and falling clock edges. The skew-adjusted clocks with near 50% duty cycle are divided for use in the rank 2 clocks, while the rank 1 clocks undergo programmable DCD adjustment to achieve a 40% duty cycle (2.4 UI tracking window) at 56GS/s ADC sampling speed. For power efficiency, the skew correction and DCD are applied to the six phases prior to a set of single to differential stages driving the T&H switches. T/H-2 clocks are generated by dividing the skew-corrected 50% duty cycle T/H-1 clocks by 6 to generate six-phase 6UI-long pulses

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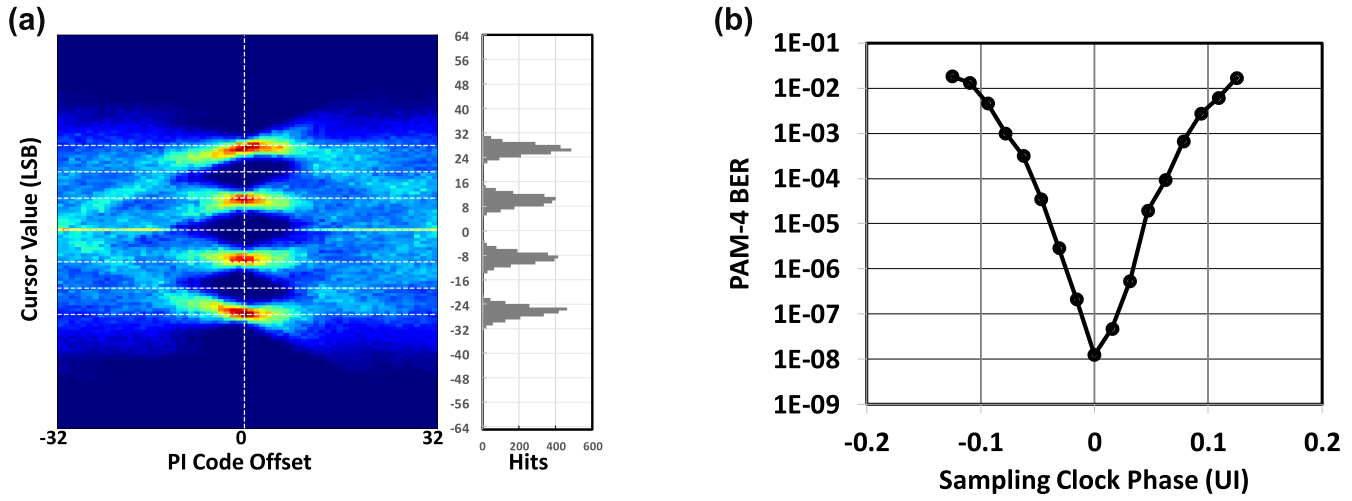


Fig. 15. Receiver performance for 112-Gb/s PRBS-31 over a long-reach channel (37.5 dB). (a) RX eye and cursor histogram. (b) BER bathtub curve.

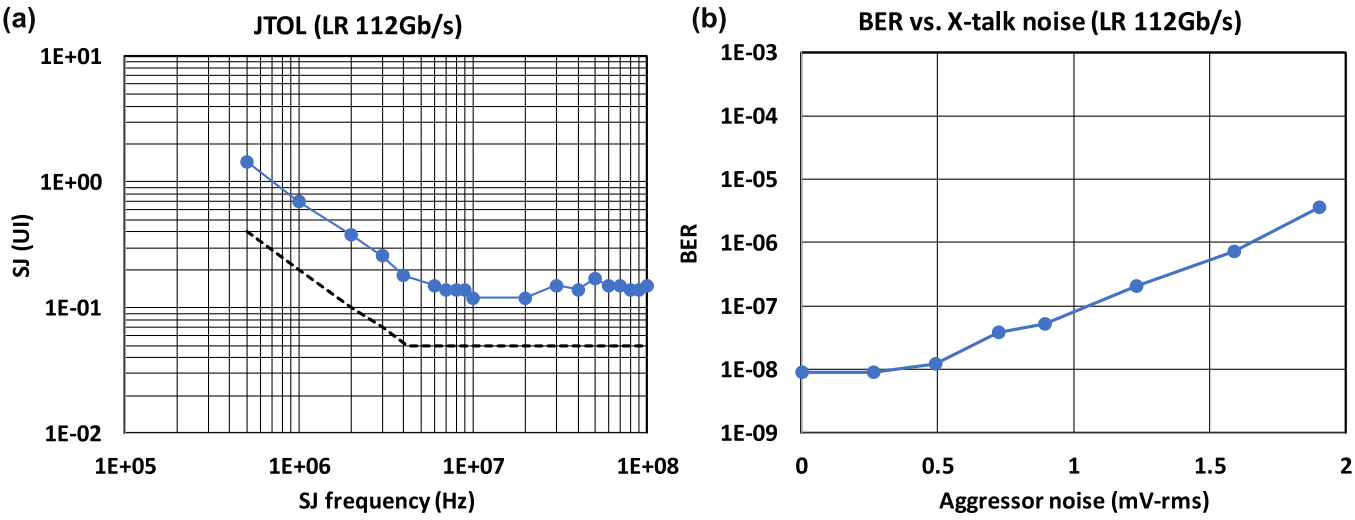


Fig. 16. (a) Sinusoidal jitter tolerance. (b) Cross-talk impact test for 112-Gb/s PRBS-31 over an LR channel.

of 36UI period. To meet the non-overlap timing condition, T/H-2 clocks are then trimmed by 2UI to have a 4UI tracking window as illustrated in the timing diagram in Fig. 11(b).

V. Bench Test

The fabricated transceiver wafer dice are assembled in ball grid array (BGA) packages, following a standard mass production VLSI packaging flow and are tested in a socket mounted on the device under test (DUT) PCB board. A bench test setup for TX-to-RX link performance over a long-reach channel is illustrated in Fig. 12. The DUT board with a DUT package socket and breakout cable connectors is controlled by another general-purpose FPGA board that fetches data and status bits from DUT and sends commands back to DUT. A control software performing PAM-4 RX adaptation and closed-loop corrections is coded and programmed onto the FPGA. An on-chip PRBS checker monitors the DSP data output to measure PAM-4 BER of the sliced data without sub-sampling. Contiguous raw ADC outputs (8-K samples) are captured and stored in the on-chip controller memory upon a trigger signal from the off-chip FPGA controller board. The FPGA controller periodically fetches the stored ADC output samples to perform continuous adaptation of data/error slice levels, FFE/DFE coefficients, CTLE peaking codes, PGA gain codes, AFE offset and baseline wandering correction DAC code, and ADC calibration codes (offset, gain, and skew). An external reference clock generator drives the on-die LC-PLL in DUT.

Fig. 13(a) and (b) shows the 112-Gb/s PAM-4 TX eye diagrams for the PRBS-7 and PRBS-31 patterns, respectively, using the built-in FFE of the Keysight DSA-X 96204Q oscilloscope. The TX output goes through 2 in of Megtron-6 PCB trace and a 24-in-long HUBER+SUHNER cable into the oscilloscope. PRBS-31 eyes show more symmetric PAM-4 eyes, albeit with reduced eye openings. The maximum TX swing is 1-V differential peak-to-peak and the measured random jitter is 140-fs rms for a clock pattern.

Fig. 14(a) and (b) plots the comparison between the measured and simulated composite transfer functions (TFs) of CTLE stages and PGA stages, respectively. The dashed lines

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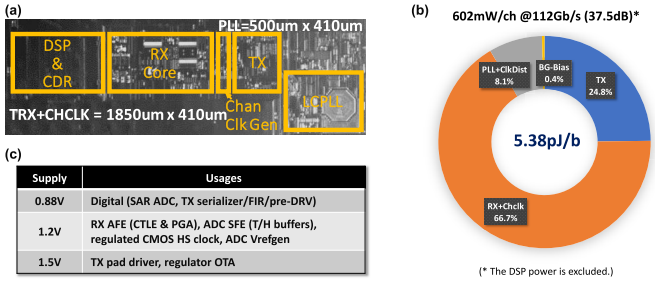
with markers represent the silicon-measured TFs for various peaking or gain code combinations. The solid lines represent the corresponding simulations. The TF mismatch between the measured data and simulation results is less than 0.5 dB across dc to 30 GHz, supporting the accuracy of our simulation methodology (e.g., parasitic extraction and field solver settings) for inverter-based CTLE and PGA circuits over a broad frequency range. The measured CTLE TF shows a max peaking of 17.5 dB at 30 GHz and its low-frequency zero is located near 300 MHz for the settings used. The measured PGA gain tuning range of 10.5 dB is observed.

To test the link performance, the TX output from the DUT board is looped back to the RX input via a channel consisting of TX/RX package traces (>20 mm total length embedded in an organic laminate substrate), two 1.5-in PCB traces, four SMK connectors, a dc-blocking cap, a pair of 12-in HUBER+SUHNER cables, two break-out boards, and a 0.8-m cable (Fig. 12). The total IL from TX to RX at 28 GHz is measured at 37.5-dB die-to-die, or 33.5-dB BGA-to-BGA excluding device package and DUT board socket, as illustrated in the SDD21 profile in Fig. 12. The measured transceiver performance is summarized as shown in Fig. 15(a) and (b). The transmitted 112-Gb/s PRBS-31 PAM-4 signal over the channel is effectively equalized by the RX at a BER <= 1E-8, as reported by the on-chip PRBS checker. The TX is configured to send a 930-mVppd signal with ~2-dB pre-cursor and ~5-dB post-cursor FFE boost.

The receiver eye in Fig 15(a) is plotted by accumulating 8-K symbol values of the equalized data slicer input in DSP while sweeping the sampling phase by overriding the PI code register by −32 to +32 offset around the CDR lock position. The corresponding cursor histogram is a vertical (ADC values measured in the unit of LSB) cross section of the RX eye taken at the PI code offset set to zero (i.e., at the CDR locking point), clearly showing equalized PAM-4 symbol levels. The measured horizontal BER bathtub in Fig. 15(b) shows a margin of >0.1UI to satisfy the OIF CEI pre-forward error correction (FEC) spec [6] of 1E-4 around the CDR locking point, where the BER reaches 1E-8 or lower.

A sinusoidal jitter (SJ) tolerance (JTOL) test in Fig. 16(a) was performed by setting a bit error rate tester (BERT) to inject SJ of varying frequency and magnitude to the DUT receiver over an LR channel. The data show an eye opening margin larger than 0.1UI at high frequency, while satisfying the OIF CEI JTOL mask [6] shown in the dashed line with a good margin across the frequency range from 500 kHz to 100 MHz. The results demonstrate the effectiveness of ISI cancellation by CTLE, FFE, and DFE as well as the jitter filtering capability of the on-chip digital CDR loop.

The cross-talk (X-talk) performance of ultra-high-speed SERDES, such as 112-Gb/s PAM-4 transceiver as reported in this article, has become very important due to the stronger aggressor-to-victim coupling resulting from sharper transition edges and the reduced SNR margin in PAM-4 signaling. The silicon characterization data of X-talk performance of such links provide valuable information to assist and guide system-level link margin budgeting and component specification. For X-talk noise margin testing, an asynchronous 56-Gb/s NRZ



(c)

Supply	Usages
0.88V	Digital (SAR ADC, TX serializer/FIR/pre-DRV)
1.2V	RX AFE (CTLE & PGA), ADC SFE (T/H buffers), regulated CMOS HS clock, ADC Vrefgen
1.5V	TX pad driver, regulator OTA

Fig. 17. (a) Die micrograph. (b) Power break-down. (c) Supply usages.

	This work	Ref [1]	Ref [2][4]	Ref [5]	Ref [3]
Technology	7nm FinFET	10nm CMOS	16nm FinFET	14nm FinFET	10nm FinFET
Data Rate	112Gb/s	112Gb/s	112Gb/s	128Gb/s	112Gb/s
TX Architecture	Quarter-rate CML 4-Tap FFE	-	Quarter-rate SST 4-Tap FFE	Quarter-rate CML 3-Tap FFE	Quarter-rate CML 3-Tap FFE
RX Equalization	CTLE 31-Tap FFE 1-Tap DFE	CTLE 16-Tap FFE 1-Tap DFE	CTLE 31-Tap FFE 1-Tap DFE	-	-
Active Area (excl. DSP)	TX: 0.136mm² RX: 0.265mm²	RX: 0.281mm²	TX: 0.38mm² RX: 0.674mm²	TX: 0.048mm²	TX: 0.03mm²
TX Power (excl. PLL)	1.34pJ/b	-	3.08pJ/b	1.0pJ/b (112Gb/s)	1.72pJ/b
RX Power (excl. DSP)	ADC: 4.8pJ/S RX: 3.6pJ/b*	ADC: 6.71pJ/S RX: 4.2pJ/b	ADC: 8.48pJ/S RX: 5.27pJ/b	-	-
PLL and Clk Distribution	0.44pJ/b	-	1.47pJ/b	-	-
Supplies	0.88V/1.2V/1.5V	0.95V/1.5V	0.9V/1.2V/1.8V	0.85V/0.9V	1.0V
Channel	37.5dB	35dB (BERT)	20dB (BERT)	-	-
BER (PRBS31)	1E-8	1E-6	2E-5	-	-

Fig. 18. Performance summary and comparison to prior published 112-Gb/s PAM-4 RX and TX with integrated PLL.

signal of varying amplitude from a BERT is injected into a neighboring channel as an aggressor. The total integrated noise power (in mV-rms) applied at the victim DUT RX input is first calibrated using a scope to measure the RMS difference of noise amplitude with and without the aggressor turned on. The IL between the scope and the X-talk test points was independently measured and accounted for. Plotted in Fig. 16(b), the BER degrades from ~1E-8 to ~1E-5 as the aggressor noise increases from 0- to 2-mV rms, leaving 10× margin from the OIF CEI pre-FEC BER spec of 1E-4.

Fig. 17(a) shows a die micrograph of the fabricated chip obtained by the back-side infrared imaging technique with major sub-blocks of the transceiver and the shared LC-PLL demarked by boxes. The measured power of 602 mW per transceiver channel for an LR channel shown in Fig. 17(b) consists of 25% of TX, 66% of RX, and 8% of amortized PLL and high-speed clock distribution contributions. A sub-ADC operating at 1.56 GS/s consumes 2.72 mW of active power, including T/H-2 clock buffers, a StrongArm slicer, SAR logic, CDAC, reference voltage, and ADC output retimers. Three power domains are used throughout the transceiver: 0.88 V for CMOS digital blocks; 1.2 V for RX AFE and ADC SFE stages and regulated supplies for LC-PLL, CMOS clock buffers, and ADC reference voltage; and 1.5 V for regulator error amplifiers and CML TX pad driver front-end.

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VI. Conclusion

Fig. 18 summarizes the transceiver performance and its comparison with prior published 112-Gb/s transmitters and receivers. The entire TX and RX dissipate 1.34 and 3.6 pJ/b, respectively, at 112 Gb/s over a long-reach channel, excluding DSP but including shared channel clock blocks. PLL and clock distribution buffers add 0.44 pJ/b per channel. The active TX and RX (excluding DSP and CDR) area are 0.136 and 0.265 mm^2, respectively.

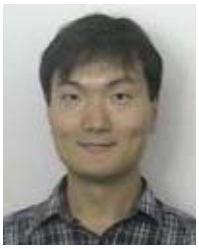
Compared with our previously reported 16-nm 112-Gb/s TX and RX [2], [4], this work achieved 60% of area and 40% of power reduction, thanks to the area-efficient all CMOS receiver front-end and clock generation and distribution circuits as well as a compact 36-way time-interleaving ADC SFE architecture. The bench test results demonstrated a superior performance over long-reach channels, satisfying the BER and JTOL requirements of OIF CEI-112G-LR-PAM4 standard [6].

Acknowledgment

The authors thank Xilinx SERDES design, layout, and verification teams for having made this project successful.

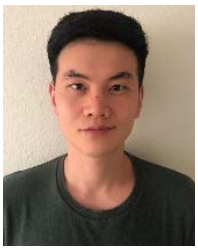
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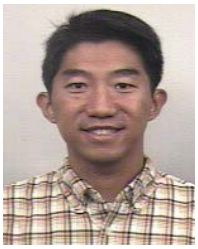
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A 56-Gb/s PAM4 Wireline Transceiver Using a 32-Way Time-Interleaved SAR ADC in 16-nm FinFET

Yohan Frans, Jaewook Shin, Lei Zhou, Parag Upadhyaya, Jay Im, Vassili Kireev, Mohamed Elzeftawi, Hiva Hedayati, *Member, IEEE*, Toan Pham, Santiago Asuncion, Chris Borrelli, Geoff Zhang, Hongtao Zhang, and Ken Chang, *Senior Member, IEEE*

***Abstract*— A 56-Gb/s PAM4 wireline transceiver testchip is implemented in 16-nm FinFET. The current mode logic transmitter incorporates an auxiliary current injection at the output nodes to maintain PAM4 amplitude linearity. The ADC-based receiver incorporates hybrid analog and digital equalizations. The analog equalization is performed using two identical stages of continuous time linear equalizer, each having a constant of ∼0-dB dc-gain and a maximum peaking of ∼7 dB peaking at 14 GHz. A 28-GSample/s 32-way time-interleaved SAR ADC converts the equalized analog signal into digital domain for further equalization using digital signal processing. The transceiver achieves <1e-8 bit error rate over a backplane channel with 31-dB loss at 14-GHz and 3.5-mV$_{rms}$ additional crosstalk, using a fixed ∼10-dB TX equalization and an adaptive hybrid RX equalization, with the DSP configured to have a 24-tap feed forward equalizer and a 1-tap decision feedback equalizer. The transceiver consumes 550-mW power at 56 Gb/s, excluding the power of the on-chip configurable DSP that cannot be accurately measured as it is implemented as part of a larger test structure.**

***Index Terms*— 56 Gb/s, ADC, PAM4, transceiver, wireline.**

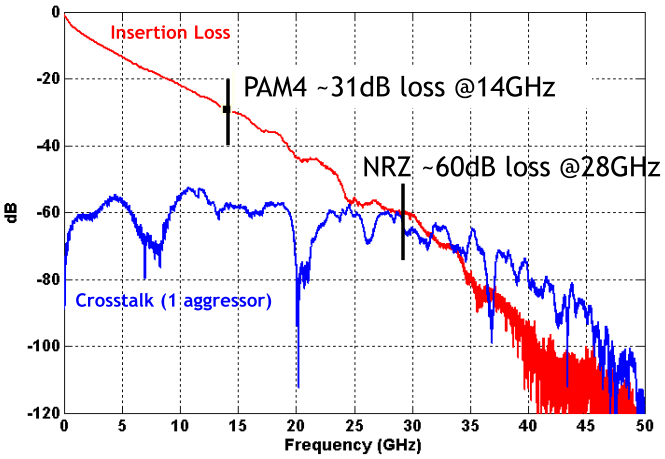


Fig. 1. IL and crosstalk profile of legacy backplane designed for 28-Gb/s NRZ signaling.

I. Introduction

THE emergence of Internet of Things and cloud computing has triggered rapid increase in bandwidth demand in data centers and telecommunication infrastructures. The increasing bandwidth demand had recently prompted the industry to propose a new electrical interface standard capable of operating up to 56 Gb/s per lane [1]. In order to avoid costly infrastructure upgrades, the interface needs to support legacy channels (i.e., backplane) designed for current generation electrical interface (e.g., up to 28 Gb/s in traditional non return to zero (NRZ) signaling). These legacy channels often have a very large insertion loss (IL) beyond 14 GHz with significant reflections. Fig. 1 shows an example of such backplane channel. If the NRZ signaling was used to operate the link at 56 Gb/s, the IL at Nyquist frequency (28 GHz) would be ∼60 dB, and the IL-to-crosstalk ratio (ICR) from one aggressor at 28 GHz would be ∼0 dB. These constraints make it very difficult to implement the NRZ signaling at 56 Gb/s with reasonable power efficiency. The PAM4 signaling [2] is a better choice for these legacy channels, since the signal frequency content can be limited to 14 GHz. For the channel shown in Fig. 1, the IL is ∼31 dB and the ICR is ∼30 dB at 14 GHz. Even though many of the PAM4 signaling challenges (e.g., smaller signal power due to transmitter's peak power constraint, sensitivity to residual ISI) still need to be addressed, these IL and ICR numbers fall within the equalization capability of the state-of-the-art transceivers.

In order to relax the raw bit-error-rate (BER) requirement, the proposed standard allows the use of forward error correction (FEC) mechanism in the targeted links/systems. Error-correcting block codes such as Reed–Solomon code (Fig. 2) is used to encode/decode the data, such that low BER (e.g., <1e-18) can still be achieved even when the raw (pre-FEC) BER is relatively high (e.g., 1e-4 to 1e-6) [3], at the expense of higher total latency in the links/systems. A raw BER of <1e-4 is currently proposed as standard [1].

This paper presents the design of a 56-Gb/s ADC-based PAM4 transceiver with a moderate target BER (e.g., 1e-6 to 1e-8) over legacy channels to be used with FEC. Section II covers the PAM4 transmitter design. The ADC-based receiver is described in Section III. The experimental results are summarized in Section IV.

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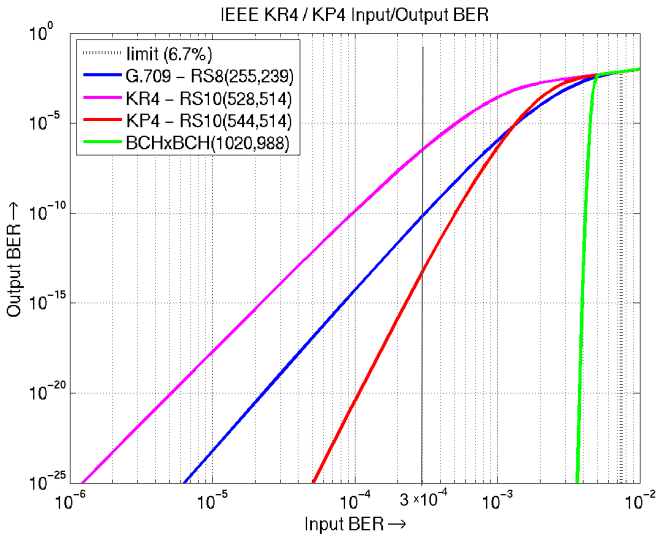


Fig. 2. BER improvement using block codes.

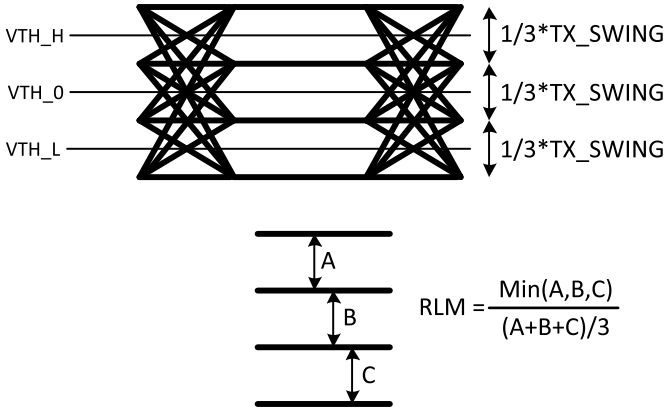


Fig. 3. PAM4 transmitter output levels and linearity definition.

II. Transmitter

There are two new challenges in designing a PAM4 transmitter [4] when compared with designing an NRZ transmitter. First, in general, a transmitter designed for PAM4 signaling must be able to deliver a higher swing, because the corresponding receiver effectively needs to be able to resolve smaller amplitude compared with NRZ signaling (in an ideal case, the receiver needs to be able to resolve one third of the transmitted swing, as shown in Fig. 3). Second, the transmitter must maintain good linearity when transmitting the four PAM4 signaling levels. The dc-level distortions due to transmitter nonlinearity directly translate into smaller effective eye height seen by the receiver. The proposed standard [1] specifies transmitter linearity requirement in terms of ratio of level mismatch (RLM), as shown in Fig. 3.

Fig. 4 shows the block diagram of the three-tap (one main-tap, one pretap, and one posttap) transmitter used in this design. A pattern generator generates either a Pseudo random bit sequence (PRBS) data sequence (PRBS-7, PRBS-15, or PRBS-31) or a programmable 128-b data pattern. The 128-b output of the pattern generator is serialized to a 16-b data that feeds a finite impulse response (FIR) generator block. The FIR generator block generates three streams of 4-b data—one data stream for each FIR tap. A transmitter

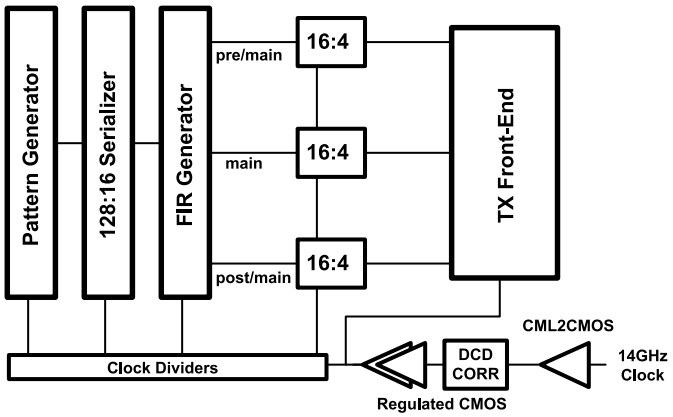


Fig. 4. Transmitter block diagram.

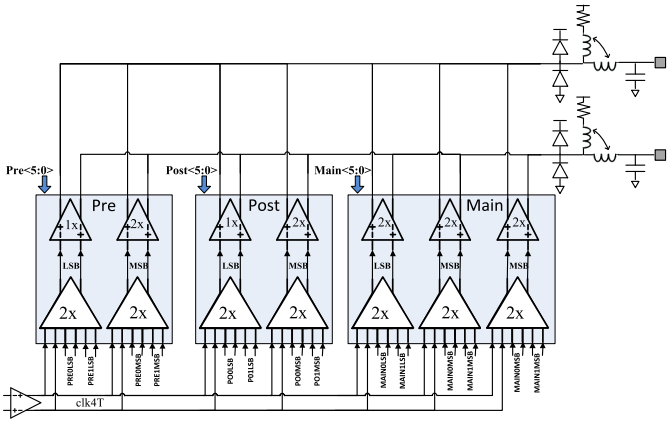


Fig. 5. Transmitter front end.

front-end block receives these 4-b data streams and generates FIR-equalized PAM4 signals. A 14-GHz differential current mode logic (CML) input clock generated by a phase-locked loop (PLL) is converted into a regulated-CMOS level and sent to the transmitter front-end block. A duty cycle detection/correction loop detects duty cycle distortion at the input of the transmitter front end and corrects the distortion in regulated-CMOS domain [5].

The transmitter front end (Fig. 5) consists of three segments (one segment for each transmitter tap) whose outputs share a common T-coil-enhanced termination network. The size of the main-tap segment is two times larger than the size of the pretap and posttap segments. Each transmitter tap segment consists of two stages: a 4-to-2 MUX/predriver stage and a final CML driver stage. The 4-to-2 MUX/pre-driver serializes a 4-b data input (the output of FIR generator) into a 2-b data (one MSB and one LSB). The final CML driver stage converts the 2-b data into PAM4 signals. The driver current of each segment can be controlled using a 5-b digital control to enable fine adjustments of the transmitter swing and/or transmitter equalizations.

Each tap of the PAM4 final CML driver (Fig. 6) is realized by the current summing of two times driver (MSB) and one time driver (LSB). The transmitter must maintain linearity between the four output levels while delivering up to 1.2 V diff-pp swing. Since the main power supply level of the driver (AVTT) is only 1.2 V, the output common mode will be too low to keep the differential pairs N0 and N1 and N2 and

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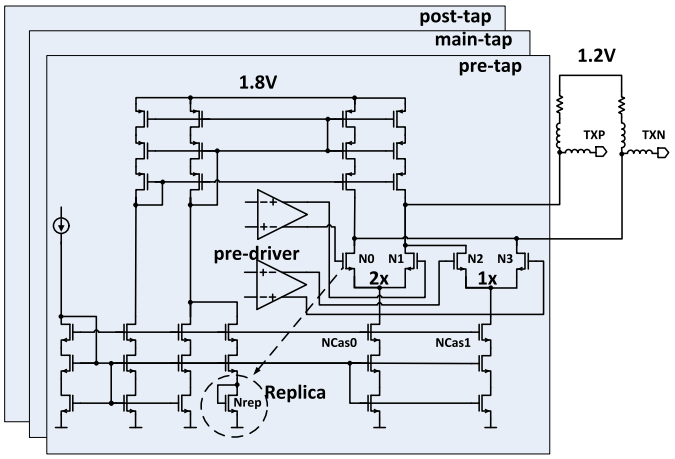


Fig. 6. CML driver.

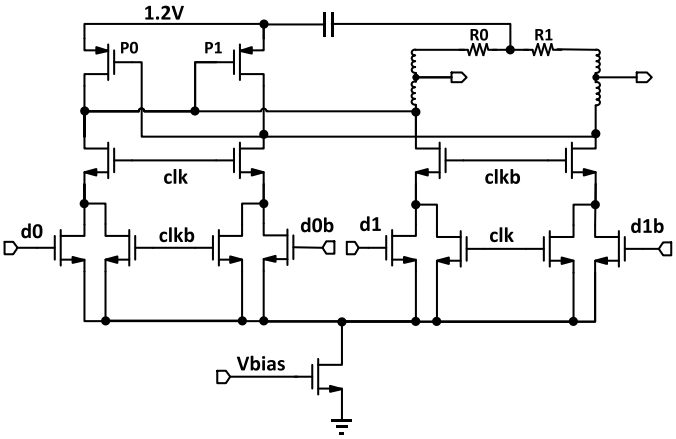


Fig. 7. 4:2 MUX/predriver circuit.

N3 in saturation when the driver delivers high current, hence distorting the output linearity. In order to raise the output common mode, some (about 25% typically) of the driver current is sourced using current sources from an auxiliary supply (1.8 V). Since only ∼25% of the driver current is sourced from 1.8 V auxiliary supply, the power penalty is only 12.5% of the total driver power. Open-loop compensation using a replica (N_{rep}) of driver input differential pair helps maintain optimum output common mode over process voltage temperature (PVT). As an example, at FF process corner and high temperature, N_{rep} provides additional current that further increase the output common mode and increase V_{ds} of N0 and N1 and N2 and N3 differential pairs to improve their saturation margins. Small cascode devices (Ncas0, NCas1) are placed above the tail current sources to increase their output impedance, which further improves dc linearity and reduces ac distortion.

The combined 4-to-2-MUX/predriver circuit (Fig. 7) incorporates a pseudo-H-bridge scheme with positive feedback. Pseudodifferential regulated-CMOS clocks (clk and clkb) are used to select the multiplexer output between pseudodifferential CMOS data d0/d0b and d1/db. The load structure of this circuit consists of cross-coupled PMOS devices (P0 and P1) in parallel with passive resistors R0 and R1. In order to extend the bandwidth, a small T-coil network is used. The circuit has a high gain at zero crossing, which improves rise/fall times and helps suppress clock switching noise at the driver output.

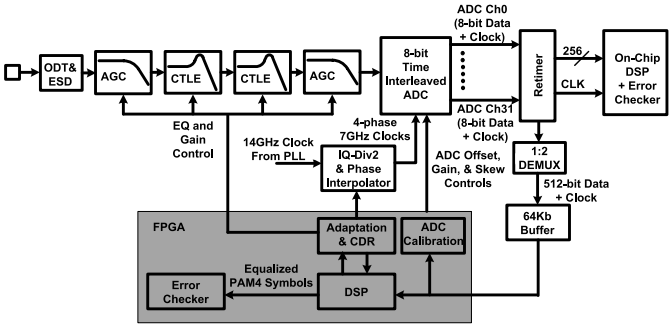


Fig. 8. Receiver block diagram.

III. Receiver

One of the main challenges in PAM4 signaling is its sensitivity to residual ISI. Since there are four possible levels, the worst case impact occurs when residual ISI from major transitions (i.e., from the lowest level to the highest level or vice versa) is superimposed on the main signal (which even in an ideal case is only one third of the major transition). In order to minimize the residual ISI, a large number of equalization taps (covering both precursor taps and post-cursor taps) are needed. However, implementing a large number of equalization taps in analog domain (at the receiver front-end) requires multistage feed forward equalizer (FFE) and/or decision feedback equalizer (DFE) summing circuits with enough gain to overcome slicer sensitivity and with good linearity over the range of PAM4 signaling levels. The design of this multistage analog front end can be very challenging at a higher line rate such as 56 Gb/s. In the proposed receiver, the large number of equalization taps is implemented in the digital domain using digital signal processing (DSP), thus avoiding linearity issues associated with the analog solution.

Fig. 8 shows the receiver block diagram. The on-die-termination (ODT) includes a T-coil structure that compensates for parasitic capacitances from the pad, ESD protection devices, and analog front-end input gates. The receiver analog front end [two stages of automatic gain control (AGC) and two identical stages of continuous time linear equalizer (CTLE)] provides signal equalization and conditioning, which reduces the resolution and full-scale-range requirements of the ADC [6], [7]. The 28-GSample/s 32-way time-interleaved (TI) ADC converts the differential analog input into 8-b digital values. The ADC outputs are then retimed to a single 875-MHz clock domain. One set of these retimed outputs is sent to an on-chip configurable DSP and error-checker block. For the testing and debug purpose, another set of the retimed outputs is sampled periodically and stored in a 64 kb (8k symbols) buffer. A replica of the on-chip DSP block (implemented in an off-chip FPGA) periodically samples these 8k symbols and generates equalized PAM4 symbols. The off-chip FPGA also performs equalization adaptation, clock and data recovery (CDR), and ADC offset/gain/skew calibrations based on the sampled ADC outputs. The FPGA adaptation logic concurrently adapts the analog front end (CTLE and AGC settings) and the DSP coefficients. The FPGA CDR logic

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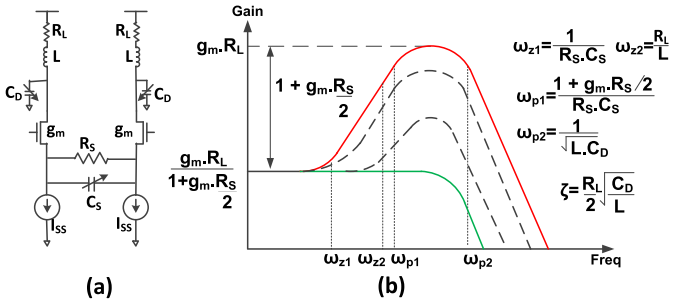


Fig. 9. Constant dc-gain and constant peaking frequency CTLE.

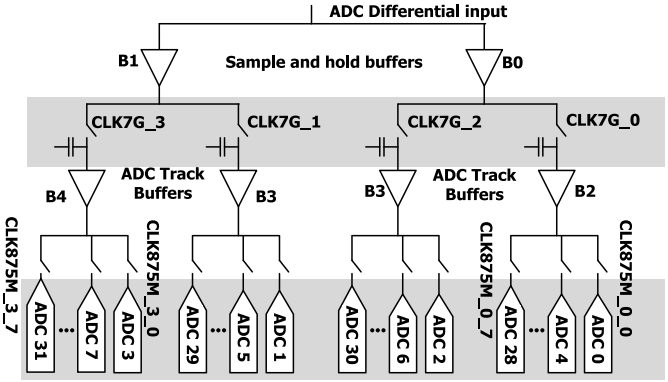


Fig. 10. 32-way TI ADC block diagram.

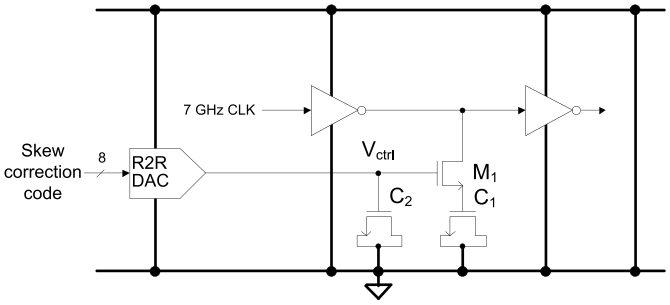


Fig. 11. 7-GHz clock skew correction circuit.

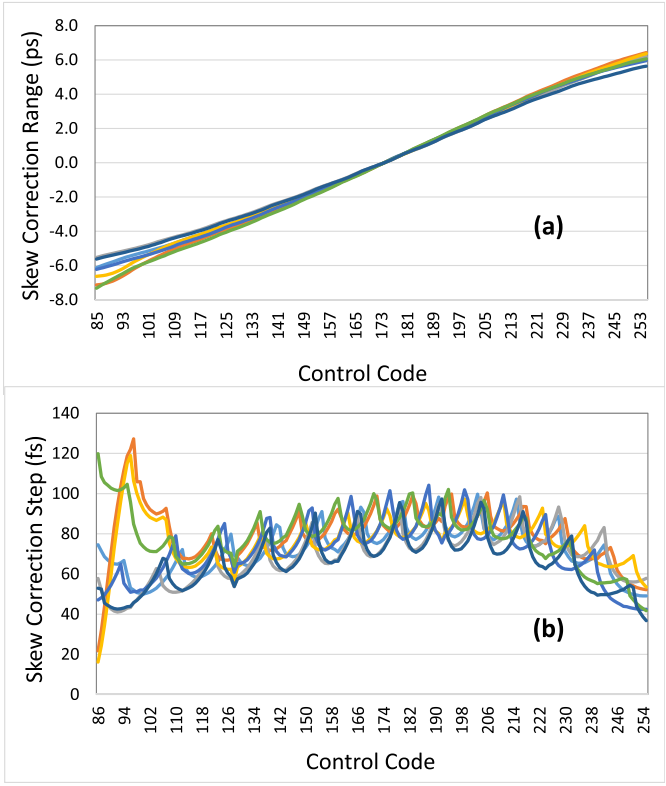


Fig. 12. (a) 7-GHz clock skew correction range and (b) correction step.

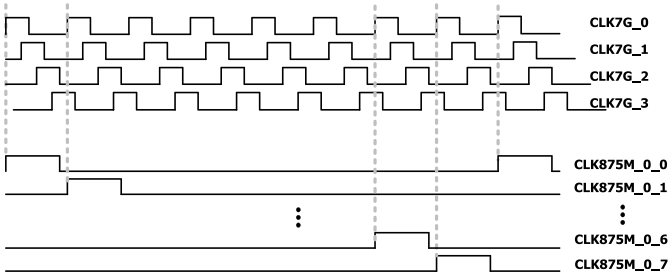


Fig. 13. ADC timing diagram.

implements a second-order baud-rate CDR using the Mueller–Muller algorithm.

The T-coil-enhanced ODT provides ∼27-GHz bandwidth and less than −8 dB return loss at 14 GHz (meeting the return loss requirement defined in [1]). Each one of the AGC stages provides up to 10 dB of programmable dc gain range. Each one of the two identical CTLE stages is designed to have a constant-dc gain (∼0 dB) and programmable high-frequency gain peaking (up to 7 dB) with constant peaking frequency at ∼14 GHz. The offset from AGC and CTLE random device mismatches is corrected at these analog front-end stages so that the effective ADC full-scale range is not compromised. Compared with a constant high-frequency gain CTLE (with programmable dc gain), the constant-dc gain CTLE can either reduce the required AGC's gain at high-loss channels and/or improve the linearity of subsequent stage at low-loss channels. Furthermore, this approach minimizes the interaction between AGC adaptation loop and CTLE adaptation loop. Constant peaking frequency around Nyquist (∼14 GHz) is chosen to suppress early post-cursor ISI as much as possible to help reduce ADC dynamic range requirement, while minimizing crosstalk and thermal noise beyond Nyquist frequency. As shown in Fig. 9, the CTLE uses an *RC* source-degeneration topology. In order to obtain constant dc gain and constant peaking frequency, the values of the source-degeneration capacitance C_s and the values of the output load capacitance C_d are set for every gain-peaking setting. Higher gain peaking is obtained by increasing the value of C_s and decreasing the value of C_d.

Fig. 10 shows the block diagram of the 8-b 28 GSample/s, 600 mV diff-pp full-scale range, 32-way TI SAR ADC used in the receiver. There are two stages of time interleavers. The first stage is a four-way time interleaver, where the input is buffered by B0/B1 buffers and sampled and held using four-phase 7-GHz sampling pulses. The sampling pulses sharing the same buffer (CLK7G_0 and CLK7G_2 sharing B0 buffer and CLK7G_1 and CLK7G_3 sharing B1 buffer) are designed to be nonoverlapping to avoid charge sharing. The second stage is an eight-way time interleaver, where each of the signals sampled by the 7-GHz sampling pulses is

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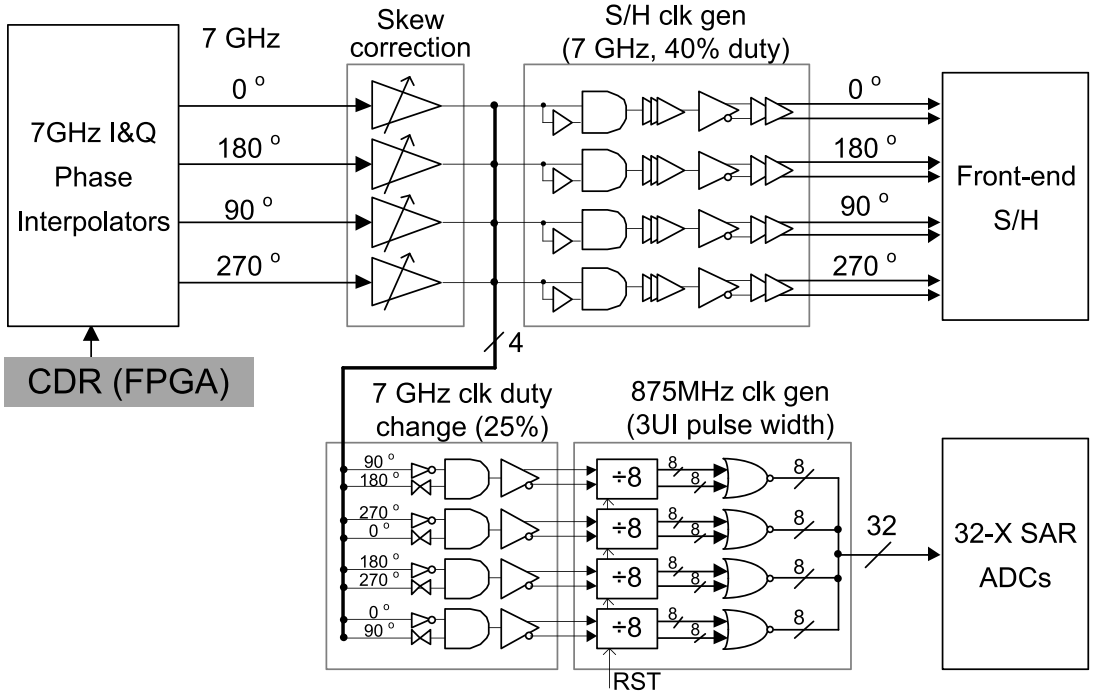


Fig. 14. ADC clocking diagram.

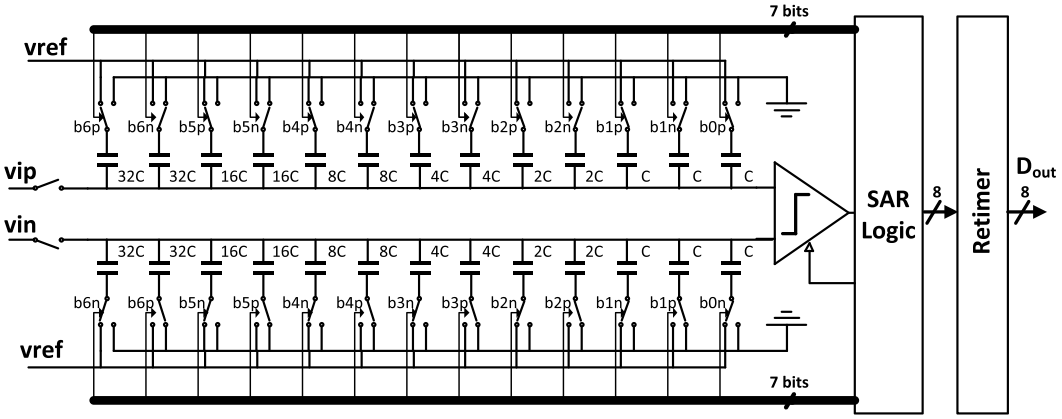


Fig. 15. 875-MHz SAR sub-ADC circuit.

further sampled and held using eight-phase 875-MHz sampling pulses and converted into digital values using eight instances of 875-MHz SAR sub-ADC. The output of the 32 instances of SAR sub-ADCs is then retimed to a single 875-MHz clock domain and sent to a 64-kb storage.

The timing skew calibration [8]–[10] of the 7-GHz clocks and the gain/offset calibration of the 875-MHz sub-ADC instances are performed using the pseudorandom data input [9], in contrast to the use of sinusoidal tones for calibration in [8]. This approach allows live data calibration where scrambled pseudorandom data are common in most high-speed interfaces. In order to keep the area of the sub-ADC circuit small, the gain and offset corrections of the sub-ADCs are performed in the digital domain at the expense of $\sim$10% reduction of the ADC full-scale range. Fig. 11 shows the circuit diagram of the 7-GHz clock skew correction block (only one clock phase is shown). The main path is composed of two stages of inverters, having a controlled *RC*-network in between them. NMOS M1 controls the charging and discharging current of the capacitor C1 (60 fF). The gate voltage of NMOS M1 (V_{ctrl}) is controlled by a digital skew correction code through the resistor-ladder (R2R) DAC. Since the R2R DAC has a high output impedance (4 kΩ, a bypass capacitor C2 (3 pF) is added to reduce noise coupling to the control voltage V_{ctrl}. The worst case differential non linearity of the R2R DAC with three-sigma over PVT variations is 0.46 LSB, so the skew correction step is always positive. Fig. 12 shows the simulation results of the skew correction knob characteristics over PVT corners. The skew correction range is more than -5.5 to 5.5 ps without any mismatch (random and systematic). When random and systematic mismatch is considered, the skew correction range shrinks down to -3.7 to 3.7 ps—still wide enough to cover clock distribution mismatches up to the first stage sample-and-holds (S/Hs). The

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skew correction step is designed to be less than 100 fs to target 38-dB signal to noise and distortion ratio (6-b ENOB) with margin for the Nyquist frequency of 14 GHz.

Fig. 13 shows the timing diagram of the TI ADC. The top four signals are front-end four-phase 7-GHz sampling pulses (CLK7G_0, CLK7G_1, CLK7G_2, and CLK7G_3). Each of the 7-GHz sampling pulse has an approximately 40% duty cycle that ensures there is no overlap between CLK7G_0 and CLK7G_2 (and similarly between CLK7G_1 and CLK7G_3). One group of 875-MHz sampling pulses (consisting of eight different phases) is generated from each phase of the four-phase 7-GHz sampling pulses (as shown in Fig. 13, CLK875M_0_0 to CLK875M_0_7 are derived from CLK7G_0). Hence, there are 32 different phases (four groups and eight phases per group) of 875-MHz pulses in total—each one corresponds to one instance of an 875-MHz SAR sub-ADC. The 875 MHz sampling pulses have a ~3UI pulsewidth in order to avoid overlaps between them. This gives ~29UI period for the sub-ADC to perform conversion. The rising edges of the 875-MHz sampling pulses are aligned with the rising edge of corresponding 7-GHz sampling pulse. If they are not aligned well, the falling edge of the previous 875-MHz sampling may occur after the rising edge of 7-GHz clock. In this case, the previously sampled signal is mixed with the current sampled signal, significantly increases signal distortion.

The block diagram of TI ADC clock generation and timing circuit is shown in Fig. 14. It generates four-phase 7-GHz sampling pulses for front-end S/H and 32-phase 875-MHz sampling pulses for the SAR sub-ADCs. A quadrature phase interpolator generates four phases of 7-GHz clocks with 50% duty cycle. The phase interpolator delay is controlled by the CDR loop. Each of the 7-GHz clock phase has a dedicated path to the front-end S/H. Skew correction block adjusts the delay of each phase of the 7-GHz clocks based on the ADC timing-skew calibration loop. The outputs of the skew correction block are split into two paths: front-end S/H path and sub-ADC path. In the front-end S/H path, a 7-GHz clock generation block generates 40%-duty-cycle sampling pulses from the four-phase clocks using AND gates. The 7-GHz clock generation block also inserts some delay in order to control timing (rising-edge alignment) between the front-end 7-GHz sampling pulses and the corresponding back-end sub-ADC sampling pulses over PVT variations. After the delay chain, the 40%-duty-cycle sampling pulses are converted into CMOS-level pseudodifferential signals to drive the front-end S/Hs. The generation of 32-phase, 3UI-wide 875-MHz sub-ADC sampling pulses starts by changing the duty cycle of the 7-GHz clocks from 50% to 25%. Two adjacent phases of 7-GHz clocks with 50% duty cycle are used to generate one phase of 7-GHz clock with 25% duty cycle. These 25%-duty-cycle clocks are then converted into CMOS-level pseudodifferential signals and sent to an 875-MHz clock generation block, which is composed of four groups of synchronous divide-by-eight block and eight NOR gates.

Fig. 15 shows the block diagram of the 875-MSa/s 8-b asynchronous SAR sub-ADC. It consists of 127 units of

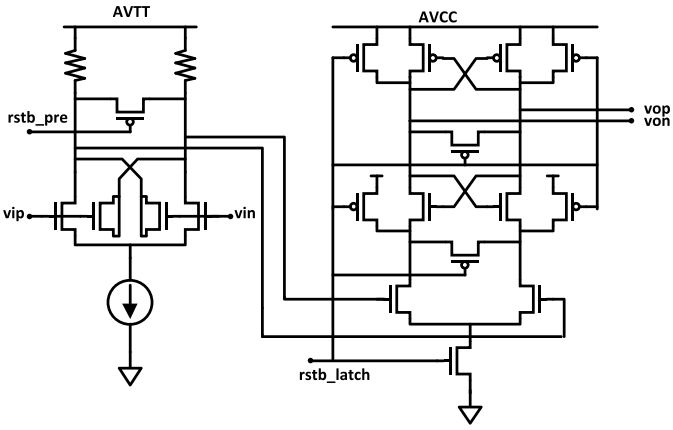


Fig. 16. Sub-ADC comparator.

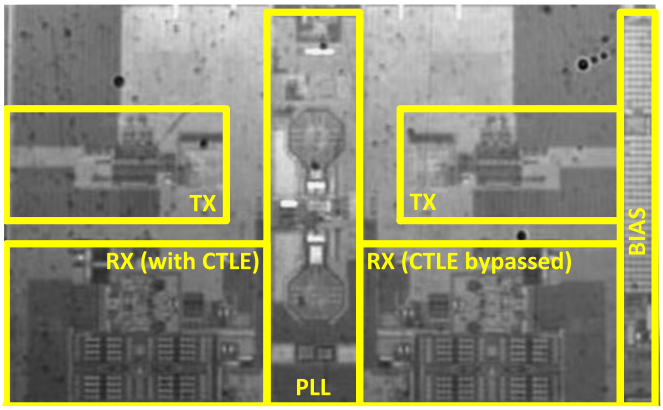


Fig. 17. Die micrograph.

capacitors, one comparator, an SAR logic block, and a retimer. A differential capacitor DAC (CDAC) structure is adopted to keep the common mode at the comparator input constant during conversion. In order to achieve high sampling rate with minimum power consumption, the size of the CDAC and any parasitic capacitors needs to be minimized. Top-plate sampling scheme [11] is adopted to reduce the CDAC requirement from 8 to 7 b, therefore reducing CDAC size by half. The CDAC unit capacitor is also sized just enough (~0.4 fF) to meet the matching requirement. Since the sub-ADC only needs to have moderate resolution (8 b), the CDAC matching requirement is not very stringent. Instead of using one of the available power supplies, a low-voltage, low-impedance reference voltage (V_{ref}) is used in the CDAC to achieve the same full-scale range without increasing the total capacitor size. The relationship between full-scale range voltage (V_{fs}) and V_{ref} is shown as follows:

$$V_{fs} = \frac{2^7 C_u}{C_{\text{tot}}} 2V_{\text{ref}}$$
$$C_{\text{tot}} = 2^7 C_u + C_{\text{par}} \quad (1)$$

where C_u is the CDAC unit capacitance, C_{tot} is the total capacitor size, and C_{par} is the parasitic scaling capacitor. From (1), for a given full-scale range and CDAC unit capacitor, lower V_{ref} value allows for lower parasitic scaling capacitor. A single programmable CDAC reference voltage (V_{ref}) is generated on-chip to drive all 32 sub-ADCs for better matching.

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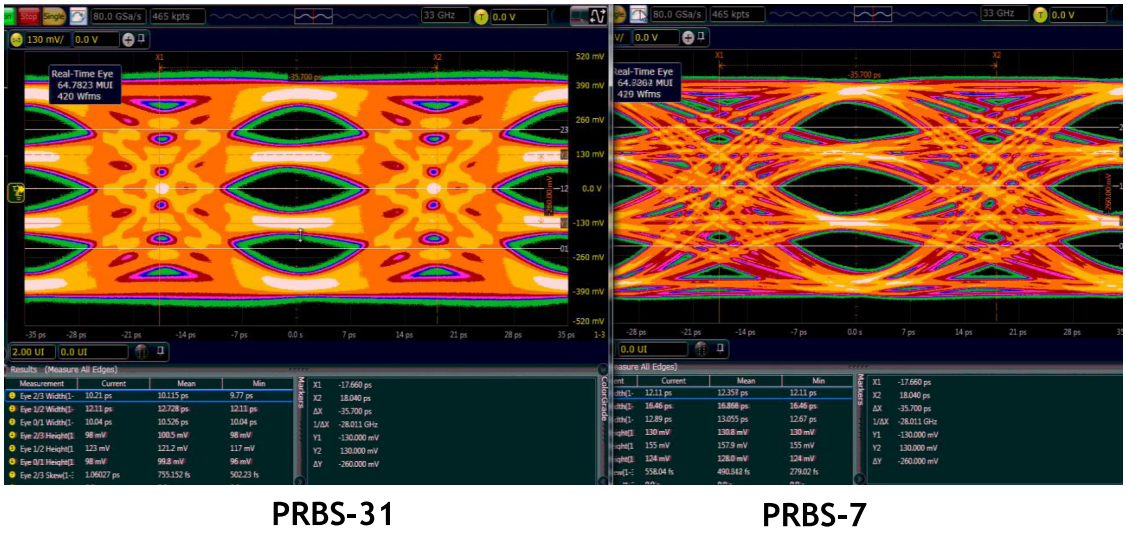


Fig. 18. Transmitter output eye diagram.

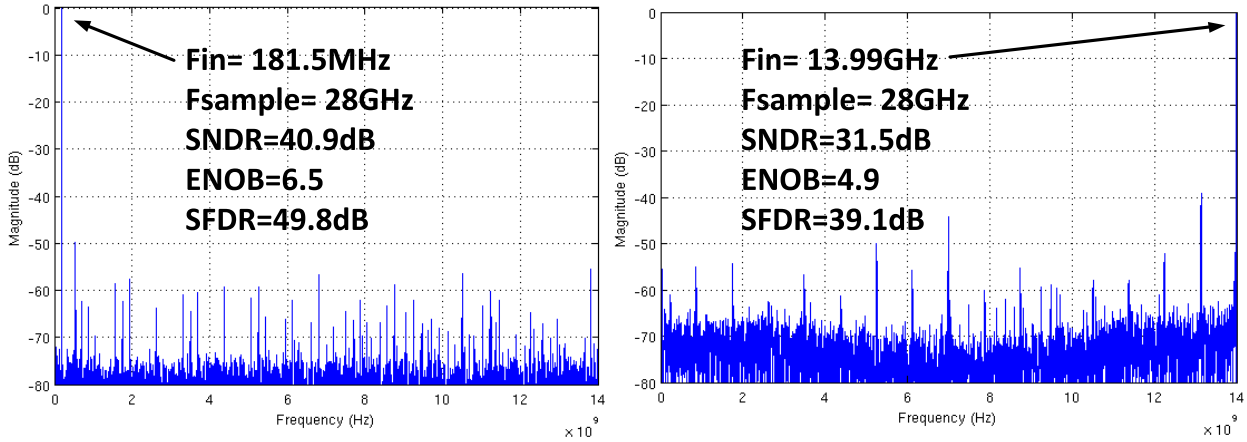


Fig. 19. Measured ADC performance with 181.5-MHz and 13.99-GHz sinusoidal inputs.

The V_{ref} output impedance is designed to be low in order to avoid disturbance from 32 DACs switching.

The comparator is composed of a preamplifier followed by a StrongArm (SA) latch (Fig. 16). The preamplifier is added to reduce the input-referred noise of the comparator to below 0.5 LSB. A reset switch is added to the preamplifier to implement overdrive recovery, which clears the memory effect from previous bit. The preamplifier is enabled earlier than the SA latch to allow the output to settle before SA latch starts. A pair of dummy NMOS with its drain and source shorted together is connected to the input and output of the preamplifier to reduce kick back from the preamplifier output to the CDAC. Together with SAR logic delay and the CDAC settling time, the comparator regeneration time constant (τ) determines the metastability error rate of the SAR ADC [12]. τ is designed to be $\sim$4 ps to achieve $<$1e-15 metastability error rate (significantly lower than the raw BER target of the link) assuming uniform distribution of the input signal.

In order to avoid distributing high frequency clock and to increase the SAR conversion speed, asynchronous SAR logic with programmable delays is implemented. To increase speed further, the SAR logic is implemented using custom-built digital circuits.

IV. Experimental Results

Fig. 17 shows the die micrograph of the testchip fabricated in 16-nm FinFET. There are two TX/RX lanes sharing a common PLL, clock distribution, and bias block. The first RX lane has full receiver functionalities. The CTLE in the second RX lane is bypassed to facilitate direct measurements of the ADC performance.

Fig. 18 shows the transmitter output eye diagram over $\sim$5-dB channel. The transmitter is configured to transmit PRBS7 and PRBS31 patterns with $\sim$4.5-dB posttap equalization. The measured random jitter is 200 fs-rms. The transmitter linearity measurement is performed by measuring average dc voltage of the four PAM4 levels and calculating RLM using the equation shown in Fig. 3. The transmitter achieves the RLM of 0.97 at 1.2 V diff-pp swing.

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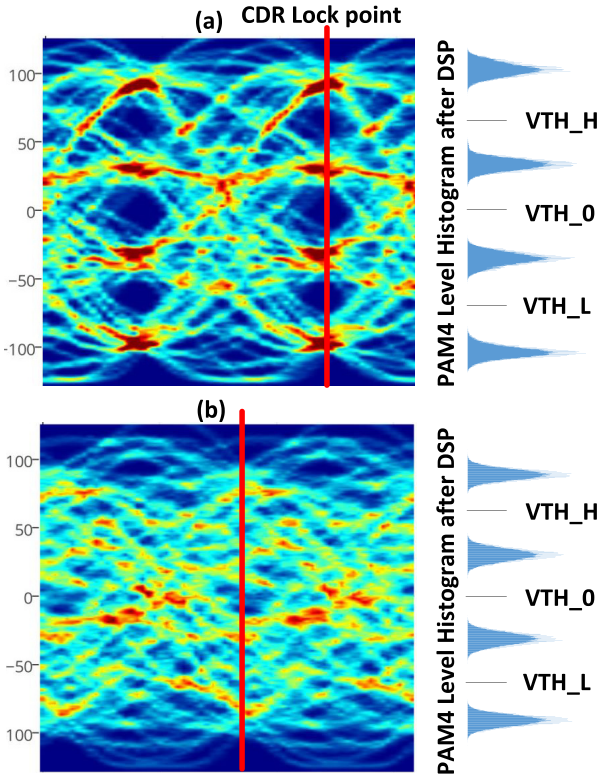


Fig. 20. Pre-DSP received eye diagram at (a) ADC Input for 6-dB channel and (b) 32-dB channel.

The stand-alone ADC performance is measured on the second RX lane with CTLE bypassed. The ADC gain, offset, and timing skew calibrations are first performed by passing PAM4 PRBS31 data sequence through the ADC. Once the calibrations are completed, the ADC inputs are connected to a differential sinusoidal signal at various frequencies—generated using an RF signal generator and a single-ended-to-differential phase splitter. FFT is performed on the ADC outputs captured in the 64-kb storage. The ADC achieves the ENOB of 6.3 at 180 MHz and 4.9 at 14 GHz, as shown in Fig. 19. The ADC performance at 180 MHz is limited by the thermal noise of the single-slice SAR ADC comparator. The ADC performance at 14 GHz is limited by residual timing skew of the four-phase 7-GHz sampling pulses.

The link functionality is tested by connecting the transmitter output to the receiver through different kinds of backplane channels. The off-chip FPGA periodically reads reading the ADC outputs stored in of the 64-kb buffers and performs ADC calibration, CDR, equalization adaptation, and DSP. Fig. 20 shows eye diagrams at the CTLE/AGC outputs (captured using the ADC as a digital oscilloscope) and the post-DSP histograms of the four PAM4 levels (sampled at the CDR lock point) over 200k symbols. The CTLE/AGC outputs show open eye with ∼6-dB channel [Fig. 8(a)] and closed eye with 25-dB channel (8b). In both cases, the DSP opens the eye in the post-DSP PAM4-level histograms. The estimated BER is <1e-8 based on extrapolation of the histograms.

In order to measure the link performance at full 56-Gb/s throughput (without data subsampling), the on-chip DSP and the on-chip error checker are used. The off-chip FPGA still performs ADC calibrations, CDR, and equalization adaptation.

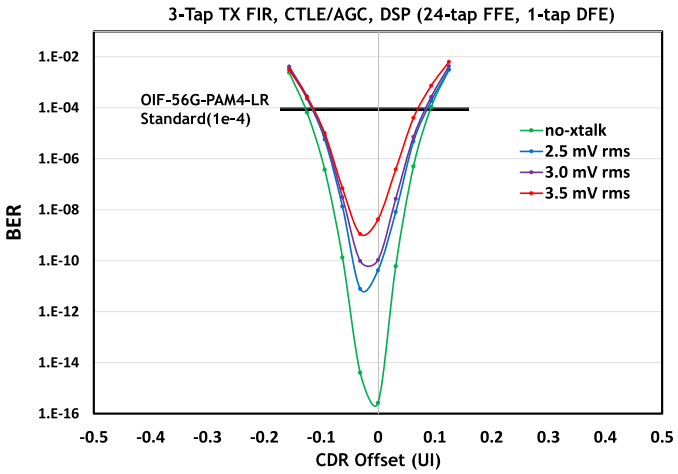


Fig. 21. BER bath-tub curve for 31-dB channels at various crosstalk levels.

TABLE I
PERFORMANCE SUMMARY

Technology	CMOS 16nm FinFET
Power Supply (V_{avcc}, V_{avtt}, V_{aux})	0.9V, 1.2V, 1.8V
Dual Transceiver Active Area	2.8mm^2
Max TX Swing	1.2V diff-pp
TX RJ (PRBS7, Major Transition)	200fs
ADC ENOB	6.5@0.18GHz, 4.9@14GHz
ADC Power (including ADC clocks)	280mW
BER at 56Gb/s (31 dB loss @14GHz, 3.5mV rms cross-talk)	<1x10^{-8}
Power per lane at 56Gb/s (Does not include DSP)	550mW (140mW TX, 370mW RX, 40mW PLL/Clock Distribution)

The resulting ADC calibration and equalization coefficients are used to set the corresponding coefficients in the on-chip DSP. The link performance measurement results are described in the form of BER bath-tub curve around the CDR lock point (Fig. 21). In this case, the ball grid array (BGA)-to-BGA channel loss is ∼31 dB at 14 GHz. The transmitter FIR settings are fixed and the receiver analog and digital equalizations are adapted. The DSP is configured to have 24-tap FFE and 1-tap DFE. Additional crosstalk is generated by sending a PRBS aggressor signal through a backplane channel next to the channel under test, measuring the rms of the crosstalk-induced voltage near the receiver, and adjusting the amplitude of the aggressor signal until the desired crosstalk rms voltage is obtained. The link BER is <1e-15 without additional crosstalk at the receiver and <1e-8 with 3.5 mV_{rms} additional crosstalk at the receiver. The timing window at 1e-4 BER (specified by the proposed standard) is >0.2UI.

Table I shows the performance summary. The transceiver consumes 550-mW total power (excluding the on-chip configurable DSP power which cannot be accurately measured as it is implemented as part of a larger test structure and not optimized in this design). The TX consumes 140 mW, the RX consumes 370 mW (of which 280 mW is consumed by the ADC), and the PLL/clock distribution/bias block consumes 40 mW.

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V. Conclusion

The transceiver described in this paper is able to achieve <1e-8 raw BER (significantly better than 1e-4 raw BER targeted by the standard [1]) over a legacy backplane channel with 31-dB loss and 3.5-mV additional crosstalk. Given the significant BER margin achieved in this design relative to the standard requirement, we will look for opportunities to further optimize power and performance in the future. Specific to the ADC-based receiver design, this will include analyzing the impact of ADC resolution to the link BER and analyzing the partition between the analog and digital portions of the equalizations.

Acknowledgment

The authors would like to thank Xilinx SerDes design and validation teams for contributing to their circuit design and silicon measurements.

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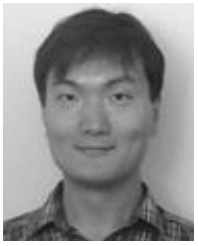
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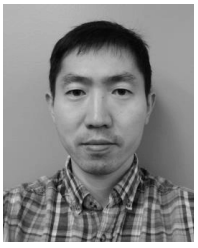
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10.5 A 12b 600MS/s Pipelined SAR and 2×-Interleaved Incremental Delta-Sigma ADC with Source-Follower-Based Residue-Transfer Scheme in 7nm FinFET

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Recently, pipelined ADCs are frequently adopted as sub-ADCs in multi-channel time-interleaved ADCs because a pipelined manner can reduce the number of interleaved channels by maximizing single-channel ADC operating speed. To reduce power consumption of residue amplifier, previous works [2,4,5] have explored power efficient residue amplifier architecture, such as dynamic amplifiers (DAs), open-loop amplifiers (OAs) and ring amplifiers (RAs). However, inter-stage gain error of power-efficient DAs, OAs, and RAs needs power-consuming gain calibration to compensate process, power supply, and temperature (PVT) variations.

This work presents a gain-calibration-free pipelined ADC architecture with the use of an open-loop residue-transferring buffer. The ADC consists of the 1st-stage successive-approximation (SAR) ADC, the 2nd-stage delta-sigma modulator (DSM), and a residue buffer. Instead of amplifying residue signal, the residue buffer transfers residue signal to the 2nd stage for pipelined operation. The non-amplified small residue signal is resolved by the 2nd stage DSM, which has wide dynamic range. A capacitor DAC (CDAC) in the 1st stage and a residue buffer are replicated in the feedback path of the 2nd stage DSM to eliminate PVT-variation-oriented inter-stage gain error without gain calibration. In addition, duration-fixed synchronous-like asynchronous clock, adaptive speed-controlled (ASC) clock, guarantees robust operation of the SAR against PVT variations. At 300MS/s, the gain-calibration-free ADC achieves a peak SNDR of 58.2dB. Moreover, the ADC has a 55.3dB SNDR with a 1Vpp swing, 500MHz (beyond Nyquist) input at 600MS/s. The ADC has superior speed over a wide input bandwidth compared with a sub-ranging SAR/DSM [1] and a noise-shaping pipelined SAR [3].

The ADC architecture and the corresponding timing diagram are shown in Fig. 10.5.1. The proposed ADC consists of a 7b single-channel SAR 1st stage, a 6.5b 2-channel time-interleaved 2nd-order continuous-time (CT) incremental DSM 2nd stage, and a residue buffer. The 1.5b redundancy can relax an inter-stage offset error requirement. The 1st stage samples the input signal with bootstrapped bottom-plate sampling. The sampling capacitance is 690fF to meet kT/C noise and residue signal accuracy. In the conversion phase, the 1st stage SAR resolves 7 bits with equally periodic 7 cycles of internal operation clock, CK_{ASC}, which triggers the comparator and SAR logic. After 7b conversion, the residue signal in a CDAC of the 1st stage is transferred to the 2nd stage by an open-loop residue buffer. The gain error of the residue signal path causes inter-stage gain error, which comes from the parasitic capacitance of a CDAC in the 1st stage and a gain error of open-loop residue buffer. The prototype ADC eliminates inter-stage gain error by matching the 2nd stage feedback path gain to residue signal path gain, rather than adopting digital calibration. The 2nd-stage CT incremental DSM operates at 300MS/s with 14 cycles of a 4.8GHz operation clock, CK_{DSMC}, which is buffered and enveloped from an externally applied CK_M. The 2-channel DSM operates in a time-interleaved manner to achieve a 600MS/s sampling rate. The linearity requirement of the time-interleaved 2nd stage is relaxed because the 2nd stage samples timing skew-less residue signal. As a result, the 2nd stage requires only offset calibration between interleaved ADCs.

The operating principle of the proposed architecture is described by the equation in Fig. 10.5.2. Quantization noise (Q-noise) consists of a gain mismatch term and a noise-shaping term. The noise-shaping term can be minimized under sufficiently high loop gain, H(s). Also, the gain mismatch term can be minimized by eliminating gain mismatch between the residue signal path and the feedback path of the 2nd stage. Fig. 10.5.2 shows the circuit implementation of residue buffer and the 2nd stage single-channel DSM. The 2nd-order noise-shaping DSM with 14 cycles of conversion clock is a proper choice in the 2nd stage to increase H(s), and it results in 6.5b resolution including 1.5b redundancy. Although the Gm-C integrator consumes less power, an active-RC integrator is widely used in CT DSM because the Gm-C integrator has a narrow linear input range. However, the coarse-converted residue signal enables the DSM to apply a power-efficient Gm-C integrator. A simple open-loop source follower is applied as a residue buffer. Since the source follower is robust to PVT variations and easy to implement, the design complexity can be relaxed. A CDAC and a residue buffer are replicated in the 2nd stage in Fig. 10.5.2 to eliminate the inter-stage gain error. For better gain matching, a comparator of the 1st stage is also replicated in the feedback path as a dummy. Well-matched replica circuits with shared-bias and adjacent layout arrangements remove inter-stage gain error from chip-to chip gain variation of the residue signal path. Moreover, the area-consuming 7b CDAC can be used in calibrating inter-stage offset error by adding offset in the feedback signal. Therefore, the comparator without offset calibration enables the SAR to enhance conversion speed.

Figure 10.5.3 shows the proposed adaptive speed-controlled (ASC) clock generation scheme. The ASC clock generator takes advantages from both the synchronous and asynchronous schemes by generating a 50% duty and duration-fixed internal operation clock (CK_{ASC}) cycles, regardless of the PVT variations, by background frequency calibration. The ASC clock generator dedicates at least a 1/2 clock period of CK_{ASC} for the internal DAC settling time. In fact, the DAC settling initiates as soon as the comparator makes the decision (just like the asynchronous operation) that dedicates more than ½ the clock period of CK_{ASC} for additional performance enhancement. At the end of the data conversion, a shift-register in the SAR logic counts the number of CK_{ASC} cycles and generates the end-of-conversion (EOC) signals. As shown in Fig. 10.5.3, the ASC clock generator calibrates the frequency of CK_{ASC} in the background by reducing or increasing the delay of the clock generator. The ASC clock generator consists of an inverter and capacitor-based ring oscillator and the delay line controller. Since the input sampling is performed by low-jitter CK_{SAMP}, the delay-controlled ring oscillator only cares to generate seven CK_{ASC} cycles with a relaxed jitter requirement for SAR operation. In addition, the duration-fixed synchronous-like asynchronous ASC clock helps the ADC easily detect the comparator meta-stability like a synchronous SAR.

The prototype ADC, implemented in 7nm FinFET CMOS technology, occupies 0.037mm^2 as shown in Fig. 10.5.7. The foreground on-chip offset calibration is performed to match the signal range between the 1st and the 2nd stage. Moreover, off-chip offset calibration is done between the 2-channel DSMs. No inter-stage gain calibration is applied with the help of the feedback path replica circuits. At 600MS/s under a 0.8V supply, the total power consumption is 13mW. The residue source follower consumes only 0.7mW. Measured performance is shown in Fig. 10.5.4. The ADC operates over a wide range (300 ~ 600MS/s) of sampling rates, although it adopts a CT DSM. The ADC achieves a peak SNDR of 58.2dB and a peak SFDR of 79.4dB at 300MS/s with a 5MHz input. At 600MS/s, the peak SNDR is 56.6dB, and it degrades less than 1.3dB up to 500MHz input. The measured SNDR at 600MS/s stays above 54dB across a 0.77~0.9V power supply range and -40~80°C temperature variations. Figure 10.5.5 illustrates the measured spectra and DNL/INL. The achieved DNL and INL are 0.71 and 1.45LSB, respectively. Figure 10.5.6 compares the ADC performance with the state-of-the-art ADCs. The proposed gain calibration-free ADC shows outstanding performance beyond the Nyquist frequency with only 1Vpp input swing.

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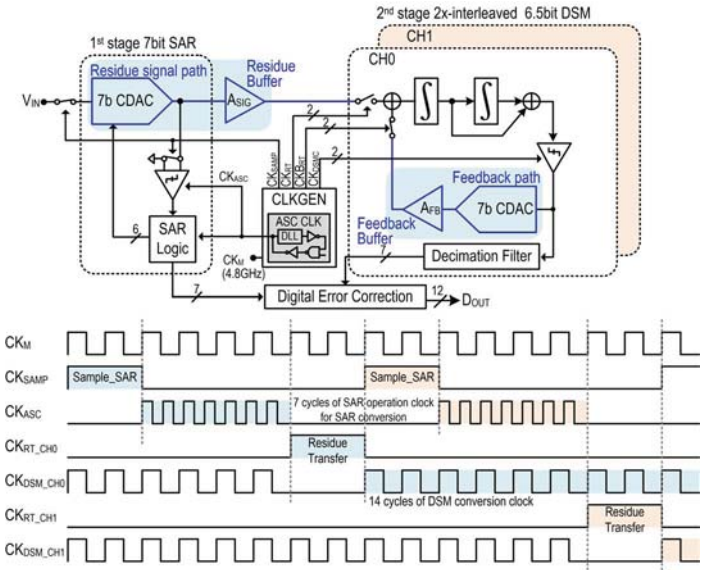


Figure 10.5.1: Block and timing diagram of the prototype 12b pipelined SAR and 2×-interleaved CT incremental DSM.

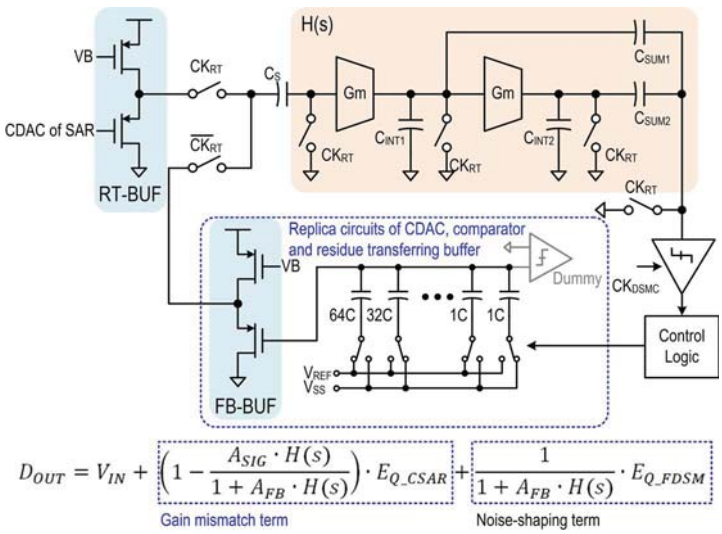


Figure 10.5.2: The circuit implementation of the 2nd stage and the operation principle of gain-error-calibration-free residue transfer scheme.

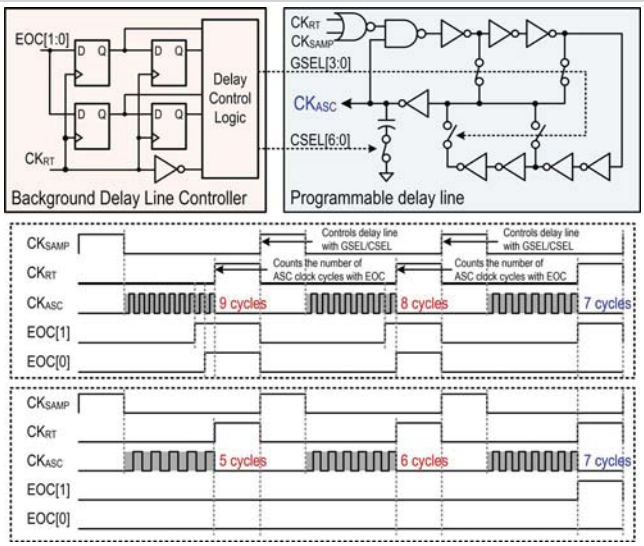


Figure 10.5.3: The circuit implementation of the adaptive speed-controlled (ASC) clock generator in the 1st stage (Top) and timing diagram of background calibration by increasing (middle) or reducing (bottom) delay.

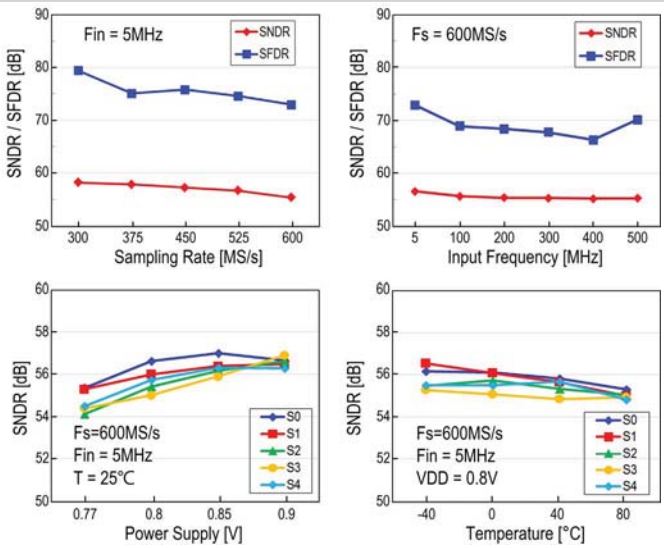


Figure 10.5.4: Dynamic performance and measured SNDR across power supply (0.77V ~ 0.9V) and temperature (-40 to 80°C) variations of 5 randomly selected chips at 600MS/s.

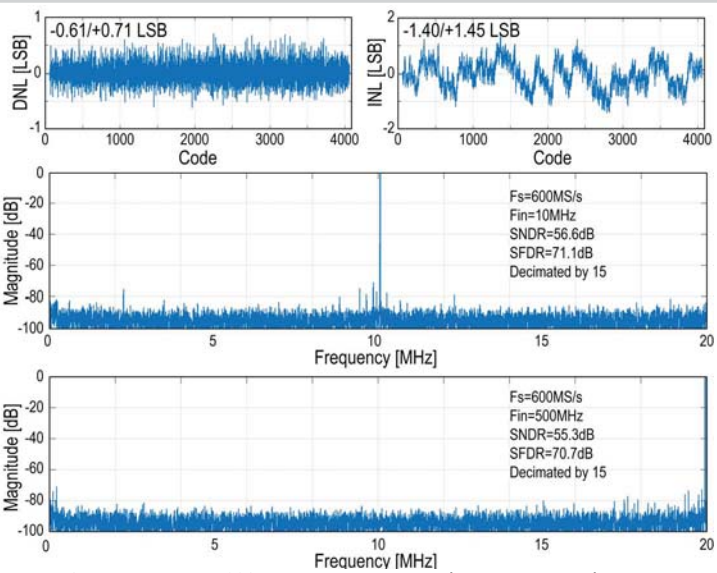


Figure 10.5.5: Measured 16384-point output spectra (decimated by 15) and DNL/INL at 600MS/s.

	This work	[1] ISSCC2016	[2] ISSCC2019	[4] VLSI2017	[5] VLSI2017
Architecture	**Pipe-SAR &TI DSM**	TI subranging SAR&DSM	Pipeline	Pipeline	Pipe-SAR
Technology	**7nm**	28nm	16nm	28nm	28nm
Resolution [bits]	**12**	12	11	12	10
Sample Rate [MS/s]	**600**	600	600	600	500
Supply Voltage [V]	**0.8**	1.2/1.5	0.85	0.9	1
Input swing [$V_{pp,diff}$]	**1**	1.5	1.5	1.6	1.6
Input Bandwidth [MHz]	**500**	265	600		250
SFDR @ Nyq. [dB]	**67.8**	-	78.3	69.2	69.2
SNDR @ Nyq. [dB]	**55.3**	58	60.2	56.3	56.6
Power [mW]	**13**	26.5	6	14.2	6
FoM_{Walden} @ Nyq. [fJ/c-s]	**45.6**	68	12	44.3	21.7
$FoM_{Schreier}$ @ Nyq. [dB]	**158.9**	158	167.2	159.5	162.8
Area [mm²]	**0.037**	0.076	0.037	0.62	0.015
Gain Calibration	**X**	O	O	O	O

Figure 10.5.6: Comparison with state-of-the-art.

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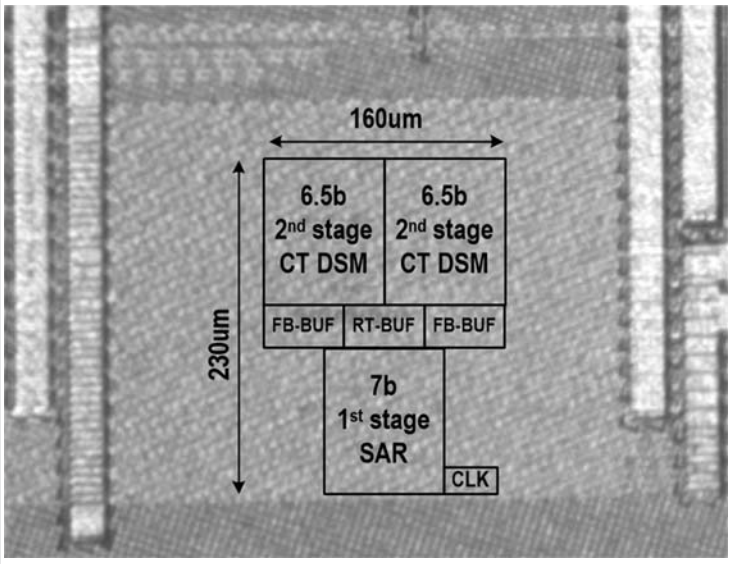


Figure 10.5.7: Die photo.

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A Hybrid Design Automation Tool for SAR ADCs in IoT

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***Abstract*— In this paper, a hybrid design automation tool for asynchronous successive approximation register analog-to-digital converters (SAR ADCs) in Internet-of-Things applications is presented. The circuit design-driven tool uses a top-down design approach and generates circuits from specification to layout automatically. A hybrid approach is introduced for different circuits of a SAR ADC: fully synthesized control logic; a script-based flow combining equations, library, and template-based design for the digital-to-analog converter; a lookup table approach combined with selective simulation-based fine tuning and template-based layout generation for the sample and hold; library-based comparator design and script-based layout generation. By balancing the automation and manual effort, the circuit design time is reduced from days down to minutes while still being able to maintain ADC performance. The proposed flow generated two ADC prototypes in 40-nm CMOS, an 8-bit 32 MS/s and a 12-bit 1 MS/s SAR ADC, and enabled excellent power efficiency. The two ADCs consume 187 and 16.7 μW at 1-V supply voltage, achieving 30.7 and 18.1 fJ/conversion-step, respectively.**

***Index Terms*— Design automation, hybrid approach, low power, successive approximation register analog-to-digital converter (SAR ADC).**

I. Introduction

TODAY, many system-on-chips (SoCs) in Internet-of-Things (IoT) applications consist of both digital and analog circuitry for a high integration level and low cost. The highly integrated chips usually consist of millions or billions of devices, and this results in a huge circuit design effort. The digital circuit design has already been automated by many mature commercialized tools and flows, from behavior description to layout generation. The analog circuits are often manually designed for both schematic and layout. As a result, although the analog circuit scale in an SoC is usually small compared to that of the digital part, the design time for analog circuits is still long. While more and more functions are moved from analog domain to digital domain, analog-to-digital converters (ADCs) as an interface between analog domain and digital domain are still required. Many wireless sensor nodes (WSNs) in IoT/Internet-of-Everything applications (e.g., Bluetooth low energy and IEEE 802.15.4) require low-power ADCs with varying resolution (8∼12 bit) and speed (∼MS/s). To achieve an optimum performance, each ADC has to be customized, increasing the design cost and time. To overcome this issue, circuit design automation is desired for ADCs to minimize the design cost while still maintaining the performance.

TABLE I
SUMMARY OF THE AUTOMATION METHODS FOR FRONT-END SYNTHESIS

	*Library based	Knowledge based	Equation based	Simulation based	Artificial intelligence algorithms
Pros	Immediately ready	Fast	Fast	Accurate	General
Cons	Large preparation effort	Lack of generality	Lack of accuracy	Long simulation time	Need many training examples

*A simulation-based method is used during library construction

The design automation for analog and mixed-signal circuits includes roughly two steps: front-end synthesis and back-end generation. Front-end synthesis can include architecture selection, translation of functional specification to subcircuit specification, and device sizing, and back-end generation can include layout generation of sub-blocks as well as floor planning and routing.

The approaches for the front-end synthesis can be roughly divided into five categories as shown in Table I: library-based approach, knowledge-based approach, equation-based approach, simulation-based approach, and artificial-intelligence-based approach. A library-based approach requires a large effort to create a new library for each block in each technology. This may be acceptable for a single simple cell but not for a complicated system, which includes many blocks. The knowledge-based approach encodes specific heuristic design knowledge from experts into a design plan that is used during the synthesis of the analog circuit [1]. Specification inputs will be translated to the topology selection and the unique solution of the circuit sizing following the

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design plan. However, a content-independent design plan is difficult to make, which limits the use of this approach. The equation-based approach uses a simplified analytic equation to formulate the performance of the circuit [2], [3]. Constrained optimization algorithms instead of a specific design plan are performed using these equations for the optimization of the circuit. Although this approach is more general, the accuracy of the result is a big problem, especially in an advanced process because the design equation has to be derived and simplified so that the optimization algorithm can be executed. To obtain higher accuracy, a full-SPICE simulation-based approach is introduced in the optimization loop [4]. The main problem with this approach is the long run time, especially if the initial search space is not well defined. In [5], an artificial-intelligence-based approach uses the idea of evolution to automate the synthesis of analog circuits. However, large amounts of well-designed examples are required for training the neural network.

For the back-end automation, four methods have been used: library-based method, script-based method, template-based method, and standard digital flow. In the library-based method, the layouts of the required components are manually designed beforehand and put together in a library, which can be directly reused later on. The layout can be precisely optimized but it takes a long design time. The script-based method [6] describes the transistors' location and interconnection in pure ASCII instead of manual place and route. The template-based method uses a template for a design and can maintain the layout shape irrespective of different sizes [7]. The standard digital flow is very mature for digital circuits' place and route but is used less often for a customized analog layout [8].

Prior-art design automation is focusing on either behavior-level, front-end synthesis or back-end synthesis and elaborates on relatively simple analog cells [1]–[5], [7], [9]–[12]. In the context of successive approximation register (SAR) ADCs, Brenna *et al.* [11] proposed a MATLAB tool that allows statistical simulation of capacitor mismatches and parasitics in the digital-to-analog converter (DAC), and a behavioral SAR ADC model is proposed in [12] which allows fast simulation. However, while these tools support with the sizing of the components and by reducing simulation time, they do not output a finalized schematic or layout of an entire SAR ADC, which is the purpose of this paper. The design automation for complete ADCs is more cumbersome due to its complicated architecture, larger scale circuitry, and more sophisticated signal processing between the analog and digital domains. Prior art tries to synthesize a whole ADC using a single synthesis flow, chosen for a specific architecture. For a stochastic flash ADC [13], the traditional analog circuitry is replaced by pure digital gates so that the whole ADC can be integrated into the digital synthesis flow. However, the resolution and performance are limited due to the stochastic flash architecture. For a $\Sigma\Delta$ ADC [8], a library-based approach and standard place and route (P&R) for layout generation have been used. Although all the analog components are still manually designed, the power efficiency is yet behind the state of the art. Wulff and Ytterdal [6] introduce a compile method for the SAR ADC but is limited to layout generation only.

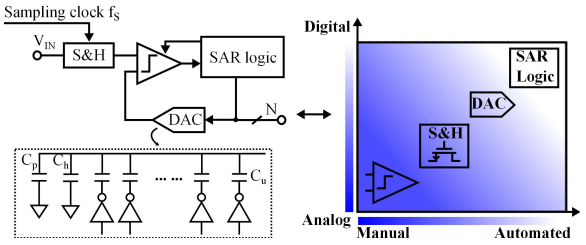


Fig. 1. Block diagram of a SAR ADC and illustration of its sub-blocks.

Huang *et al.* [14] introduce a systematic design method on a schematic level for a SAR ADC but large differences between simulation and measurement results are observed ($>10\times$).

Alternatively, instead of synthesizing all the circuits using an identical approach, a design tool based on a hybrid automation approach is introduced in this paper, allowing different automation approaches for the individual subcircuits [15]. Both the circuit and layout levels are included. With the assistance of this tool, the designer can better balance automation and customization and reduce the design time due to repetitive labor work while maintaining the ADC performance. This approach is elaborated for a SAR ADC, as the SAR ADC is popular for WSNs thanks to its excellent power efficiency [16]. The design automation approach is verified by generating two prototype ADCs. The measurements show a good agreement with the simulation results.

The remainder of this paper is organized as follows. Section II introduces the proposed hybrid automation approach applied to the SAR ADC. The specification translation of the automation approach is shown in Section III. The implementation details of the automation tool and the circuits are presented in Section IV. The measurement results are shown in Section V, and the conclusion is drawn in Section VI.

II. Hybrid Design Automation Applied to a SAR ADC

A conventional N-bit SAR ADC usually includes four main blocks as shown in Fig. 1: sample and hold (S&H), comparator, control logic, and DAC. In each SAR conversion, the S&H samples the analog input voltage and stores it on the capacitor arrays of the DAC. The DAC output will approximate the sampled input voltage in N comparisons through a successive-approximation algorithm. This is performed by the feedback loop that consists of the comparator and the SAR logic. In a synchronous SAR ADC, a high-frequency clock ($N \times f_s$) is needed but this can cause a significant overhead in power consumption. Therefore, the asynchronous operation is used in this paper [16]. Thanks to the dynamic operation of all the ADC building blocks, the power consumption is inherently scalable with the sampling frequency. Furthermore, the topologies of the S&H, comparator, and DAC are fixed to a bootstrapped sampling switch [17], a dynamic comparator [18], and a top-sampling fully binary-weighted C-DAC with a monotonic switching scheme [19], [20], as it has been shown that such topologies can already cover a large performance range in terms of speed and resolution.

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TABLE II
DESIGN TIME ANALYSIS OF A SAR ADC

Manual Design time	Front-end (inc. optimization)	Back-end (inc. optimization)	Simulation
SAR logic	days	days	minutes
DAC	hours	days	minutes
S&H	hours	hours	minutes
Comparator	days	hours	hours
ADC top	hours	days	hours

TABLE III
SUMMARY OF THE PROPOSED HYBRID AUTOMATION METHODS

	Front-end generation				Back-end generation		
	Standard digital flow	Equation based	Simulation/ knowledge based	Library based	Standard digital flow	Template based	Library based
SAR logic	√				√		
DAC		√		√		√	
S&H			√			√	
Comparator				√			√
ADC top		√	√			√	

Note that the monotonic switching scheme is not the most energy-efficient one, but thanks to the small capacitors used in the proposed DAC layout, the power consumption of the DAC is nonetheless relatively insignificant compared to that of the ADC (<10%). While the tool currently only implements the above-mentioned topologies, it could be expanded with other topologies in a similar way.

As mentioned, to ensure the performance of the ADC, a manual design is usually applied to all the four blocks, which is very time consuming (Table II). In particular, for the SAR logic, both the front-end and back-end design time are in the order of days, thus both need to reduce. For the DAC design, the design time for the back end is much more than for the front end, and therefore, reducing the back-end design time improves the total design time more. For the S&H, the front-end design and the back-end design consume a similar design time. For the comparator, most of the design time is spent in finding the proper sizes for the devices and reducing the back-end design time helps less. For the ADC top level, most of the design time is required for the back end, as a large number of analog and digital connections have to be made in the layout.

To define the best design approach, the four basic blocks of a SAR ADC are analyzed first from two aspects: analog/digital-oriented and manual/automated aspects (Fig. 1). Analog-oriented aspect implies that the circuit is physically constrained (by noise and mismatch) and expertise is needed to properly optimize the circuit, which may require manual effort. Digital-oriented aspect implies that the circuit is less physically constrained and can be abstracted and synthesized more easily without losing performance. In this paper, different approaches are adopted for the front-end and back-end designs, comprising the four basic blocks, and the top-level design of a SAR ADC as follows (Table III).

1) *SAR Logic:* Both the front end and the back end of the control logic can be synthesized using a standard synthesis flow. This has the advantage of flexibility in reconfiguring the number of bits, the operation speed, and the timing of the control signals. The design time is much shorter, up to a few hours, compared to that of the manual design, and when redesigned, only a few key parameters need to be tuned. The redesign time, including front end, back end, and simulation, can be reduced to less than 1 h. In addition, it features the possibility to integrate digital calibration logic in the future as will be discussed later. While the power efficiency of manually designed dynamic logic [16] is still better than synthesized logic, a standard synthesis flow is adopted here thanks to its advantages in design time and reusability.
2) *DAC:* The front-end design of the DAC capacitor array can be automated as follows. The DAC capacitance value is computed automatically according to the noise and matching requirements through an equation-based method as shown later. The DAC driver is simply an inverter, and therefore uses a library-based approach. Compared to the front end, the DAC back-end design usually takes much longer due to the strict requirements, e.g., symmetry and optimization of parasitics at attofarad (aF) level. Fortunately, thanks to the regularity of the DAC layout, its design can be automated through template-based programming as shown later, which reduces the majority of the DAC design time.
3) *S&H:* The S&H circuit, together with the ADC input capacitances, influences the ADC bandwidth. This effect can be easily checked by simulating the spurious-free dynamic range (SFDR). Considering the short simulation time, a simulation-based approach is introduced for the front-end design. In addition, to further optimize the S&H circuit while minimizing the manual effort, a knowledge-based approach is used by tuning only the critical devices as shown later. For the back end, a template-based layout generation through programming is used to reduce the relayout time during optimization or redesign.
4) *Comparator:* The ADC performance is very sensitive to comparator nonidealities (e.g., offset error, noise, and layout asymmetry), and therefore, the comparator design needs more attention. In the elaborated design, the comparator design is library based for both the front end and the back end.
5) *ADC Top Level:* For the front end, the ADC top-level design uses equations to translate the top-level specifications into subcircuit requirements at the beginning of the design, and a simulation-based approach to assure that the performance is met. For the back end, thanks to the regular layout geometry of a SAR ADC, a template-based layout generation is introduced, which reduces the layout time from days to minutes.

The discussed design steps are embedded in a design flow to automate a SAR ADC design in six steps (Fig. 2). The flow first takes the ADC specifications as input, then translates the input into parameters for the sub-blocks. Afterward, each sub-block is generated automatically using their corresponding method. Finally, the layout of all the sub-blocks will be assembled automatically, and the performance of the generated ADC is verified through simulations using an extracted view.

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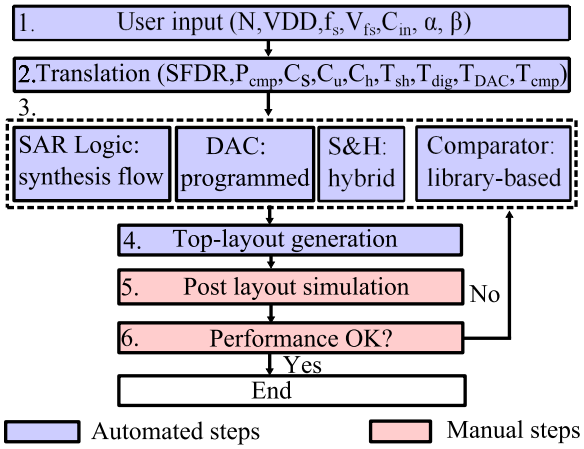


Fig. 2. Design flow of the proposed method.

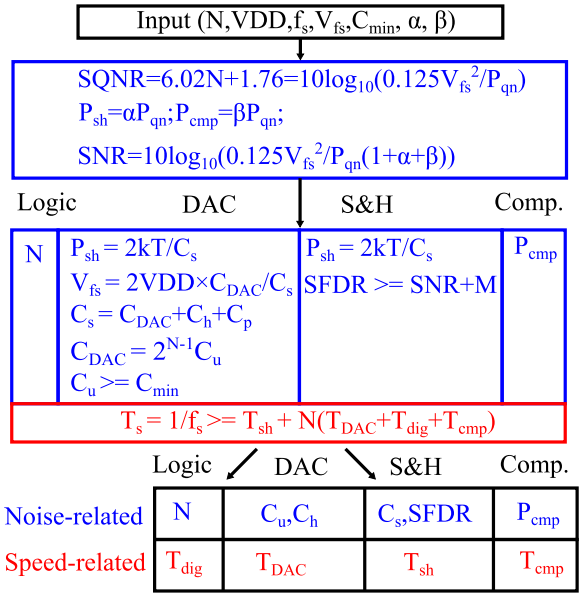


Fig. 3. Translating from ADC specifications to block parameters.

The design process will stop when the simulated ADC performance reaches the design target. Otherwise, further optimization is still possible by fine tuning the parameters of sub-blocks. In this paper, the steps 1–4 in Fig. 2 are automated while the simulation and performance checks are manual.

III. Specification Translation

The translation of the overall ADC specifications into the sub-block parameters is automated through an equation-based method, in step 2 of Fig. 2. The equations are divided into two categories: noise related and speed related, as shown in Fig. 3. The input resolution N determines the signal-to-quantization-noise ratio, which leads to the ADC quantization noise floor P_{qn} (Fig. 3), based on a provided input signal range V_{fs}. The noise requirements of the sub-blocks (the sampling noise P_{sh} and the comparator noise P_{cmp}) are computed based on this quantization noise, as shown in Fig. 3. As a result, the signal-to-noise ratio (SNR) degradation due to the extra circuit noise is $10\log_{10}(1+\alpha+\beta)$ dB. For instance, with $\alpha=\beta=(1/2)$, a 3-dB SNR degradation [equivalently 0.5-bit effective number of bits (ENOB) loss] will occur due to the noise from sampling and comparator but other choices are possible dependent on the user preference.

Then, the derived noise requirements are used to compute the circuit parameters for each sub-block. In particular, the total input capacitance C_s of the ADC can be computed from the sampling noise P_{sh} requirement. At the same time, according to the equations to calculate the input signal range of the ADC ($2\text{VDD}\cdot C_{\text{DAC}}/C_s$), the DAC capacitance C_{DAC} and the attenuation capacitance C_h are computed. From this, assuming a monotonic DAC switching scheme, the minimum unit capacitance C_u of the DAC can be computed based on the noise constraint. The mismatch constraint of the capacitors is not considered as it is expected to be tackled by mismatch calibration as in [21] as discussed later. For the S&H, the required SFDR should be M dB higher than the targeted SNR, where M is typically set to 10 dB but can be overruled by the user. For the comparator, the calculated noise requirement P_{cmp} will decide which comparator is selected from the library.

Meanwhile, the operation speed of the comparator, DAC, and logic is related to the overall ADC operation speed f_s (Fig. 3). The total time of the S&H, comparator, DAC, and logic should be no more than the period of the sampling clock T_s. By default, half of the period is used for the S&H to track the analog input signal, and the other half is used for the conversion, which consists of $N\times$ operation of the comparator, DAC, and SAR logic. In this paper, the logic and DAC are first designed and simulated, and the remaining time is for the comparator.

IV. Block Implementations and Top-Level Layout

As discussed, the requirements of the sub-blocks are derived from the equations in Fig. 3 and will be used for the design of each sub-block, respectively, and for the top-level layout generation.

A. Fully Synthesized Control Logic

The digital control logic of the asynchronous SAR ADC performs the binary search algorithm and also generates an internal clock for the asynchronous ADC. In particular, a monotonic switching scheme [19] is implemented to improve the power efficiency of the ADC. Thanks to the well-developed digital synthesis flow, most of the design effort is now taken over by the digital design tools. For example, when redesigning for a different speed, the designer only has to change the timing constraint parameter of the synthesis flow. The delay of the digital control logic will be optimized automatically by the synthesis flow, to meet the required operation speed. This is more convenient and faster compared to the iterative manual optimization method.

B. Hybrid DAC Design

The charge-redistribution DAC is often chosen for SAR ADCs thanks to its good power efficiency. It consists of DAC drivers and DAC capacitors, which are designed automatically through a script-based automation flow combining

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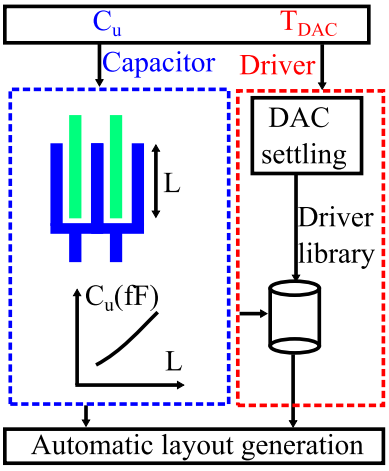


Fig. 4. DAC design automation flow.

a mix of equation-based, library-based, and template-based designs (Fig. 4). An equation-based method is first adopted to size the unit capacitor C_u based on the noise constraint as shown in Fig. 3. On the other hand, when the ADC resolution goes beyond 10 bits, C_u tends to be limited by the capacitor mismatch and thus has to be sized relatively large, degrading the power efficiency. Alternatively, to save power, C_u can be sized to just meet the kT/C noise, and calibration can be employed to compensate the capacitor mismatch errors [21]. Calibration is not yet present in the current tool but can be easily integrated into the digital circuit synthesis. In case mismatch is dominant while calibration is not used, the sizing process of C_u becomes more difficult. In that case, an assistance tool such as that in [11] could be inserted to determine appropriate sizing.

It can be calculated that for medium- and low-resolution ADCs with a rail-to-rail input range, the noise-constrained C_u is unpractically small. For example, C_u for an 8-bit ADC can be as small as 40 aF based on noise requirements, which will be much smaller than the parasitics. This is undesired since it will reduce the input signal range and make the input signal range more vulnerable to parasitics. To avoid this, the designer can optionally set a lower boundary ($C_{\min}$) for C_u, e.g., 600 aF. After selecting C_u, the DAC driver needs to guarantee that the DAC output can settle in time T_{DAC}. The selection is made from a library of drivers with various strengths, matching the selected C_u and the required operation speed f_s. The DAC driver inputs are the load of the digital control logic. When using larger size DAC drivers, the loading of the corresponding digital control bit should be adapted for the synthesis.

The DAC layout is very structured with a repetitive geometry, and therefore can be automated efficiently to save layout design time. In this paper, a programmable cell (Pcell) is used for the DAC design automation and its layout generation. By programming the key parameters of the DAC, this tool enables the users to provide the specification, automate the device sizing, and generate the layout accordingly. C_u is a customized finger capacitor as shown in Fig. 4. This allows the employment of a small unit capacitance (sub-fF) and to reduce the DAC area and power consumption at the same time. In addition, the unit capacitance can be approximated as

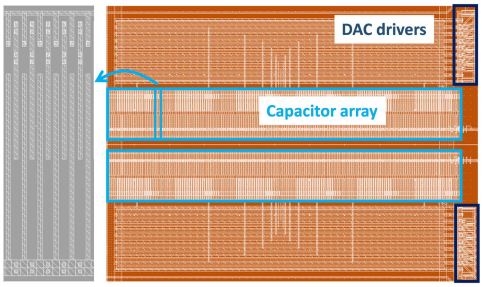


Fig. 5. Generated DAC layout example for an 8-bit ADC.

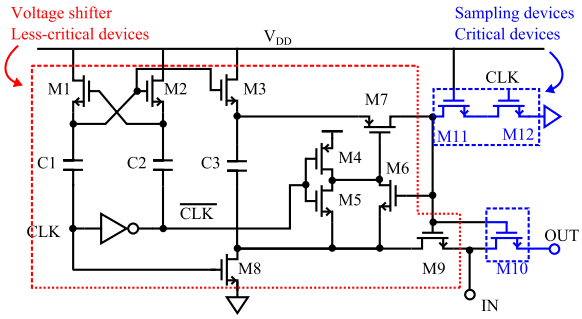


Fig. 6. S&H circuit.

linearly proportional to the overlapping length L between the top plate and bottom plate of the capacitor. Therefore, the unit capacitance can be precisely controlled by simply programming the unit capacitor length L dependent on the required value of C_u (Fig. 4). As shown in Fig. 5, the unit capacitor value C_u changes the length L, which is in the vertical direction, while the resolution N changes the total number of capacitors placed beside each other in the horizontal direction. As such, L and N can be independently modified without significantly changing the overall DAC floorplan. The DAC layout is automatically generated from the Pcell, which is described in SKILL language.

C. Hybrid S&H Design

For a given DAC capacitance, the ON-resistance R_{on} of the sampling switch determines the bandwidth of the S&H. In modern CMOS technologies, the supply voltage drops below 1 V, making it difficult to reach the desired bandwidth (∼MHz) for the ADCs in WSNs if directly controlled switches are used. Therefore, a bootstrapped circuit [17] is used in this paper, as shown in Fig. 6. As mentioned, a simulation-based approach is introduced for the S&H circuit thanks to the short simulation time for SFDR. However, noting that there are multiple devices in the S&H circuit (Fig. 6), the optimization time for the sizing would be too long to be acceptable.

To reduce the optimization time while still maintaining the good performance for the S&H circuit, a knowledge-based approach is added as follows. The circuit can be roughly divided into two parts: the devices for sampling (performance critical) and for the voltage shifter (performance less critical). A voltage multiplier (M1–M9) performs the function of boosting the gate voltage of the sampling switch M12. These devices are relatively less critical as long as the function of voltage boosting can be achieved. Therefore, the size of the

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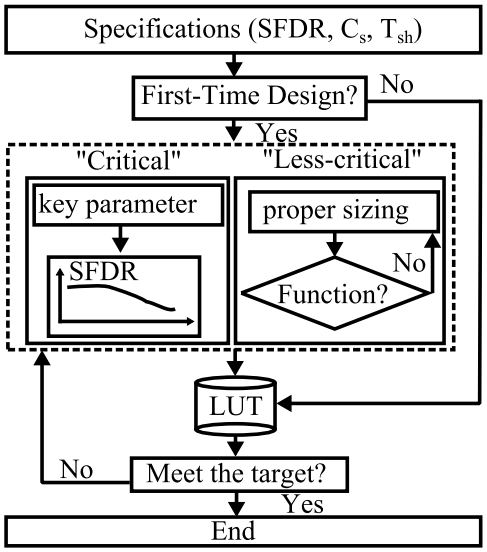


Fig. 7. S&H design automation method.

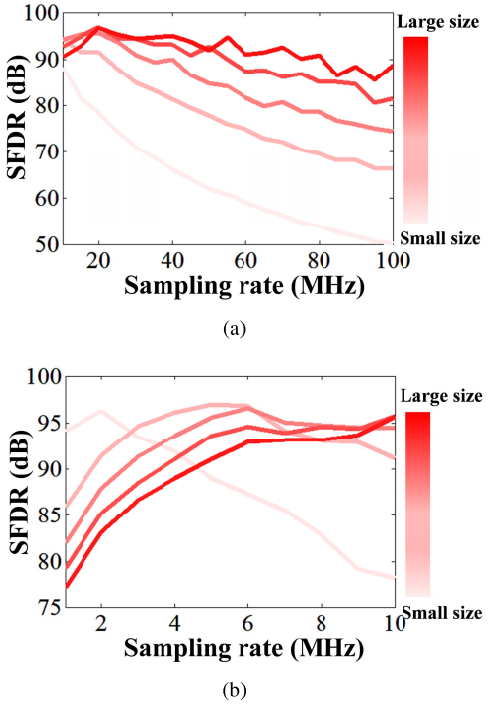


Fig. 8. Simulated SFDR of S&H with various device sizes for relatively (a) high and (b) low sampling frequency, respectively.

noncritical devices is fixed for all designs in this automation tool. M10–M12 are critical for the S&H performance, and therefore need to be carefully treated.

Therefore, a hybrid design approach, consisting of simulation- and knowledge-based approaches, is proposed for the front-end design of the S&H circuit (Fig. 7). One S&H is implemented at first (including layout) and reused as a template when redesigned for a different specification. When redesigned, the less critical devices are fixed while the critical devices are tuned for optimization. As a result, the total number of iterations needed to reach the optimum point is significantly reduced, enabling the employment of a simulation-based lookup table (LUT) approach for the S&H circuit. The LUT can be constructed based on the simulation results for various device sizes as shown in Fig. 8. In this way, in a next design, an approximately optimized device size can be directly selected from the LUT. It is crucial to choose the device dimension properly from the LUT according to the required SFDR and the ADC sampling rate. Larger devices do not always result in better SFDR but can degrade the SFDR for low-speed ADCs due to device leakage. Moreover, further optimization is still possible by further fine tuning the critical devices. To save the optimization time, one parameter, which is the width of the devices, is used for sizing the three critical devices since they are usually tuned in the same direction. Furthermore, it is worthwhile to mention that the ADC input capacitance C_s influences the SFDR of the S&H circuit. The bandwidth of the S&H is determined by an RC constant, where R is determined by the size of the sampling transistor M10 and C is C_s. This impact of C_s is accommodated indirectly by scaling R in an opposite direction to maintain a constant RC value. For the back-end design, considering that most of the devices are fixed and only three transistors are programed, a template-based layout generation through SKILL is introduced for the S&H, similarly as for the DAC, which further reduces the layout redesign time.

TABLE IV
ILLUSTRATION OF LIBRARY-BASED METHOD FOR COMPARATOR

P_{cmp} / T_{cmp}	<=P_0	(P_0, P_1]	(P_1, P_2]	(P_2, P_3]	... ...
(0, T_0]	CMP1.1	CMP1.2	CMP1.3	CMP1.4	... ...
(T_0, T_1]	CMP2.1	CMP2.2	CMP2.3	CMP2.4	... ...
(T_1, T_2]	CMP3.1	CMP3.2	CMP3.3	CMP3.4	... .
(T_2, T_3]	CMP4.1	CMP4.2	CMP4.3	CMP4.4	... .
(T_3, T_4]	CMP5.1	CMP5.2	CMP5.3	CMP5.4	... .

D. Library-Based Comparator Design

In WSNs, the comparator performance is crucial for SAR ADCs, as it can either dominate the overall ADC power consumption or degrade the ADC ENOB. Therefore, even for an experienced ADC designer, it usually takes days to optimize the speed, noise, and power efficiency of the comparator. However, it is not straightforward to correlate the comparator performance with the device sizing due to its relatively complex circuitry, nonlinear operation, and sensitivity to layout imperfections. Therefore, a library-based approach is introduced, in which comparators with different performance combinations can be selected by the computer-aided design tool when users input the specifications. The comparators in the library will have a 2-D index: input-referred noise and comparison time. The probability of metastability can be controlled by assigning a more conservative comparison time to build in some extra buffer time. According to these two requirements, a proper device in Table IV will be selected. In this paper, a dynamic comparator is used thanks to its efficiency and frequency scalability [16], [18].

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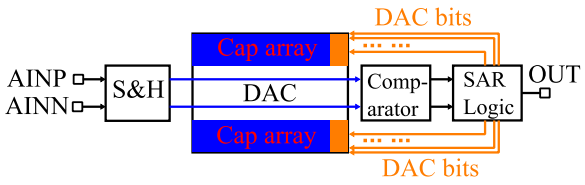


Fig. 9. Template of the ADC top layout.

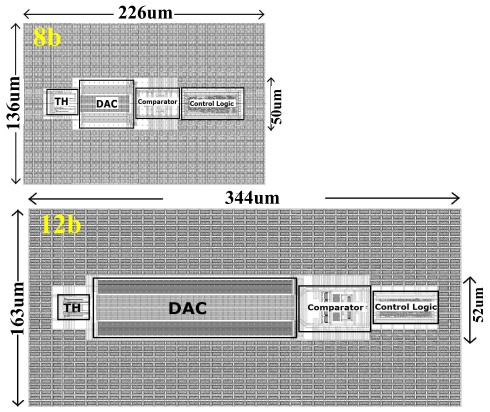


Fig. 10. Generated ADC layouts using the proposed flow.

E. Top-Layout Generation

The traditional P&R in the standard digital flow minimizes the manual layout effort for digital circuits by automatic P&R [8]. However, this can cause problems for analog circuits, which are sensitive to layout nonidealities (e.g., parasitics, asymmetries, and IR drops), and thus may degrade the ADC performance. Alternatively, a template-based layout generation is introduced [22]. This layout generation is used for well-understood designs with sufficient regularity or geometric templates. It only adapts the size for devices that are critical to optimization while the remaining devices are relatively fixed. The interconnections between the devices are adapted automatically.

The aforementioned tool for the design of the sub-blocks is developed using SKILL language and the Pcell tool in Cadence. The layout generations of the ADC top are automated using a geometrical template-based method. This method allocates each sub-block according to a template and connects the pins between each block through scripting (Fig. 9). The geometrical coordinates of all sub-blocks are parameterized, and the pin locations of each block are automatically adapted accordingly. This greatly saves the design time for layout, and as a result, it takes only a few seconds for the program to integrate into a design rule check and layout versus schematic clean compact layout (Fig. 10). Then, simulations based on an extracted view can be executed to verify the performance of the ADC. Note that it is still possible to fine tune each sub-block individually to optimize the overall ADC performance using the implemented design flow.

V. Verification With Prototypes

To verify the proposed design flow, two prototype chips with different specifications for speed and resolution have been implemented in a 40-nm CMOS: an 8-bit 32-MS/s SAR ADC and a 12-bit 1-MS/s SAR ADC (Fig. 11). The compact layout leads to a small core area of the two ADCs (Fig. 10) of 0.011 and 0.016 mm^2, respectively. Including decoupling capacitors, they occupy 0.031 and 0.056 mm^2.

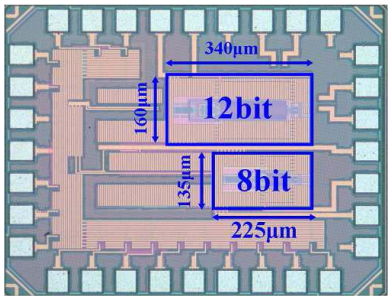


Fig. 11. Die photograph of the prototype ADCs in 40-nm CMOS.

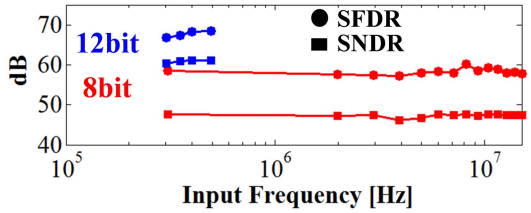


Fig. 12. Dynamic performance of the two ADCs.

The 8-bit ADC consumes 187 μW at 32 MS/s and 1-V supply and achieves 47.4-dB SNDR up to the Nyquist frequency (Fig. 12). This leads to a 30.7-fJ/conversion-step figure of merit (FoM) as defined in 1, which is comparable to the simulated 25.1 fJ/conversion-step, considering that simulation results did not include nonidealities, e.g., noise and mismatch-esMing Ding

$$\text{FoM} = \frac{\text{Power}}{2^{\text{ENOB}}\min(f_s, 2\text{ERBW})} \tag{1}$$

The 12-bit ADC consumes 61.1 μW at 1 MS/s and 1 V, which is close to the simulation results. Considering that the DAC capacitors are sized for kT/C noise instead of mismatch, large integral nonlinearity (INL)/differential nonlinearity (DNL) errors are measured, as shown in Fig. 13, thus limiting the ADC performance. However, this error is assumed to be solved by the DAC mismatch calibration [21]. The measured input-referred noise of the 12-bit ADC is 0.47 LSB$_{\text{rms}}$. This indicates that the noise-limited ADC SNDR is above 11 bit once the DAC mismatch-induced error is corrected.

In addition, both ADCs achieve stable SNDR/SFDR up to the Nyquist frequency (Figs. 12 and 14), indicating a sufficient performance for the S&H circuits. Overall, the measured power consumption and speed match with the simulated numbers. We should note that the simulated numbers are not estimated by the proposed tool but by postlayout simulation of the layout which is generated by the tool. The measured FoM for the 8-bit ADC is close to expectation (as shown later). For the 12-bit ADC, the FoM correspondence is worse compared to the 8-bit ADC due to the mismatch-induced errors. However, the correspondence is better than the prior art.

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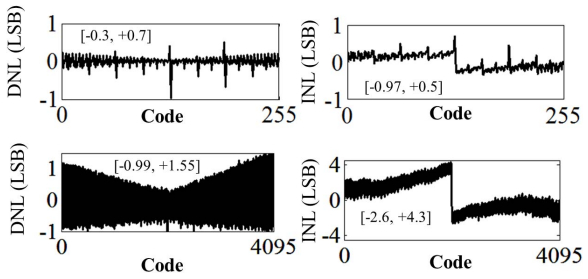


Fig. 13. Measured INL/DNL of the two ADCs.

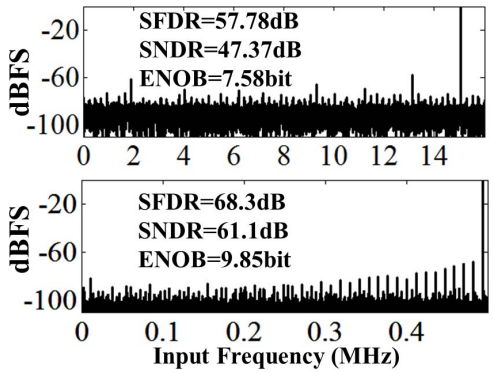


Fig. 14. Measured spectrum of the two ADCs with a near-Nyquist tone.

TABLE V
ADC DESIGN TIME OF THE PROPOSED FLOW

Manual Design time	Front-end (inc. optimization)	Back-end (inc. optimization)
SAR logic	days → minutes	days → minutes
DAC	minutes → seconds	days → seconds
S&H	hours → minutes	hours → seconds
Comparator*	days → seconds	hours →seconds
ADC top	minutes → seconds	days → seconds

*** Design time for the library is days**

The automated design flow successfully reduces the total ADC design time down to minutes level (Table V), after initial construction of the libraries, templates, and code. In particular, both the front-end and back-end design times of the SAR logic are reduced significantly from days to minutes. The time to prepare the digital design flow is at minutes level. For the DAC, the design time is reduced down to minutes level thanks to the automated layout generation. For the S&H circuit, it takes around three iterations for the front-end design through the LUT and fine tuning, and the layout can be generated automatically. The layout template preparation time for the DAC and the S&H is in few hours. The comparator library preparation time is still in days since it needs to be manually designed but this has to be done only once. The ADC top integration is reduced from days to seconds. Overall, thanks to the automated generation of all the subcircuits, the design time needed for one iteration is greatly reduced. In this way, it speeds up the design time considerably, even for a designer with less expertise. At the same time, the performance of

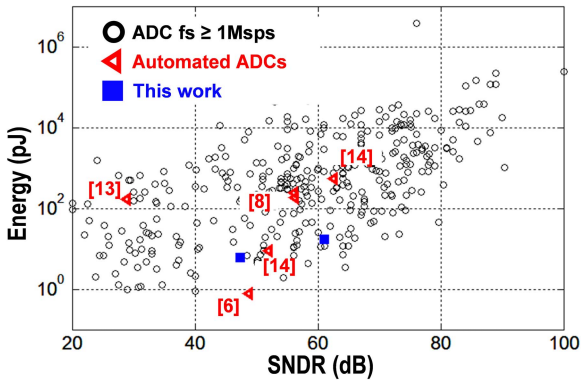


Fig. 15. Benchmark with the state of the art.

TABLE VI
PERFORMANCE SUMMARY OF THE TWO ADCS

	[13]	[14]				[6]	This work			
Experiment	Meas.	Meas.	Sim.	Meas.	Sim.	Meas.	Meas.	Sim.	Meas.	Sim.
Automation level	Schematic Layout	Schematic				Layout	Schematic Layout			
Architecture	Stochastic	SAR				SAR	SAR			
Technology (nm)	90	180		90		28	40			
Resolution (bit)	-	12		10		9	8		12	
Speed (MSps)	210	0.6		50		20	32	32	1	1
Supply voltage (V)	1.2	3.3		1		0.69	1	1	1	1
INL (LSB)	-	2.79	0.1	1.41	0.06	-	0.7	-	4.3	-
DNL (LSB)	-	5.24	0.22	2.81	0.1	-	0.97	-	1.55	-
Nyquist SNDR (dB)	29	62.7	73.9	52.1	61.9	48.84	47.37	49.3	61.1	71.6
Power (uW)	34800	328	53.5	440	155	15.87	187	192	16.7	16.8
FoM (fJ/c.s.)	7943	500	22	26.7	3.1	3.51	30.7	25.1	18.1	5.4
Meas. vs. Sim. ratio	-	22.7		8.6		-	1.2		3.4	

the ADC is maintained (Fig. 15), showing a good balance between the design time and ADC performance compared to the state of the art [23]. Compared to other synthesized ADC approaches (Table VI), this paper is one of the few to automate both schematic and layout design while it achieves good power efficiencies and the best matching between the predicted performance and measured results.

At present, the proposed tool aims for design assistance of SAR ADCs at medium speed and resolution. In the future, capacitor-mismatch calibration as in [21] can be integrated so that higher resolution ADCs can be generated. For high-speed ADCs, probably more attention is required to obtain a high-speed layout, the logic may need to be custom designed, and other circuit topologies might be required. Beyond SAR ADCs, the hybrid automation method could also be applied to other ADC architectures but it is most useful to relatively digital-intensive architectures. The most time-consuming part would be to build the layout templates and the component library where applicable. The more complex analog blocks are required, the less likely it is to benefit from the proposed tool.

VI. CONCLUSION

In this paper, a circuit design-driven tool with a hybrid automation approach for SAR ADCs is proposed. Compared to the prior art, instead of simplifying or modifying the circuit architecture to enable synthesis, our hybrid performance-selective approach allows the employment of appropriate circuits for performance and automation for design

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time reduction. In this way, the design time is reduced without compromising the performance, meanwhile still keeping the possibility to control performance-critical devices through programming, enabling a balance between automation and customization. The implemented prototypes achieve a performance that is sufficiently competitive with full manual designs while strongly reducing the design time and having an accurate control over the performance goal.

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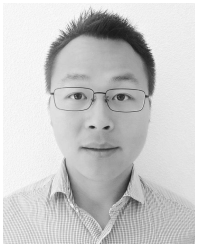
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Understanding the Regenerative Comparator Circuit

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***Abstract*—The regenerative comparator circuit which lies at the heart of A/D conversion, slicer circuits, and memory sensing, is unstable, time-varying, nonlinear, and with multiple equilibria. That does not mean, as this paper shows, that it cannot be understood with simple equivalent circuits that reveal its dynamics completely, and enable it to be designed to specifications on static and dynamic offset and noise. The analysis is applied to the StrongArm latch.**

I. Introduction

Flip-flops used as regenerative amplifiers are found everywhere in electronic circuits. It is well-understood that they have two stable states, and that given sufficient time, the circuit will regenerate an input voltage unbalance to reach one of these states.

The latched comparator must be symmetric by its very nature, since its binary states are symmetrical. Practical unbalances in the circuit arising from transistor and load mismatch lead to uncertainty in the regenerated binary output when a small analog input is applied. This problem of offset is well-known to circuit designers, and since the observed offsets can be much larger than in simple linear amplifiers, a certain mystery attends to "dynamic offsets" that appear only in a latched comparator.

This paper clears much of the mystery by visualizing regeneration. By using one trajectory as a frame of reference, circuit unbalances can in most cases be modelled by a sequence of linear, time-varying equivalent circuits that capture, piecewise, the variation of the circuit with time. Thus, simple expressions are obtained for all offsets, and a straightforward method emerges to null them to a desired accuracy at very small cost in power or chip area.

II. The Static Latch

Any discussion of a regenerative amplifier must start with the CMOS static latch (Fig. 1). This is a classic circuit: simple, uncluttered, and therefore easily understood. The only choice lies in the method whereby a small analog input is coupled into the latch without disturbing regeneration. Here, the two load capacitors C_L are pre-charged to a common voltage, on which is superposed a small differential voltage. That is, the voltage V_{O1} is slightly larger than V_{O2}, while their average voltage, in this example, is chosen slightly lower than V_{DD}. The pre-charged capacitors are switched into the latch (it will be seen that there are *only* two nodes in the circuit), and the circuit regenerates this difference to the voltage rails. The regeneration waveform (Fig. 1) is familiar to almost everyone who designs circuits. The existing literature does not give a satisfactory explanation for the one striking feature of every

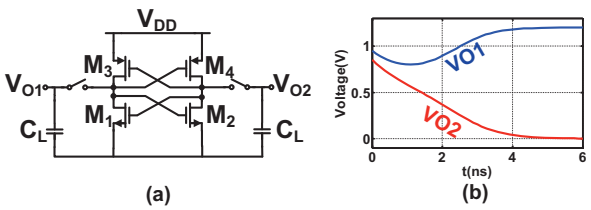


Fig. 1. (a) Schematic of a static latch; (b) Voltage waveforms in a static latch.

regenerative waveform: Why do V_{O1} and V_{O2} fall together, and then at some critical time, they peel away and separate? To find the answer we create a phase portrait of this circuit [1, Ch. 11]. The independent capacitors define two state variables, but using the capacitor voltages as state variables can obscure the essential properties. Instead, we take the average of the capacitor voltages as one state variable, the common-mode voltage V_{OC}, and their difference V_{OD} as the other. These are orthogonal quantities, in the sense that one can change while the other remains constant. The phase plane has been used before to investigate latch dynamics [2], [3], but not defined by the circuit modes. As will soon be clear, this choice makes all the difference.

A. Phase Plane

The phase plane is defined by axes V_{OC} vs. V_{OD} (Fig. 3). It is covered by a *vector field* that is signified at every point (V_{OD}, V_{OC}) on the plane by an arrow with the magnitude and direction of the ratio $(dV_{OC}/dt) \div (dV_{OD}/dt)$, where each time derivative at that point is obtained from the circuit equations. Starting from any initial condition in the plane, there develops by connecting the vector field a unique integral curve which depicts graphically how the state variables will change with time.

The vector field that fills the phase plane can equally well describe a nonlinear differential equation, or a linear one. Sometimes linearity will apply in a limited region of the plane. In that region any integral curve can be decomposed into a superposition of two eigenvectors. In turn, each eigenvector can be associated with the natural response of some linear circuit. For the static latch, as it turns out, linearity is a very good assumption over much of the excursion of V_{O1}, V_{O2}, particularly over the *one* integral curve that will interest us. If $\beta_N = \beta_P$ where $\beta \triangleq \mu C'_{ox} W/L$ for the CMOS inverters, then the overall G_m of the inverter remains almost constant except near $\frac{1}{2}V_{DD}$. At low V_{DD} this appears a small local deviation. The equivalent circuit that is produced by replacing each CMOS inverter with a constant G_m voltage-controlled current source Fig. 2 resembles a differential amplifier, but the cross-coupled controlling voltages identify it as a flip-flop. C_p

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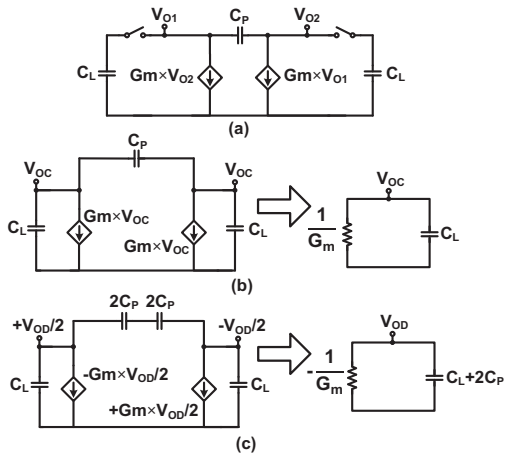


Fig. 2. (a) Equivalent circuit of a static latch; (b) Common-mode equivalent circuit; (c) Differential mode equivalent circuit.

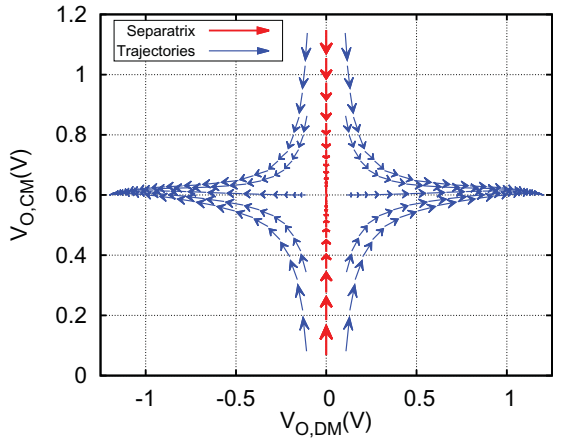


Fig. 3. Phase plane plot of a static latch. V_{DD}=1.1 V.

is a net cross-coupling capacitor between the two nodes arising from FET capacitance. We have deliberately left out the output conductance g_{ds} of the FETs since it makes little difference to the analysis that now follows.

B. Equivalent Circuits for Modes

A decomposition into common-mode and differential mode will also guide the search for meaningful equivalent circuits. Since these modes are independent, any response of this linear circuit can be decomposed into a superposition of the two modes. If the symmetric equivalent circuit (Fig. 2) is operating purely in common mode, then $V_{O1}(t) = V_{O2}(t)$. Since by symmetry no currents will flow through C_p, the circuit can be bisected into two half circuits. It is clear that the natural response of the two halves of the circuit in common mode is stable, with a pole located at $s_C = -G_m/C_L$. On the other hand, if the symmetric circuit is operating purely in differential mode, then $V_{O1}(t) = -V_{O2}(t)$. Symmetry now dictates that the mid-plate potential in the capacitor C_P remains zero. After bisection into half circuits, each half consists of a capacitance $C_L + 2C_P$ across a negative conductance $-G_m$. Now the natural response of the two halves is *unstable*, defined by a pole located at $s_D = +G_m/(C_L + 2C_P)$.

Each mode will be stimulated by its own initial condition. Therefore the time response to any initial condition $V(0) = V_{OC}(0) + V_{OD}(0)$ may be expressed as a superposition of the common and differential modes that are stimulated:

$$V_{O1,2}(t) = V_{OC}(0)\exp\left(-\frac{G_m t}{C_L}\right) \pm \frac{1}{2}V_{OD}(0)\exp\left(+\frac{G_m t}{C_L + 2C_P}\right) \quad (1)$$

This expression lends understanding to the example integral curves plotted on the phase plane. The vertical trajectories start from an initial condition $V_{O1} = V_{O2}$ that does not stimulate the differential mode. The trajectories lead into an equilibrium point at the coordinates $(0, \frac{1}{2}V_{DD})$. On the other hand if the output nodes are initialized so that $V_{OD} \neq 0$ but $V_{OC} = \frac{1}{2}V_{DD}$, that is, the circuit is released from an initial condition that is in purely differential mode, the horizontal trajectory in the phase plane accelerates away from this equilibrium point. If the circuit were truly linear this trajectory head towards $V_{OD} \to \pm\infty$. But in the actual circuit the node voltages cannot exceed the supply or ground, so as these trajectories approach the coordinates $(+V_{DD}, \frac{1}{2}V_{DD})$ and $(-V_{DD}, \frac{1}{2}V_{DD})$ they slow down to come to rest at one of these two equilibrium points. These last two equilibria are *stable*. However, the equilibrium at $(0, \frac{1}{2}V_{DD})$ is *metastable*: only if the circuit is initialized along the common-mode axis will it approach the equilibrium; otherwise, for any other initial condition, it will be deflected away from it towards one of the two stable equilibria. In phase plane terminology, the metastable equilibrium defines a saddle point.

Using the phase plane, we are able to understand the characteristic time-domain waveforms of a static latch. In a typical use, the latch is initialized with both nodes connected to, or biased close in voltage to, the power supply. This defines a large initial common mode. A small differential voltage is superimposed to direct the latch regeneration. On the phase plane this may correspond to the initial condition of the trajectory on the upper right. The large initial common mode will decay towards the metastable point, but the small differential mode will grow exponentially. The resulting trajectory is a superposition of the stable eigenvector which lies on the vertical axis, and the unstable eigenvector on the horizontal axis. Because of the large initial condition, the stable eigenvector dominates at first causing V_{O1} and V_{O2} to decay together. Then, as the unstable eigenvector grows, the differential voltage becomes dominant, causing the two voltages to split apart until the circuit reaches a stable equilibrium point. The turnaround point in the $V_{O1}(t)$ waveform, defined by its minimum value, corresponds to that point on the phase plane trajectory where the slope of the vector field is 0.5 in magnitude.

C. Circuit Imbalances

Offsets are important when the latched comparator is used in analog-to-digital conversion. Indeed, even in memory sense applications, offsets in the sense amplifier can be so large as to pose a threat to reliable readout. In a nominally symmetric circuit, offsets arise from parameter mismatch in corresponding pairs of elements, such as in the threshold voltage V_t of

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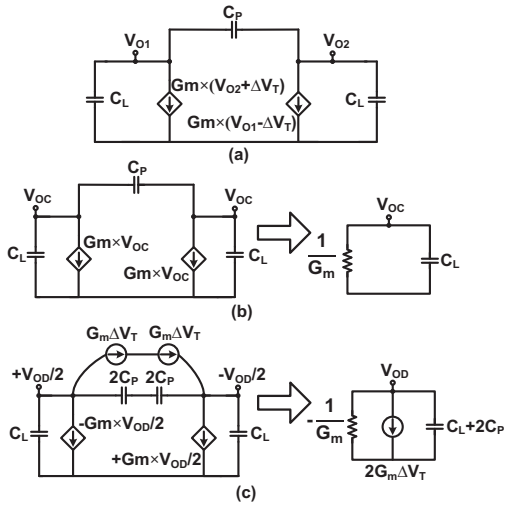


Fig. 4. (a) Equivalent circuit of a static latch with ΔV_T; (b) Common-mode equivalent circuit; (c) Differential mode equivalent circuit.

the NMOS pair, or in the capacitance of the loads C_L. All unbalances appear as an input-referred offset voltage, which we now seek to estimate by simple analysis.

As we investigate the effects of unbalances, it is desirable to continue analysis in terms of half circuits. Middlebrook shows how to treat parameter unbalances in *static* symmetric circuits [4] as equivalent half circuits; we extend this to *dynamic* circuits.

1) Mismatched Trip Points: Let us denote the nominal trip point of each inverter as V_T. This is the point on its static I/O characteristic at which the input and output voltages are equal. In a well-designed inverter, $V_T = ½V_{DD}$. Suppose that due to random spreads in threshold voltages, the trip points of the two inverters are unequal. Without loss of generality, we ascribe a deviation $+\Delta V_T$ to the trip point of one inverter and $-\Delta V_T$ to the other. The latch comprising these mismatched inverters has its metastable point at $V_{OC} = ½V_{DD}$ but $|V_{OD}| = 2\Delta V_T$. That is, the metastable point is translated from its nominal position on the phase plane. Next we show how this mismatch will affect the vector field on the entire phase plane. First we identify the *separatrix* trajectory that leads into the metastable point. This we can do with the aid of equivalent circuits. To the first order the transconductance G_m of the two inverters remains matched for small ΔV_T. Now the linear equivalent circuit is as shown in Fig. 4. If the outputs are separated into modes, this circuit is equivalent to two circuits, one for the common mode and the other for the differential mode. The main point is that $2\Delta V_T$ appears *only* in the circuit for the differential mode an *independent* current source. By definition, the separatrix is that trajectory on the phase plane along which the unstable mode is *not stimulated*. The unstable mode, which results in a growing exponential, arises in this circuit when the negative resistor $(-1/G_m)$ exchanges energy, or interacts, with a capacitor. But suppose the capacitors C_L were precharged (with the appropriate sign) to $2\Delta V_T$. Then the independent current source would find a return path through the two negative resistors, and no current flows through the capacitors. The unstable mode is not excited. Therefore, if the circuit

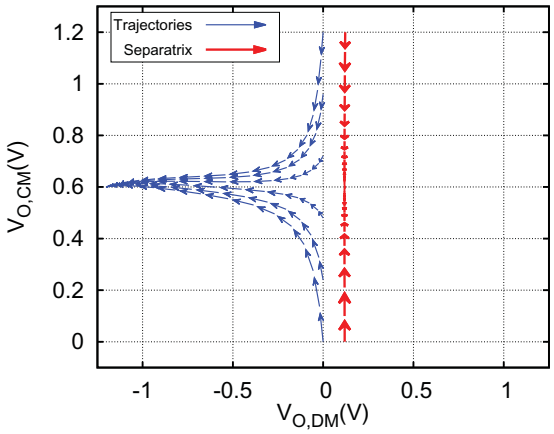


Fig. 5. Phase plane of a static latch with V_T mismatch

is released from an initial condition with any common-mode voltage but a differential voltage of $2\Delta V_T$, the common mode will decay into the metastable point and the differential voltage will remain constant for all time. In short, the separatrix that in the balanced circuit was vertical and coincident with the axis $V_{OD} = 0$ is now, in the presence of mismatched trip points, *translated* by $2\Delta V_T$. An offset of this amount has appeared in the circuit.

2) Mismatch in G_m: Random spreads in FET β will cause mismatch in the transconductance of the two inverters. Writing the transconductances as $G_m \pm \Delta ½G_m$, we use equivalent circuits to examine the effects of this mismatch. We will assume that the trip points are matched, which means that the metastable point remains at the same location on the phase plane as for the balanced circuit.

In the common-mode equivalent circuit (Fig. 6), this mismatch introduces an error current $\Delta ½G_m V_{OC}(t)$ connected in a way that is itself clearly *not* in common-mode. The error current source belongs in the differential mode circuit. Removing this current source restores symmetry to the common-mode circuit, and it is readily seen that if the circuit is initialized with some common mode voltage $V_{OC}(0)$, this will decay into the metastable equilibrium value with a time constant set by the real pole $s_C = -G_m/C_L$.

The current owing to the mismatch ΔG_m appears in the differential mode equivalent circuit as an *independent* source because it is not affected by any of the variables in this circuit. It acts to *cross-couple* the modes [4]. The source waveform must follow the decay of the common mode. We ignore the small perturbation of ΔG_m on $-G_m$. The differential mode's natural response is unstable, caused by a negative conductance charging the shunt capacitance $C_L + 2C_P$. However if at every instant the independent current source carries exactly the sum of the currents through these two elements, then they will not interact and the unstable mode will not be excited. The states of the circuit will follow the separatrix into the metastable point. This requires that the differential voltage is initialized to a $V_{OD}(0)$ such that

$$(-G_m - s_C(C_L + 2C_P))\,V_{OD}(0)e^{-s_C t} = \Delta G_m V_{OC}(0)e^{-s_C t}. \quad (2)$$

Assuming $2C_p \ll C_L$, this condition relates the initial condi-

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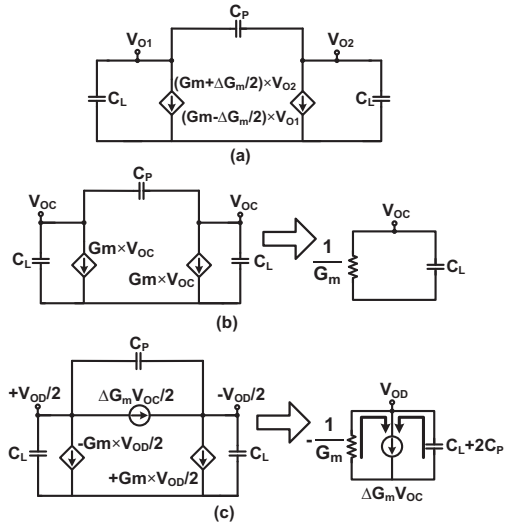


Fig. 6. (a) Equivalent circuit of a static latch with ΔG_m; (b) Common-mode equivalent circuit; (c) Differential mode equivalent circuit with coupled source from common mode circuit.

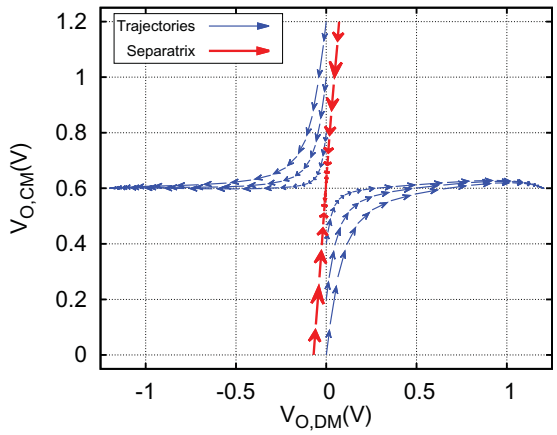


Fig. 7. Phase plane of a static latch with G_m mismatch

tions:

$$V_{OD}(0) \simeq (\Delta G_m/2G_m)\, V_{OC}(0) \tag{3}$$

If the output nodes are initially pulled up to the supply voltage, then $V_{OC}(0) = V_{DD}$. For the latch not to regenerate to '1' or '0', an initial differential voltage given by (3) must be applied at the same time. *This is the offset caused by mismatch in* G_m. It is called a *dynamic* offset because it changes with the initial common mode forced at reset, which here depends on the supply voltage. On the phase plane, this means that the separatrix is *rotated* from a vertical line (Fig. 7).

3) Mismatch in Capacitors: Suppose all corresponding pairs of FETs are matched, but the load capacitors are mismatched. This is modelled by unequal capacitors $C_L \pm \frac{1}{2}\Delta C_L$ connected to the two outputs.

Again, this capacitor unbalance will introduce a cross-coupling current source that cross-couples the common mode waveform into the differential mode with a magnitude of $s\Delta C_L V_{OC}(s)$. Its effect, just like for G_m unbalance, is to rotate the separatrix, and to introduce an offset which depends on

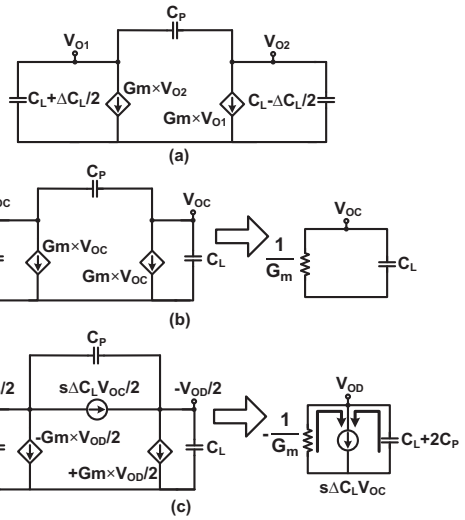


Fig. 8. (a) Equivalent circuit of a static latch with ΔC_L;(b) Common-mode equivalent circuit;(c) Differential mode equivalent circuit with coupled source from common mode circuit.

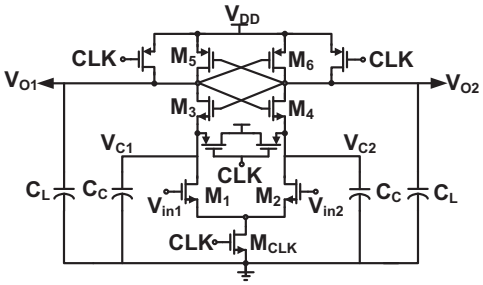


Fig. 9. Schematic of the StrongARM latch.

the initial value of the common-mode voltage as determined by the reset action:

$$\frac{V_{OD}(0)}{V_{OC}(0)} \simeq \frac{\Delta C_L}{2C_L + 2C_P} \tag{4}$$

This is another dynamic offset. (4) is consistent with [5], but here this result is arrived at much more straightforwardly.

III. StrongArm Latch

This widely used latching comparator circuit in Fig. 9 was originally presented as part of a suite of low-power digital circuits [6]. It offers a convenient method to couple a voltage to be regenerated into a pair of cross-coupled inverters—the main weakness of the CMOS static latch—while guaranteeing zero static power consumption when regeneration is complete. It gained widespread attention after it was used in the StrongARM microprocessor.

A survey of the literature suggests that in spite of widespread use, the detailed action of the StrongARM latch is, even now, poorly understood. Without a full understanding it cannot be used properly as a low offset comparator. Therefore, we will first explain how, in correct operation, the circuit traverses two phases over which the applied voltage is amplified before regeneration will start. When poorly designed, an internal regeneration can be triggered as early as in the

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second phase, but with the undesired consequence of a larger offset.

A. Overall Operation

We will describe the circuit in terms of how its operation is most widely understood.

The difference between two input voltages, as measured with respect to ground, is coupled into the latch through the NMOS pair M1-M2. The pair is activated by the tail FET, M_{CLK}, which is assumed to act like a switch. The input voltage common-mode (V_{IC}) must lie above the threshold voltage of M1,M2 to set the bias current.

The bias current also flows through the cross-coupled inverters, M3-M5 and M4-M6, that are stacked in series. Differential current produced by M1-M2 "unbalances" the inverter, causing it to regenerate on this unbalance. Regeneration forces one FET in each inverter to turn OFF, thus choking off a current flow path through both M1 and M2. Thus M1, M2, and the tail current FET are all forced into deep triode with $V_{DS} = 0$ where they no longer conduct current. The comparator consumes no static power in its regenerated state.

This description is sufficient to see why the StrongARM latch is popular today. A closer analysis is needed to understand aspects such as the circuit's inherent latency before it regenerates, and how unbalances in the circuit elements will cause static and dynamic offsets.

B. Operational Phases

The circuit's operation should be divided into three discrete phases, with regeneration understandably taking place in the last phase. In the first two phases, sampling and propagation, the circuit amplifies the applied differential voltage on to internal nodes, as we now explain.

1) Reset State: The phases are most clearly identified when the comparator is released from a well-defined state. The circuit is defined by four state variables, the voltages on two grounded capacitors C_C and on two load capacitors C_L. But it is a time-varying circuit, and in the second and third phases the capacitors exchange charge, collapsing the number of states to two. Be that as it may, the circuit must be initialized with all four states at a predetermined and fixed value so as to erase memory of the previous regeneration. A convenient initialization is to precharge all four capacitor voltages through FET switches to the supply voltage V_{DD}. This initializes the source and drain terminals of M3-M6 all to the same potential.

2) Sampling Phase: The input voltages V_{in1} and V_{in2} are being applied when the tail current transistor M_{CLK} is initialized. The average input voltage sets the bias current (I) through M1 and M2. This common-mode current will be forced to discharge C_{C1} and C_{C2}. M3 and M4 will remain OFF until the capacitors have discharged by an amount equal to the threshold voltage V_{tN}. This period of time defines the sampling phase,

$$T_{smp} = C_C V_{tN} / I \tag{5}$$

Over this period the difference in the input voltages V_{ID}, which creates a differential current $g_{m1}V_{ID}$, integrates a differential voltage across the capacitors C_C of

$$v_{smp} = (g_{m1} V_{ID} \times T_{smp}) \div (½ C_C) \tag{6}$$

This differential voltage serves as the initial condition for the next phase, propagation.

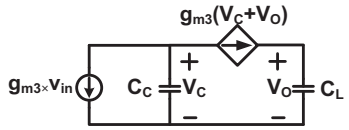


Fig. 10. Differential half circuit during propagation phase of the StrongARM latch.

3) Propagation Phase: In propagation phase, M3 and M4 will turn ON, and by the end of this phase the common mode (bias) current flowing through them will have discharged both output voltages V_{O1} and V_{O2} by V_{tP} to turn on the cross-coupled PMOS pair M5, M6. During this phase M1,M2 and M3,M4 are ON. The voltages on capacitors C_C and C_L will ramp down together, separated by the constant difference of $V_{GS3} (= V_{GS4})$. Thus,

$$T_{prp} = (C_L + C_C)|V_{tP}|/I \tag{7}$$

Although the gates of M3,M4 are cross-coupled to the drains, in common mode the pair of gates follows the same voltage waveform as the pair of drains, as if each FET was diode connected. The cross-coupling becomes evident in differential mode. A cross-coupled pair of transistors will only regenerate if the loop gain is greater than one. This is not so when, as we have assumed, M1,M2 remain in saturation through the propagation phase, and $C_C < C_L$. This is readily proved by replacing M3,M4 with a transconductance g_{M3} in the linearized equivalent circuit of Fig. 10, and representing M1,M2 with a constant current source. The cross-coupling shows in the control variable of g_{M3}, which instead of being the familiar difference of two voltages, each measured with respect to ground, is now their sum. The voltages v_C and v_O are the differential voltages across C_C and C_L, respectively. The two capacitors in this equivalent circuit are in series and it has a single pole in the left half s-plane at

$$s_{prp,dm} = -g_{M3} \div ((C_L C_C)/(C_L - C_C)) \tag{8}$$

When $C_C < C_L$, this cross-coupled NMOS pair will act as a stable amplifier. This simple circuit of Fig. 10 can amplify because the cross-coupling introduces a controlled amount of positive feedback, which, as we have just shown, stops short of outright regeneration.

Provided M1,M2 remain in saturation, the dominant effect over the propagation phase arises from the transimpedance that converts the input pair current, $g_{M1}v_{in}$, to the output voltage v_O. This is defined by integration of a current on to the load capacitor C_L over a time window defined by T_{prp}, with a boost provided by the cross-coupled NMOS pair M3,M4. In addition there is a redistribution of the charge acquired at the end of the sampling phase, through the cross-coupled M3.M4 into C_L. This last effect accounts for the coefficient of 2 in the numerator of the expression for output voltage:

$$V_{OD} = \left(\frac{C_L + 2C_C}{C_L - C_C}\right)\left(\frac{V_{tN}}{I} g_{M1}\right) V_{ID} \tag{9}$$

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4) Regeneration Phase: At the end of the propagation phase, the cross-coupled PMOS pair, M5,M6, turns ON and, since the sources are shorted, it will regenerate the differential voltage present at the output nodes. This voltage is, to a very good estimate, given by (9). The pole for PMOS pair lies in the right-half s-plane, and is given by

$$s_{reg} = +g_{M5}/C_L \tag{10}$$

The regenerated output voltages will grow to control all the FETs in the entire latch. At one of the two stable equilibria, one output voltage will reach V_{DD} and the other ground. No static current will flow on either side of the circuit. If the PMOS pair were to dwell at the metastable equilibrium and the supply voltage was large enough, then the output voltages would be equal $V_{O1} = V_{O2} = V_{DD} - V_{GS}(M5,6)$.

5) StrongARM Latch in Summary: This detailed analysis shows that the StrongARM latch, with all internal nodes initialized to V_{DD}, undergoes two phases of common-mode discharge before it regenerates. These phases amount to a delay of $(T_{smp} + T_{prp})$ before the circuit regenerates with the time constant given by the inverse of the pole frequency (10). Others have noticed that the StrongARM latch regenerates after some latency, or waiting period [7]. But our analysis shows that over this period of latency a useful internal amplification will take place. For the applied input voltage V_{ID}, the gain to the point of onset of regeneration is

$$\frac{V_{OD}}{V_{ID}} = 2\frac{C_L + C_C}{C_L - C_C}\frac{V_{tN}}{V_{P1,2}} \tag{11}$$

where $V_{P1,2}$ is the pinchoff, or overdrive, gate voltage on M1,M2. By substituting typical values into the two terms above, this gain is about $5 \sim 10$.

C. FET Mismatch

The internal gain is important for estmating offset because it amplifies unbalances in the input differential pair most, usually to the point that they will dominate all other FET imbalances in the circuit. This helps to identify the principal source of offset, and then to focus design effort for its mitigation. Both V_t and β mismatch in the input pair will appear at the comparator input as an effective V_{ID}.

D. Capacitor Mismatch

C_C and C_L, both grounded capacitors, are vulnerable to mismatch. To model their contribution to dynamic offset in the simplest way, we follow the method shown in the static latch analysis where capacitor mismatch induces a coupling from common-mode to differential-mode. Since the StrogArm latch operates by discharging both C_C from their reset condition by V_{tN} over the sampling mode, and then, by discharging C_L and C_C by $|V_{tP}|$ over the propagation phase, we expect small mismatch between corresponding pairs of these capacitors to induce significant (differential) mismatch over the large excursions of common mode.

The analysis is simplified considerably if the FETs M1,M2 are assumed to act like current sources throughout the propagation phase, because then we are dealing with a circuit with a cutset of capacitors only, which forms a dissipation-less conservative system. Conservative circuits are fully specified by the initial and final state, without regard to how the state changes in time. Thus, at the end of the sampling phase, because mismatch in the internal capacitors C_C introduces coupling between common and differential modes, a differential offset voltage appears across the C_C's of $(\Delta C_C/C_C)V_{tN}$. Then, during propagation, charge will exchange with the C_L capacitors through the cross-coupled pair M3,M4, and when $C_C < C_L$ (as is usual), the output will settle stably to a *voltage* that is amplified by $(1 - C_C/C_L)^{-1} > 1$, while total *charge*, of course, is conserved.

Thus, assuming $V_{tN} = |V_{tP}| = V_t$, the final offset due to capacitor mismatch alone is

$$v_{OD} = \frac{V_t}{(1 - C_C/C_L)}\left[\frac{\Delta C_L + 2\Delta C_C}{C_L}\right] \tag{12}$$

The term V_t at the head of the left-hand side reminds us that this is a dynamic offset induced by a common-mode excursion across an unbalanced circuit. The term $2C_C$ arises because C_C undergoes a common-mode voltage excursion of $V_{tN} + |V_{tP}|$, almost twice the excursion of the voltage on C_L. Fortunately this circuit rejects typical bounce in supply voltage, so its regeneration time, static offset, and dynamic offset all remain independent of V_{DD}.

E. Offset Compensation

With this background, we can explain comprehensively, and for the first time, how the offset calibration strategies first proposed in [8] for latched comparators will work. We have established that with zero differential input, mismatch in the threshold voltage and β of the pair M1,M2 will usually create the offset voltage at the output nodes according to (9), dominating contributions from mismatch in all other FETs. Now if the capacitances C_C and C_L are fine tuned in closed loop under digital control to create an almost equal offset but opposite in sign according to (12), the algebraic sum of the two offsets will cancel. So by introducing a deliberate and measured mismatch in the capacitors, the comparator may be made to appear offset-free even with random mismatch in the FETs .

The process is readily automated by applying a common-mode voltage to the two inputs of the comparator (zero differential input), which, due to inherent mismatches will cause the comparator to repeatedly produce either a '0' or a '1', then searching through binary-weighted arrays of small capacitors attached to C_C and C_L until the output of the latch changes state. That setting can be held in a register dedicated to each latch.

The expression (12) also shows a benefit of resetting the initial voltages across C_C to V_{DD}. In many practical cases the internal nodes at the drains of M1,M2 are not reset, but are instead pulled up to $V_{DD} - V_{tN}$ by the output nodes when they are reset. Indeed, resetting these nodes to V_{DD} prolongs the latency period by T_{smp} (see (5)), which seems undesirable. But (12) reveals that as a result of this delay ΔC_C is twice as effective in compensating offset. Thus, at the expense of

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a larger latency, a lighter capacitor loading may be used for offset compensation.

But doesn't the greater latency annul the benefit of a lighter capacitor loading? To answer this question, we examine the critical comparator in an A/D converter. A comparator is in critical condition when it is resolving an input so close to a threshold that regeneration may not complete in the allotted clock phase; that is, the converter would make a metastability error. This is limited by the regeneration time constant, seldom latency. In the StrongARM latch it is the PMOS pair M5,M6 that regenerates, and in the initial part of the regeneration transient it is loaded only by C_L. When C_C shoulders the larger burden of compensating latch offset, the latch regeneration time constant is essentially unchanged. When the offsets are large, a better balance between latency and regeneration may require that the calibration arrays are more evenly distributed between the C_C and C_L nodes.

F. Range of Input Common Mode

Our survey of the published uses of the StrongARM latch shows that in most circuit realizations, the input common mode voltage is chosen poorly. This can degrade offset quite considerably for the reasons that we will now explain. This was observed experimentally in an early use of the latch as an SRAM sense amplifier [7].

Over the sampling and propagation phases, the input common-mode voltage (V_{IC}) sets the bias current through M1-M4, but also determines whether or not these FETs operate in saturation. This is best understood in a perfectly balanced, offset-free latch with zero differential input, which upon release from reset should travel into its metastable state. It is sufficient to examine the state of the circuit when the regenerative pair M5,M6 starts to conduct, i.e. when $V_{O1} = V_{O2}$ reach $V_{DD} - |V_{tP}|$. Since this excursion is purely in common mode, M3,M4 will behave as if they are diode connected and therefore they will operate in their saturation region. But for the input pair M1,M2 to operate in saturation, V_{IC} must not exceed the upper limit

$$V_t < V_{IC} < V_{DD} - V_{GS}(\text{M3,M4}) \tag{13}$$

What if V_{IC} is larger? Then at some point during the propagation phase M1,M2 will be pushed into their triode region. Now consider that an input differential voltage, V_{ID}, is being applied—this may well represent an equivalent offset—and divide the propagation phase into two sub-phases: over the first sub-phase M1,M2 appear like a differential current source $g_m V_{ID}$; and over the second sub-phase they appear like a resistor $1/g_m$. This is a piecewise simplification of a circuit that, over time, is pushed from saturation into deep triode, and it is captured by two different equivalent circuits over the sub-phases (Fig. 11). They show that the benefits of an internal amplification in the latch, as we have argued above, are now eroded in two ways: the integration window while M1,M2 remain in saturation is now only a fraction of T_{prp}, which lowers the amplification; and the integrated voltage stored on C_C, also shared by C_L, leaks away through the resistor during the second sub-phase. Now mismatches in M3,M4 and M5,M6

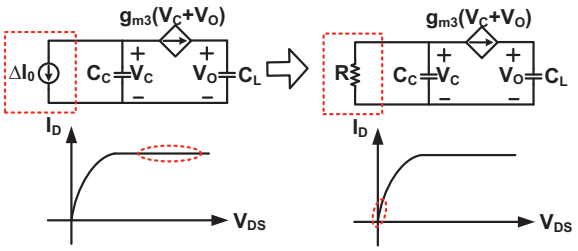


Fig. 11. Equivalent differential circuit during propagation phase when $M1\,\&\,2$ transit from saturation to triode region.

will add significantly to the mismatches of M1,M3, causing the total offset to double or more.

IV. Thermal Noise

Comparators are mainly limited by random or systematic offsets. But long-standing methods to circumvent offsets, such as overranging and digital error correction in multi-step A/D converters, are now running up against limits posed by low supply voltages. And although it might not be apparent today that thermal noise poses a bottleneck to performance, we expect it to be so once offset compensation in comparators is better understood and being widely practiced.

There is some published work to model noise in the StrongARM latch [9], [10]. [10] identifies the operational phases of this circuit correctly, but it uses a method of noise analysis that is too indirect to yield some key insights: for instance, that the input common mode must be chosen correctly (see Section III-F above) in order to minimize noise.

Our analysis of offset in the StrongARM latch, on the other hand, extends straightforwardly to a prediction of thermal noise. But first we must describe how noise will randomly trigger regeneration around the metastable point. In a perfectly balanced comparator with zero input applied, the noise current in the input pair M1,M2 will integrate on C_C during the sampling phase, and continue to integrate and amplify on C_L during the propagation phase. The random voltage integrated on C_L will trigger regeneration to '0' or '1'. Now integration of a white noise current with single-sided spectral density S_{in} on a capacitor C over a known time window T, for example T_{prp}, produces a voltage with variance [11, p. 331]

$$\langle v_n^2 \rangle = S_{in}\left(T/2C^2\right) \tag{14}$$

If the input common-mode does not exceed the limit in (13), the noise of the input pair will experience the largest amplification and it will account for most of the input-referred noise. Otherwise, for two reasons, the input-referred noise will grow larger: first, because of the lower g_m of M1,M2 since they are in triode region; and second, because M3,M4 will now contribute substantially to the total noise.

V. Comparison with Measured Data

All the expressions relating to static and dynamic offsets closely match the offset derived from transient simulations of the StrongARM latch as its input is swept. On the other hand, it is surprisingly difficult to find measured data in the many publications on this comparator in a form that could

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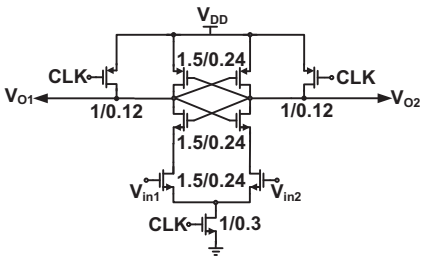


Fig. 12. Test circuit in [7]. All dimensions in μm, V_{DD}=1.5 V.

be used to validate the analysis developed in this paper: [7] is the exception, because it provides histograms measured across 45 samples of a latch realized in 130-nm CMOS. Notably the measured offset grows by more than $2\times$ when the input common-mode exceeds the limit specified by (13). The circuit in question is shown in Fig. 12. Table I shows that the measured RMS offset at two common-mode levels agrees very well with predictions from our analysis. This is more than a matter of making the numbers come out close; our predictions were made to verify certain hypotheses. Although Infineon fabricated the circuit being measured, we use publicly available mismatch coefficients taken from the TSMC 130 nm process.

For $V_{IC} = 0.7V_{DD}$ which lies within the range in (13), we assume that offset arises *only* from V_t mismatch and β mismatch in M1,M2. Since our prediction is very close (for this size of population) to the total measured offset, it verifies the hypothesis that due to internal amplification, M3-M6 will not contribute appreciably to mismatch. For $V_{IC} = V_{DD}$ the analysis is somewhat more complicated because the gain $V_{O1,2}/V_{ID}$ changes over the two sub-phases described in Section III-F. We use the simple mathematical device of averaging the gain over the two sub-phases, and including the offset from V_t mismatch in M3,M4 (β mismatch doesn't matter because these FETs are in series with M1,M2). The prediction in this case is accurate. In spite of the lower internal amplification, mismatch in M5,M6 still does not contribute. However, it is clear that too large a V_{IC} will worsen the comparator's offset considerably.

VI. Conclusion

We have developed a simple, physically-based analysis for the internal workings of a regenerative comparator. It recognizes the symmetry of the circuit, by identifying common mode and differential mode. The large-signal behaviour is readily understood by plotting equilibria and trajectories in the phase plane *defined by these modes*. It is most fruitful to consider comparator offset and noise as perturbations on the separatrix, the phase plane trajectory that leads from the initial reset condition into the metastable equilibrium. This brings clarity to long-debated distinctions between static and dynamic offsets in regenerative comparators. Simple equivalent circuits are shown to capture all time-varying aspects. For small offsets, nonlinearity in FET g_m does not change the shapes of the trajectories.

These insights guide design of the widely used StrongARM comparator. Following reset, three phases of operation are identified, with the role of pairs of FETs. An internal amplification is revealed, and from it follows the need to choose the input common-mode voltage correctly, for uses where the offset and noise should be kept very small. Most importantly, it is shown how by using a small array of switched capacitors, a dynamic offset can be made to cancel the static offset with little penalty on speedt. This calibration is robust against supply fluctuations.

Predictions from the analysis are validated against measurements taken on a prototype StrongArm comparator. The measurements illustrate the major design considerations to emerge from the analysis. Our analysis of offset and its implications on design is much more direct than previous work on the topic, such as [12].

We are grateful to Professor Bernhard Wicht of Reutlingen University, Germany, who shared sizes of the FETs in the SRAM sense amplifier that he had reported in a perceptive early work.

	$V_{IC} = 1.05V$	$V_{IC} = 1.5V$
Measured Offset	8.5 mV	19.0 mV
Calculated Offset	9.5 mV	19.1 mV
Due to $\Delta V_{t1,2}$	5.8 mV	5.8 mV
Due to $\Delta\beta_{1,2}$	7.4 mV	14.0 mV
Due to $\Delta V_{t3,4}$	2.0 mV	11.6 mV

TABLE I
Measured RMS offset vs. calculated. FET mismatch parameters: $A_{Vt} = 3.5\text{mV}\cdot\mu\text{m}$, $A_\beta = 2.5\%\cdot\mu\text{m}$.

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Behzad Razavi

The StrongARM Latch

The StrongARM latch topology finds wide usage as a sense amplifier, a comparator, or simply a robust latch with high sensitivity. The term "StrongARM" commemorates the use of this circuit in Digital Equipment Corporation's StrongARM microprocessor [1], but the basic structure was originally introduced by Toshiba's Kobayashi et al. [2]. The StrongARM latch has become popular for three reasons: 1) it consumes zero static power, 2) it directly produces rail-to-rail outputs, and 3) its input-referred offset arises from primarily *one* differential pair. In this column, we study the circuit and its properties.

Basic Operation

Figure 1(a) shows the original StrongARM latch, reported in [2] without M_8 and in [1] with M_8. The circuit was later modified as depicted in Figure 1(b) [3]. We first study the latter and then point out the differences among these versions.

The latch of Figure 1(b) consists of a clocked differential pair, M_1–M_2, two cross-coupled pairs, M_3–M_4 and M_5–M_6, and four precharge switches, S_1–S_4. The circuit provides rail-to-rail outputs at X and Y in response to the polarity of $V_{in1} - V_{in2}$. We describe the operation in four phases.

In the first phase, CK is low; M_1 and M_2 are off; nodes P, Q, X, and Y are precharged to V_{DD}; and the circuit reduces to that shown in Figure 2(a).

In the second phase, CK goes high, S_1–S_4 turn off, and M_1 and M_2 turn on, drawing a differential current in proportion to $V_{in1} - V_{in2}$. With M_3–M_6 initially off, this current flows from C_P and C_Q [Figure 2(b)], thereby allowing $|V_P - V_Q|$ to grow and possibly exceed $|V_{in1} - V_{in2}|$. That is, this phase can provide voltage gain. We call this phase the amplification mode. Since the tail current is fairly constant during this period, we can write $|V_P - V_Q| \approx (g_{m1,2}|V_{in1} - V_{in2}|/C_{P,Q})\,t$, where $g_{m1,2}$ denotes the small-signal transconductance of M_1 and M_2, and $C_{P,Q} = C_P = C_Q$.

As V_P and V_Q fall to $V_{DD} - V_{THN}$, the cross-coupled NMOS transistors turn on (third phase), allowing part of the drain currents of M_1 and M_2 to flow from X and Y [Figure 2(c)]. The amplification mode therefore lasts for approximately $(C_{P,Q}/I_{CM})\,V_{THN}$ seconds, where I_{CM} is the common-mode (CM) current drawn from each capacitance. The voltage gain in this mode is roughly given by [4]

$$A_v \approx \frac{g_{m1,2}\,V_{THN}}{I_{CM}}. \qquad (1)$$

The behavior of the latch in the third phase can be analyzed with the aid of the equivalent circuit shown

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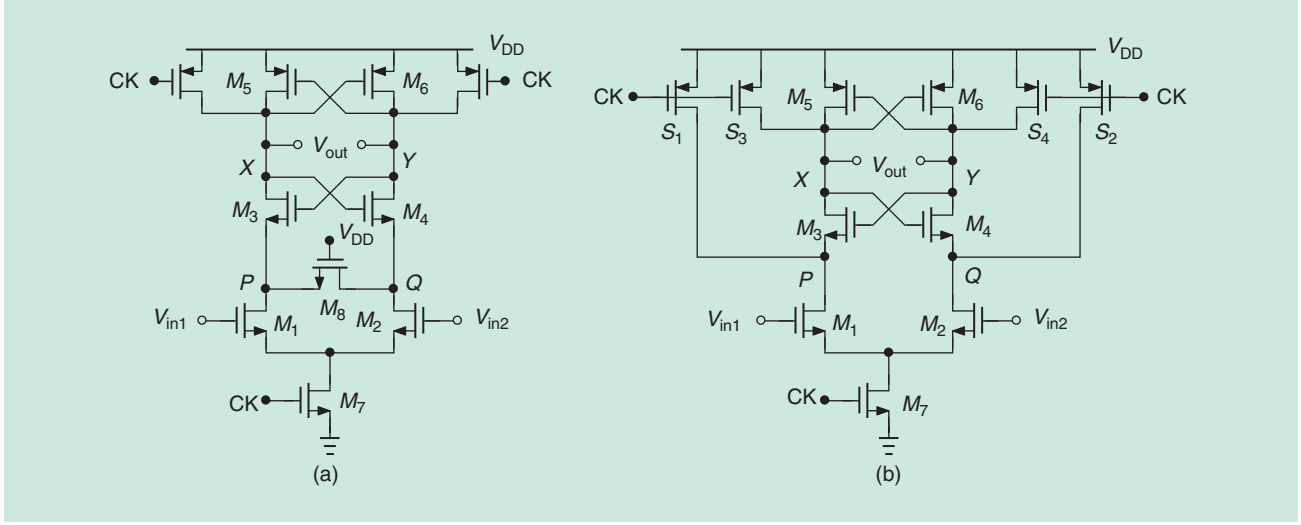


FIGURE 1: (a) The original and (b) modified StrongARM latch topologies.

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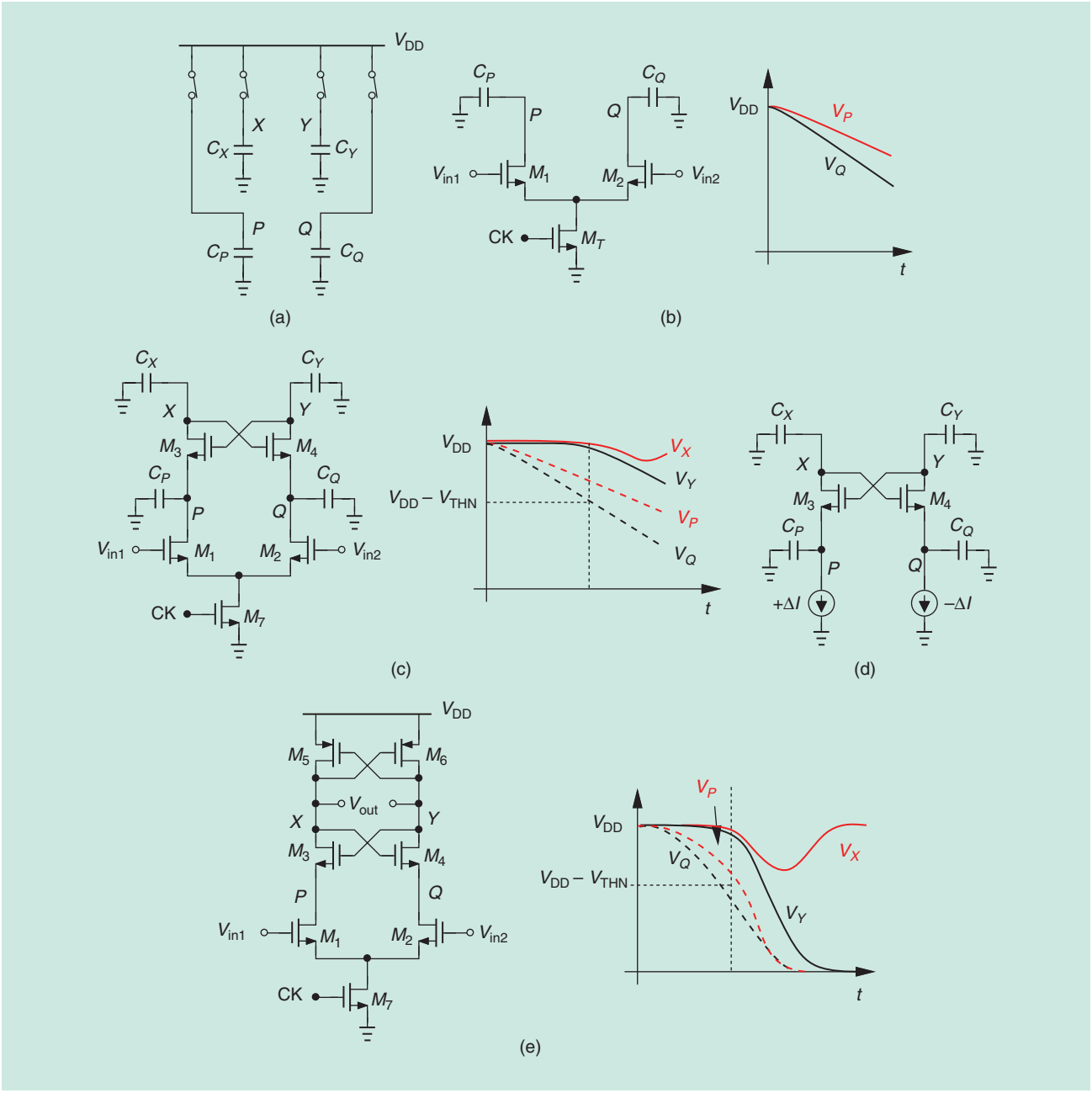


FIGURE 2: Latch operation phases: (a) precharge, (b) amplification, (c) turn-on of cross-coupled NMOS pair, (d) equivalent circuit of (c), and (e) turn-on of cross-coupled PMOS pair.

in Figure 2(d), where $+\Delta I$ and $-\Delta I$ represent the differential current produced by M_1 and M_2. Summing currents at the four nodes yields

$$-C_X\frac{dV_X}{dt} = g_{m3}(V_Y - V_P) \tag{2}$$

$$-C_Y\frac{dV_Y}{dt} = g_{m4}(V_X - V_Q) \tag{3}$$

$$-C_P\frac{dV_P}{dt} = C_X\frac{dV_X}{dt} + \Delta I \tag{4}$$

$$-C_Q\frac{dV_Q}{dt} = C_Y\frac{dV_Y}{dt} - \Delta I. \tag{5}$$

We subtract the second equation from the first, obtaining

$$-C_{X,Y}\frac{d(V_X - V_Y)}{dT} = g_{m3,4}(-V_X + V_Y - V_P + V_Q). \tag{6}$$

Integrating both sides of (4) and (5) and combining the results, we have

$$C_{P,Q}(V_Q - V_P) = C_{X,Y}(V_X - V_Y) + 2\Delta It, \tag{7}$$

which, upon substitution in (6), gives

$$C_{X,Y}\frac{d(V_X - V_Y)}{dt} - g_{m3,4}\left(1 - \frac{C_{X,Y}}{C_{P,Q}}\right)(V_X - V_Y) = -2g_{m3,4}\frac{\Delta I}{C_{P,Q}}t. \tag{8}$$

If $g_{m3,4}$ is assumed relatively constant, this equation reveals a natural response of the form $\exp(t/\tau_{reg})$, where τ_{reg} is the regeneration time constant and expressed as

$$\tau_{reg} = \frac{C_{X,Y}}{g_{m3,4}(1 - C_{X,Y}/C_{P,Q})}. \tag{9}$$

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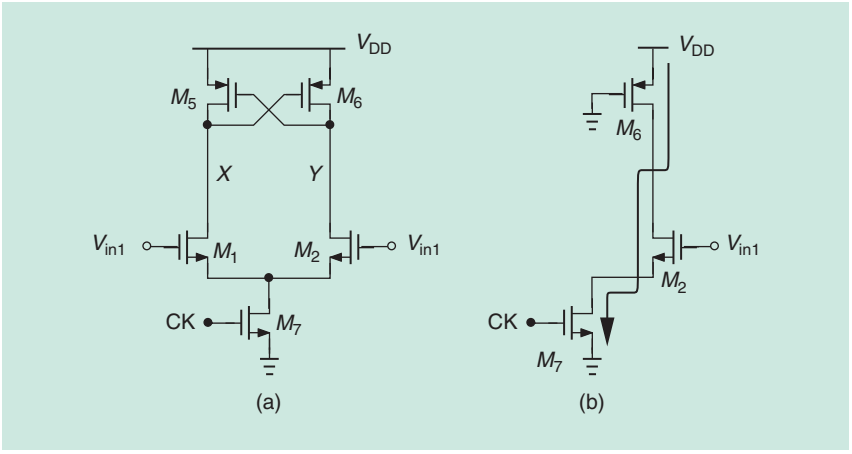


FIGURE 3: (a) Latch without ac cross-coupled NMOS pair and (b) the resulting static current.

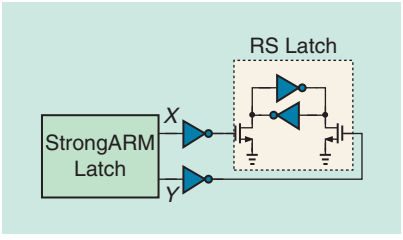


FIGURE 4: The StrongARM latch followed by the RS latch.

Interestingly, the degeneration caused by C_P and C_Q raises τ_{reg} by a factor of $1 - C_{X,Y}/C_{P,Q}$. Since, in practice, $C_{X,Y}$ includes the input capacitance of the stage following the comparator and is hence greater than $C_{P,Q}$, the cross-coupled NMOS transistors provide little regeneration in this phase.

The output voltages V_X and V_Y continue to fall until they reach $V_{DD} - |V_{THP}|$, at which point M_5 and M_6 turn on [Figure 1(e)] and the circuit enters the fourth phase. The positive feedback around these transistors eventually brings one output back to V_{DD} while allowing the other to fall to zero.

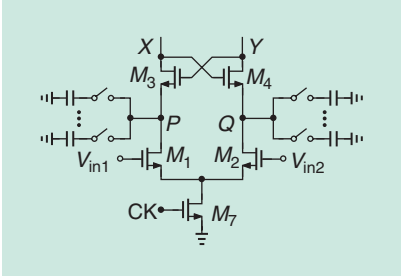


FIGURE 5: Offset cancellation by programmable capacitors.

It is important to appreciate the role of each transistor in the StrongARM latch of Figure 1(b). Besides M_1–M_2 and M_7, the remaining devices also serve critical purposes.

- Transistors M_3–M_4 cut off the dc path between V_{DD} and ground at the end of the fourth phase, avoiding static power drain. To understand this point, let us omit M_3 and M_4 as shown in Figure 3(a) and assume a differential input, $V_{in1} - V_{in2}$, of about 100 mV around a common-mode (CM) level near $V_{DD}/2$. When the latch is clocked, V_X falls, V_Y rises, and M_5 turns off. Consequently, the circuit reduces to that in Figure 3(b), drawing a static current from V_{DD}. (This does not occur for rail-to-rail inputs.)
- Transistors M_5 and M_6 principally restore the output high level to V_{DD}; without them, the CM discharge at X or Y would yield a degraded high level (if $|V_{in1} - V_{in2}|$ is small).
- Switches S_1 and S_2 play two roles: a) remove the previous states at nodes P and Q, suppressing dynamic offsets, and b) establish an initial voltage of V_{DD} at these nodes, allowing amplification before M_1 and M_2 enter the triode region. Both of these points distinguish the topology of Figure 1(b) from that in Figure 1(a). The original StrongARM latch fails to equalize V_P and V_Q accurately because M_8 turns off near the end of the precharge mode. Without M_8, the dynamic offset would prove even more serious. Moreover, the circuit has little voltage gain in the amplification mode for V_P and V_Q begin at $V_{DD} - V_{THN}$. Since in this case M_3 and M_4 turn on before significant gain accrues, they contribute a greater offset.
- Switches S_3 and S_4 precharge X and Y to V_{DD}, ensuring that M_5 and M_6 remain off during the initial amplification and negligibly raise the offset.

The StrongARM latch generates invalid outputs ($V_X = V_Y = V_{DD}$) for about half of the clock cycle. For the subsequent logic to interpret the outputs correctly, an RS latch must follow the circuit. Figure 4 shows a typical arrangement where inverters serve as buffers between the two latches and allow the RS latch to toggle only if V_X or V_Y falls.

The power consumed by the StrongARM latch of Figure 1(b) arises from primarily the charge and discharge of the capacitances. It is therefore roughly equal to $f_{CK}(2C_{P,Q} + C_{X,Y})V_{DD}^2$, where f_{CK} is the clock frequency and the factor of 2 accounts for the discharge of both P and Q to near ground in every cycle.

Offset

If operating as a sense amplifier or a comparator, the StrongARM latch must achieve a sufficiently small input-referred offset voltage. As explained in the previous section, the precharge action of S_1–S_4 in Figure 1(b) keeps M_3–M_6 off initially, thereby reducing their offset contribution. In a typical design, the mismatches between M_3 and M_4 are divided by about a factor of $A_v \approx 4$ when referred to the input, and those between M_5 and M_6 by about a factor of ten (because these transistors turn on only near the end). Thus, M_1 and M_2 become the dominant contributors.

Since the amplification mode provides voltage gain by the flow of charge from C_P and C_Q, one can create asymmetry by making $C_P \neq C_Q$ and hence cancel the circuit's offset. Illustrated in Figure 5 [5], [4], the idea is to establish different discharge rates at P and Q. Writing

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the drain current of each transistor as the sum of a component proportional to $V_{in1} - V_{in2}$ and a CM component, I_{CM}, we have

$$V_P = V_{DD} - \frac{g_{m1}(V_{in1} - V_{in2})}{2C_P} t - \frac{I_{CM}}{C_P} t \quad (10)$$

$$V_Q = V_{DD} + \frac{g_{m2}(V_{in1} - V_{in2})}{2C_Q} t - \frac{I_{CM}}{C_Q} t. \quad (11)$$

It follows that

$$V_P - V_Q = -\frac{g_{m1,2}}{2} \cdot \frac{C_P + C_Q}{C_P C_Q} (V_{in1} - V_{in2})\, t + \frac{C_P - C_Q}{C_P C_Q} I_{CM}\, t. \quad (12)$$

We observe that during amplification $V_P - V_Q$ accumulates an offset equal to $(C_P - C_Q)/(C_P C_Q) I_{CM} t$, which can cancel the latch's random offset. The amplification mode ends roughly when V_P and V_Q fall below $V_{DD} - V_{THN}$, and its duration is given by $t \approx V_{THN}(C_P + C_Q)/(2I_{CM})$, where $(C_P + C_Q)/2$ is used as an approximation. The built-in offset is therefore equal to $V_{THN}(C_P/C_Q - C_Q/C_P)/2$.

To perform offset cancellation, the main inputs are shorted together, the circuit is clocked, and the output decision drives a register that controls the values of C_P and C_Q [5], [4]. Of course, to reduce the offset from a high value (e.g., 30 mV) to a low value (e.g., 1 mV), a large number of small unit capacitors must be attached to P and Q, degrading the speed *and* raising the power dissipation. Another offset cancellation method for the StrongARM latch is described in [6].

Electronic Noise

From the foregoing offset studies, we can predict that the precharge action of S_1–S_4 in Figure 1(b) also reduces the electronic noise contributed by M_3–M_6. Most of the input-referred noise originates from M_1 and M_2 and the kT/C noise deposited by S_1 and S_2 because the other transistors come into play only after significant gain has accrued. In the amplification mode, the equivalent circuit of Figure 2(b) behaves as an integrator, generating output noise from the noise of M_1 and M_2. The variance of this voltage (a quantity akin to the mean square value) grows with time as [4], [7], [8]

$$E(V_{PQ}^2) = \frac{8kT\gamma g_{m1,2}}{C_{P,Q}^2} t. \quad (13)$$

Since the amplification mode lasts about $(C_{P,Q}/I_{CM}) V_{THN}$ seconds, we compute the final output noise variance due to M_1 and M_2 in this mode as

$$\sigma_{1,2}^2 = \frac{8kT\gamma}{C_{P,Q}} \cdot \frac{g_{m1,2} V_{THN}}{I_{CM}}. \quad (14)$$

Adding the kT/C noise contributed by S_1 and S_2, dividing the result by the square of the voltage gain, and writing $g_{m1,2} \approx 2I_{CM}/(V_{GS} - V_{THN})_{1,2}$, we obtain the total (integrated) input-referred noise observed in this mode as

$$\overline{V_{n,in}^2} = \frac{(V_{GS} - V_{THN})_{1,2}}{V_{THN}} \cdot \left[\frac{4kT\gamma}{C_{P,Q}} + \frac{(V_{GS} - V_{THN})_{1,2}}{V_{THN}} \frac{kT}{2C_{P,Q}} \right]. \quad (15)$$

The first term within the square brackets represents the noise due to M_1 and M_2 and is typically four to eight times greater than the second. Other sources of noise are quantified in [4].

While not specific to the StrongARM latch, the simulation of noise in comparators poses interesting issues. Unlike small-signal analog circuits, a comparator does not directly provide an output noise and a gain by which the noise should be divided. For simpler topologies, one can place the comparator in a metastable condition and perform a small-signal analysis, but the StrongARM latch completes switching actions and noise injections even *before* the output begins to change. A methodical simulation proceeds as follows. Suppose a comparator with a zero offset and a zero differential input is clocked many times (we assume the simulator includes noise in transient simulations). Then, the Gaussian noise within the circuit allows eventual recovery from metastability, producing ones and zeros at the output with equal probabilities [Figure 6(a)].

In the next step, we apply a small differential input (a few millivolts) as shown in Figure 6(b) and repeat the simulation. Since V_S skews the comparator decisions, the ones and zeros occur with unequal probabilities; zeros appear only if the input-referred noise is more negative than $-V_S$.

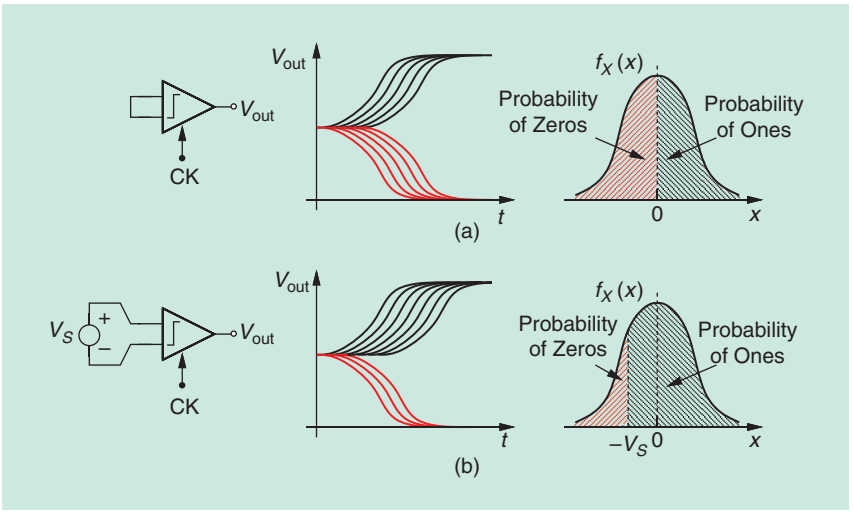


FIGURE 6: The behavior of noisy comparator with a (a) zero and (b) finite input differences.

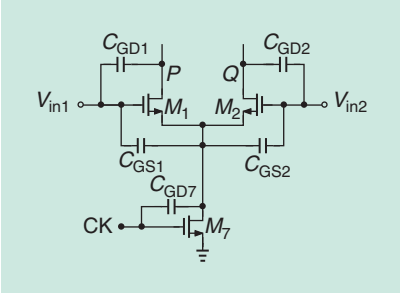


FIGURE 7: Kickback noise paths.

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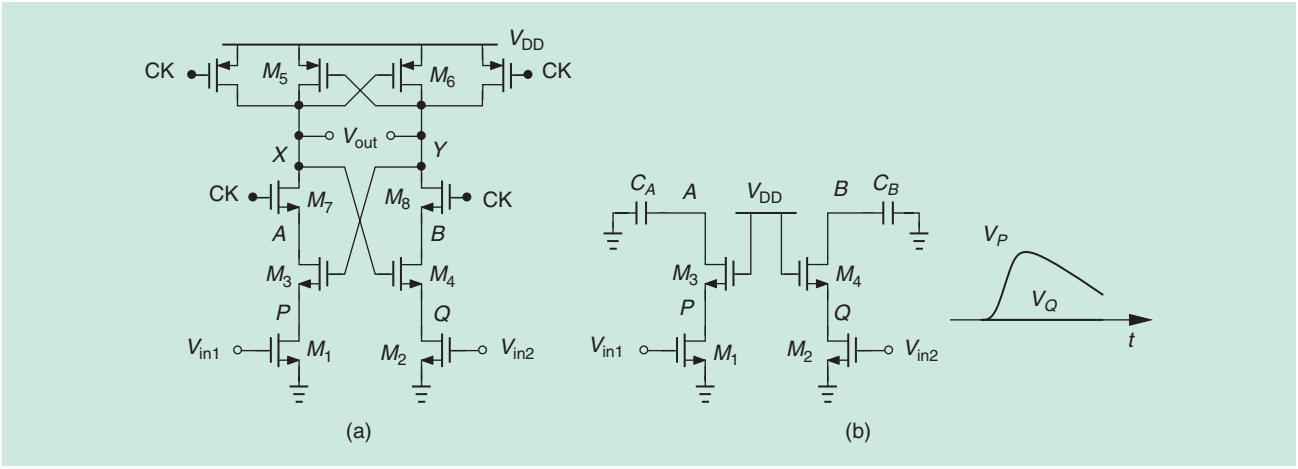


FIGURE 8: (a) An alternative topology for lower kickback noise and (b) behavior in the precharge mode.

For a large number of clock cycles, therefore, we predict that the number of zeros at the output, n_0, is proportional to the area under the Gaussian probability distribution function, $f_X(x)$, from $-\infty$ to $-V_S$; the number of ones, n_1, is proportional to the area from $-V_S$ to $+\infty$. Based on the numbers observed in the simulations, we can write

$$\frac{\int_{-\infty}^{-Vs} f_X(x)\,dx}{1-\int_{-\infty}^{-Vs} f_X(x)\,dx} = \frac{n_0}{n_1} \qquad (16)$$

and hence compute the variance of $f_X(x)$, which corresponds to the input-referred noise voltage squared. The value of V_S must be chosen large enough to ensure n_0/n_1 substantially departs from unity but not so large that n_0 or n_1 is excessively small and statistically insignificant.

Kickback and Supply Transients

The StrongARM latch draws high transient currents from the inputs and the supply. These transients become troublesome if a large number of comparators operate in parallel, as in a flash analog-to-digital converter.

The "kickback" currents drawn from the inputs stem from several mechanisms (Figure 7), exhibiting both differential and CM components. The former appear mostly as V_P and V_Q fall toward ground at unequal rates and couple to the inputs through C_{GD1} and C_{GD2}. This effect becomes more pronounced as M_1 and M_2 enter the triode region and their gate-drain capacitances increase. The CM kickback noise currents are much greater and occur when M_7 turns on, initially drawing its drain current from C_{GS1} and C_{GS2}, and when it turns off, with CK coupling through $C_{GD7}(\approx C_{GS7})$ to C_{GS1} and C_{GS2}.

The StrongARM latch draws high transient currents from the inputs and the supply.

It is possible to reduce the kickback noise by clocking the input devices through their drain path rather than their source path. Depicted in Figure 8(a) [9], such a topology incorporates M_7 and M_8 to control the latch. However, the kickback noise is lowered at the cost of a higher input offset because M_1 and M_2 now operate in the triode region during the amplification mode. This issue can be avoided by making M_3-M_4 and M_7-M_8 wide; but, as illustrated in Figure 8(b), the slow discharge at A or B in the precharge mode leads to significant imbalance between V_P and V_Q and hence a large dynamic offset.

The supply transient currents originate from the precharge action of S_1-S_4 in Figure 1(b). If CK falls fast, three of S_1-S_4 momentarily enter the saturation region (the fourth one is in the triode region because its drain voltage is equal to V_{DD}) and pull a large current from V_{DD}. The key point here is that designs consuming a low average power may still draw high peak currents from the supply, dictating a low supply impedance.

Questions for the Reader

1) Do V_P and V_Q in Figure 1(b) reach 0 V at the end of the regeneration phase?
2) Explain why M_3 and M_4 in Figure 1(b) can be omitted if the inputs have rail-to-rail swings.
3) Explain why the coupling through C_{GD7} in Figure 7 is less on the rising edge of CK than on the falling edge of CK.

You can share your thoughts with me by sending an e-mail to razavi@ee.ucla.edu.

Answers to Last Issue's Questions

1) Can we use a negative impedance converter (NIC) in a PA predriver to cancel the input capacitance of the output stage?

   Since an RF predriver typically uses a resonant load, the NIC would cause oscillation. If injection-locking is desired in this stage, a simple cross-coupled pair suffices.

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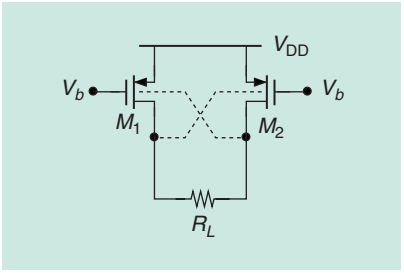


FIGURE 9: A cross-coupled pair using bulk terminals.

2) How does the thermal noise contributed by M_1 and M_2 in Figure 9 to V_{out} compare to that by a regular XCP?

This circuit produces a noise voltage across R_L equal to $\sqrt{2}\, g_m R_L / (2 - g_{mb} R_L V_n)$, where V_n denotes the gate-referred noise of each transistor and the noise of R_L is neglected. For a regular cross-coupled pair, the output noise is given by $\sqrt{2}\, g_m R_L / (2 - g_m R_L V_n)$. For a fair comparison, the total resistance seen at the output must be the same for the two topologies, thus yielding the same output noise.

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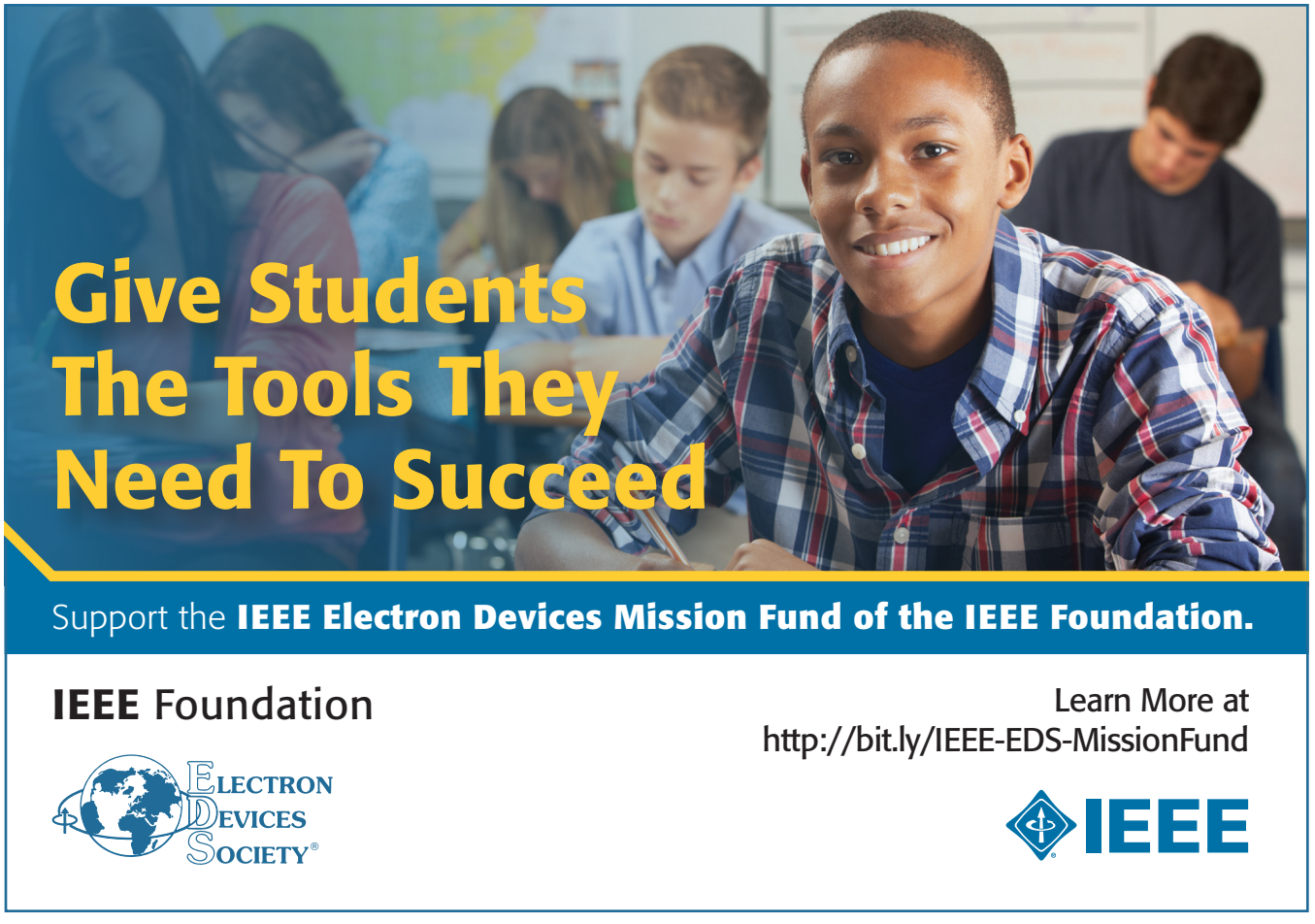
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